

AetherArray

Self-Recalibrating Reconfigurable RF Array for Phased-Array Calibration
and ISAC Research

Master Technical Reference

Version 0.2
October 2026

Repository revision: 415cb685a34be02bb3de3a589be7b1a02ccbae80
Typst source compiled from: main.typ, converted from docs/aetherarray-master-reference.md

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AetherArray: master technical reference

- Status: draft, first issue, for review by the project owner
- Last reviewed: 2026-10-04
- Document version: 0.2: second pass making integrated sensing and communication (ISAC) a first class technical concept, section 10bis; first issue was version 0.1 at commit de61a40
- Repository baseline: main@33086b9395f57f4056a3205e7e8174d50defd500, committed 2026-10-04, “Add SIM-001 analysis tooling”. This is one commit newer than e772a25, the baseline named in the request for this document, so the newer commit is the one described.
- Canonical stack-up: reva-stackup-r1:6363d8ab0f2b, decision 0009
- Test suite at this baseline: 160 tests collected and passing, `cd tools && python -m pytest rfkit/tests -q`, Python 3.12 with `requirements.txt`. This count is a snapshot of 2026-10-04, not a property of the project; rerun the command for the current figure.
- How this document was produced: written from the repository contents at the baseline above, with external literature and the IEEE contest rules researched on 2026-10-04; the ISAC pass of version 0.2 found no newer repository evidence, and the official IEEE and publisher sites were still unreachable from this environment, so no citation could be upgraded to a primary read. It records no new decision and changes no existing one. Where it proposes something, it says so.

The one paragraph to read if you read nothing else. AetherArray is a four element, 2.44 GHz, phase only phased array that has been **designed but not built**. Its purpose is scientific rather than commercial: to measure, on real hardware, whether the history of how an RF array drifts can be used to recalibrate it with fewer new physical measurements than calibrating it from scratch. The architecture, the working frequency, the controller, the printed circuit board stack-up and the acceptance rules for the first simulations are all decided and recorded. No board has been fabricated, no full wave simulation has produced a result, no drift has been measured and no learning model exists. The 2027 IEEE AP-S Student Design Contest, whose theme is reconfigurable receiving antennas for integrated sensing and communication, is the intended external test of the idea; that ambition is described here but is not yet recorded in any decision of the repository.

How to use this document

Who it is for

Reader	Start with	Then read	You can skip
RF or microwave professor	Part 0, Part III, Part IV, Part XI	Part V, Part VI, Part X	Part I sections 1 to 4
Antenna professor	Part 0, sections 6, 7, 9, 10, 10bis, Part IX	sections 22, 58, Part VI	sections 17 to 20
Signal processing or machine learning professor	Part 0, sections 10bis, 11 and 12, Part VIII	Part II, Part XV, section 57	Part IV
IEEE AP-S or MTT-S mentor, competition jury	Part 0, section 10bis, Part IX, Part X, Part XXI	Part VIII section 44, Part XVII	Part I
Student joining the project	Part I in full, then Part 0 again	Part III, Part V, Appendix L	nothing
Hardware or RF recruiter	Part 0, Part X, Part XII	Part III, Part V	Part XV

How claims are labelled

The repository has several vocabularies for evidence, each in the place it was needed. This document uses all of them and never mixes them up. They are explained in section 34; a short version is here so the labels make sense from the first page.

Label	Meaning in this document
[observed]	somebody stood in front of the instrument or hardware and read it; date given

Label	Meaning in this document
[documented], written [inventory] where the repository uses that word	listed in an inventory or an older record; present state unknown
[listing]	stated only by a distributor or aggregator, not by the manufacturer
[vendor]	stated by a manufacturer for a part or model family, not measured here
[analytical]	computed from a closed form model or a repository script; not a simulation of the real geometry and not a measurement
[simulated]	produced by a full wave or circuit solver on a modelled geometry. No such result exists yet
[measured]	produced by an instrument on project hardware. No such result exists yet
[synthetic]	generated from a formula to test software; never evidence about hardware
[decided]	fixed by an accepted decision record in decisions/
[planned]	specified in the repository but not done
[proposed here]	suggested by this document; not in the repository; needs a decision before it counts
[assumed]	a working hypothesis, stated so it can be invalidated
[to verify]	an open question, with the method that would settle it where known
[snippet only]	an external fact read only through search engine excerpts of the official page, because the page itself could not be opened from this environment

Internal facts are cited by repository path or identifier, for example decision 0007, experiments/EXP-005-repeatability-floor.md, or uncertainty I24 in docs/uncertainties.md. External sources are cited by bracketed identifiers listed in the References at the end. The identifiers that already exist in docs/references/bibliography.md (A6, V8, T1 and so on) are reused unchanged.

Numbers that can go stale

Every table in this document that sits between `stackup:begin` and `stackup:end` markers is **generated** from `hardware/rev-a/stackup/rev-a-stackup.json` by `python -m rfskit.cli stackup --write-docs`, and the test suite fails if one of them no longer matches the canonical file. Numbers quoted in prose are copied from those tables or from the decision records, and are dated by the baseline above. If a prose number and a generated table disagree, the generated table is right.

Figures

Figure 19, in section 10bis.7, is the signature figure: one aperture, two ISAC functions, one calibration layer. Diagrams are written as Mermaid or as text inside code fences, so that they live in the repository as source. GitHub renders Mermaid directly; a PDF conversion needs a Mermaid filter. The three plotted figures in `docs/figures/` are produced by `python tools/docs/master_reference_figures.py`. All three are **[analytical]** illustrations of the ideal array factor: no coupling, no element pattern, no simulated or measured data.

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Part 0. Executive overview

0.1 What is AetherArray?

AetherArray is a four channel, 2.44 GHz, phase only, switched line phased array research platform, designed so that the drift of a real RF array can be observed and so that recalibration with fewer new physical measurements can be tested against conventional recalibration. Its first hardware revision, Rev A, is two printed circuit boards: a passive board of four microstrip patch antennas, and a beamformer board joined to it by four coaxial jumpers. On the beamformer board each channel has an enable switch that either passes the signal or terminates the channel in 50 ohm, followed by three switched line phase bits of 45, 90 and 180 degrees built from PE4259-63 RF switches; the four channels meet in a four way Wilkinson network whose common port is switched either to a vector network analyser or to an on board AD8318 logarithmic power detector, and two MCP9808 sensors log temperature. An external Terasic DE1-SoC drives the sixteen beam state lines: its FPGA applies each state on a single clock edge, holds the control lines still while the detector is sampled and timestamps everything from one clock, and its processor stores the records and runs the inference. The design flow goes from analytical seeds through full wave simulation in Ansys HFSS, scripted with PyAEDT, optionally through a circuit model in Keysight ADS or scikit-rf, to measurement on a Rohde and Schwarz ZVL analyser, with every result exchanged as a Touchstone file and read by rfkit, the project's library on top of scikit-rf. Calibration is posed as a Bayesian inverse problem: an explicit physical forward model supplies the likelihood, and a prior learned from the array's own drift history is meant to make recalibration cheaper. A competition facing demonstrator for integrated sensing and communication would use the calibrated array as a reconfigurable receiver that steers towards a wanted commercial transmitter, suppresses an interfering one, and senses changes in a room lit by ambient Wi-Fi or Bluetooth signals: a shared aperture, opportunistic form of ISAC, defined and bounded in section 10bis.

Sources: decisions 0003, 0004, 0005 and 0009; docs/architecture/rev-a-rf-architecture.md; docs/architecture/control-architecture.md; docs/mathematics/inverse-calibration.md; docs/architecture/rf-data-layer.md. The demonstrator is [proposed here], from the project owner's brief of 2026-10-04; the repository records no decision about it (section 45).

Three things are easily confused, and this document keeps them apart throughout.

	What it is	Where it is recorded	State on 2026-10-04
The physical platform	the two Rev A boards, the controller, the analyser and the software around them	decisions 0003, 0005, 0009; hardware/rev-a/	designed, schematic captured; control section needs re-capture; nothing fabricated
The scientific question	whether a learned prior over drift reduces the new measurements needed to recalibrate	decision 0002; docs/architecture/ml-calibration.md; EXP-015	formalised; gated by G2, which needs the built array
The AP-S demonstrator	an ISAC receiver built on the platform, to show what calibration is worth	not in the repository	proposed; contest rules confirmed only through search excerpts

0.2 The central scientific question

Can a prior learned from the temporal drift of a specific RF array reduce the number of **new** physical measurements needed to recover that array's state, while reaching the same final calibration accuracy, and therefore the same RF performance, as a conventional recalibration from scratch?

This is decision 0002, point 1, written in the vocabulary of this document. In the repository it appears as experiment EXP-015 and specification ML-B in docs/architecture/ml-calibration.md section 6.

The figure of merit is a count, not an accuracy:

$$M_{\text{required}(\text{method})} = \min\{k : \Pr(\Delta(\mathbf{H}_t) \leq \delta \mid y_{1:k}, \mathbf{x}_{1:k}, \mathcal{D}) \geq 1 - \alpha\}$$

Symbol	Meaning	Unit
k	number of new physical measurements taken in this recalibration session	count
$\mathbf{H}_t$	the true array state, unknown; the probability is taken over its posterior after the k measurements	complex, dimensionless

Symbol	Meaning	Unit
$\Delta(\mathbf{H}_t)$	the error the beam chosen from the current estimate would have if the true state were $\mathbf{H}_t$: pointing error in the repository; null depth or a sensing figure in the demonstrator	deg, or dB
δ	the target value of Δ	as Δ
α	accepted risk of missing the target	dimensionless
$y_{1:k}, \mathbf{x}_{1:k}$	the readings and the commanded beam states that produced them	V or complex; 16 bit words
$\mathcal{D}$	the array's history: earlier calibrations, times, temperatures	data

This is the stopping rule of docs/mathematics/inverse-calibration.md section 3.1. A “measurement” here means one commanded beam state, applied, settled and read once, with a fixed integration time per reading. Fixing the integration time matters: without it, a method could appear to need fewer measurements simply by averaging each one longer.

Why a count, and not “how accurately does the model predict the drift”? Because the scarce resource is physical. A measurement means applying a state, waiting for it to settle and taking a noisy reading, at the analyser or through the detector; the computation afterwards takes milliseconds (benchmarks/metrics.md section 3). A drift model can predict well and still save nothing, if the measurements needed to confirm its prediction are as many as those needed to estimate the state without it. M_{required} measures the saving directly, and it is falsifiable: if the history carries no information, the method and the from scratch baseline need the same k .

Two parameters of this definition are not yet fixed. No numeric pointing target δ exists anywhere in the repository; decision 0007 says so explicitly and refuses to invent one. And no value of α is recorded. Both must be written down, with their reasoning, before EXP-015 runs. For the demonstrator, Δ would more naturally be a null depth or a signal to interference ratio, and the same rule applies: the target is fixed before the data.

0.3 Why the project exists: the commanded beam is not the real beam

A beamformer is commanded with a vector of complex weights, one per channel. What the hardware actually applies is different, and the difference changes with time:

$$\mathbf{w}_{\text{real}(t)} = \mathbf{H}_t \mathbf{w}_{\text{cmd}}$$

Symbol	Meaning
$\mathbf{w}_{\text{cmd}} \in \mathbb{C}^N$	the weights the controller asks for: amplitude and phase per channel
$\mathbf{w}_{\text{real}(t)} \in \mathbb{C}^N$	the weights the RF hardware really applies at time t
$\mathbf{H}_t \in \mathbb{C}^{N \times N}$	the array state: per channel gain and phase errors on the diagonal, coupling between channels off it
N	number of channels, 4 for Rev A

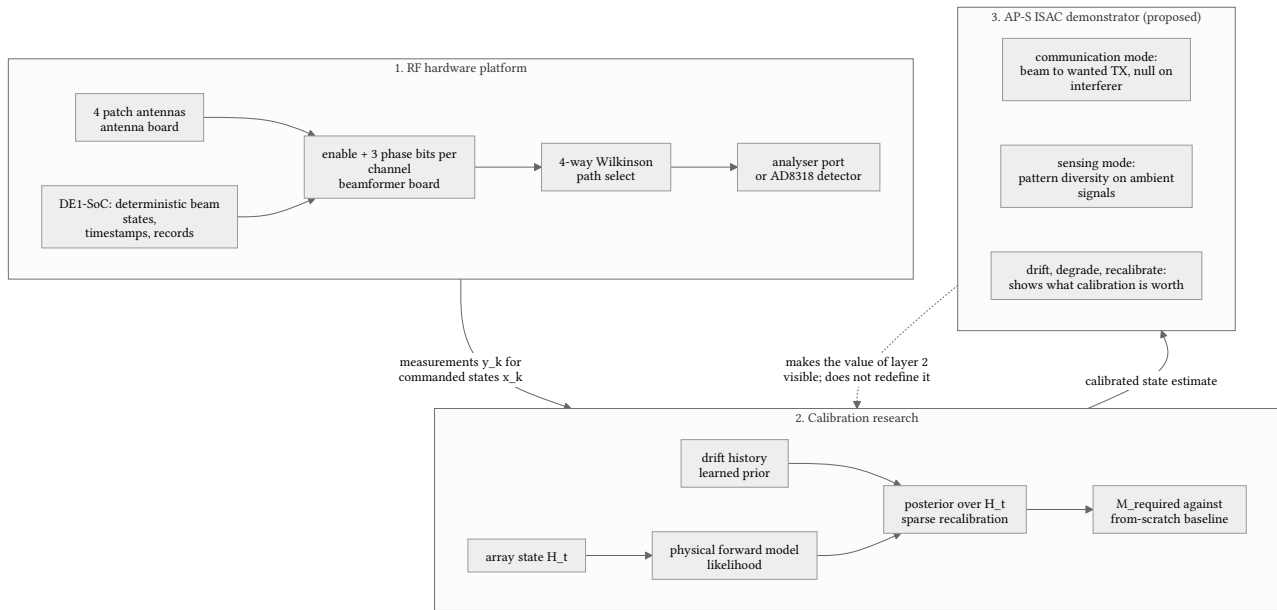
The repository writes the same relation as $\mathbf{y} = \mathbf{H}\mathbf{x}$ (docs/mathematics/formulation.md section 5). $\mathbf{H}_t$ collects many physical effects: different cable and trace lengths, component tolerances, switch path differences, printed circuit board permittivity varying from place to place and from one order to the next, the coupling of each antenna to its neighbours, and slow changes with temperature, time and handling. If $\mathbf{H}_t$ were the identity, the array would do exactly what theory says. It never is.

The consequences are visible and practical:

- the main beam points in the wrong direction;
- the gain in the wanted direction falls;
- the nulls, the directions of near total cancellation used to reject interference, fill in;
- the sidelobes rise;
- a pattern used as a sensing signature changes even though the environment did not;
- a calibration done yesterday becomes wrong today, because $\mathbf{H}_t$ has moved.

Calibration means estimating $\mathbf{H}_t$ and correcting the command. Each estimate costs measurements, and since $\mathbf{H}_t$ drifts, the cost recurs. That recurring cost is the object of the research.

0.4 Three layers



Layer 1 is hardware and measurement: what exists to be calibrated and how it is observed. **Layer 2** is the research: the array state, the inverse problem, the temporal prior and the measurement count. **Layer 3** is a demonstration: it uses the calibrated array for adaptive reception and sensing, and it makes the value of calibration visible as a null that fills in when the hardware drifts and comes back when the array is recalibrated. Layer 3 is a way of showing layer 2, and it must not redefine it. In particular, the demonstrator does not turn the project into “machine learning beamforming”: beam synthesis on Rev A is an exact enumeration (section 38), and the learned component stays a prior over drift. Figure 19 in section 10bis.7 shows the same structure from the ISAC side: one aperture, two functions, one calibration layer.

0.5 Where the project stands, in one paragraph

Nine decisions are accepted (Appendix G). The Rev A beamformer schematic is captured in KiCad by a generator script and passes its electrical rule check, but its control interface must be re-captured for the DE1-SoC controller. The stack-up is chosen. The first full wave simulation, SIM-001, a 50 ohm microstrip line, is fully specified and its model builder and analysis script exist, but its first execution on 2026-10-03 failed before any geometry was created, because AEDT Student opened no scripting session; **no solver result exists**. The analyser audit, EXP-004, has fixed the working frequency from direct bench observation but still lacks the exact instrument model, the calibration kit and the option list. The acquisition experiment EXP-005 Phase A is ready to run on owned hardware and has not been run. Nothing has been purchased or fabricated. No gateware exists. The `rfkit` library and its 160 tests run on synthetic data only. The learning track is formalised and entirely unimplemented, and it depends on gate G2, whether drift exceeds the measurement floor, which can only be answered on the built array. Part X gives the full status matrix with evidence.

Part I. Course and theory refresher

This part builds, from the beginning, every idea the rest of the document uses. Each section follows the same pattern: the physical intuition, the mathematics, what it means for AetherArray, how it will be simulated or measured, and what is known today. A reader who already knows microwave engineering can skip to section 7; a reader who knows phased arrays can skip to section 11.

1. Electromagnetic waves

1.1 Intuition

A radio wave is a travelling disturbance of two linked fields. The **electric field** $\mathbf{E}$, in volts per metre, pushes on charges; the **magnetic field** $\mathbf{H}$, in amperes per metre, is produced by moving charges and pushes on them in turn. A changing $\mathbf{E}$ produces $\mathbf{H}$ and a changing $\mathbf{H}$ produces $\mathbf{E}$, which is why the pair can sustain itself away from any source. Far from the antenna that launched it, the wave is locally plane: $\mathbf{E}$, $\mathbf{H}$ and the direction of travel are mutually perpendicular, and the direction of travel is given by the **propagation vector** $\mathbf{k}$, whose magnitude is the wavenumber.

1.2 Frequency, wavelength, phase velocity

At a fixed point the field oscillates at the **frequency** f , in hertz. At a fixed instant it repeats in space every **wavelength** λ , in metres. The pattern moves at the **phase velocity** v_p , so that one wavelength passes in one period:

$$\lambda = \frac{v_p}{f}, \quad v_p = \frac{c}{\sqrt{\epsilon_r \mu_r}}, \quad \lambda_0 = \frac{c}{f} \text{ in vacuum}$$

Symbol	Meaning	Unit
c	speed of light in vacuum, 299 792 458 m/s exactly	m/s
ϵ_r, μ_r	relative permittivity and permeability of a uniform medium; $\mu_r = 1$ for every material in this project	dimensionless
λ_0	free space wavelength	m

At the AetherArray working frequency, $f_0 = 2.44$ GHz (decision 0004):

$$\lambda_0 = \frac{299\,792\,458 \text{ m/s}}{2.44 \times 10^9 \text{ Hz}} = 0.1229 \text{ m} = 122.9 \text{ mm}$$

This is exact to the figures shown. Half of it, 61.4 mm, is the element spacing of the antenna board, so the four element array is about 184 mm across its element centres.

In a uniform dielectric the wave is slower and the wavelength shorter by $\sqrt{\epsilon_r}$. On a printed circuit board the wave travels partly in the board and partly in air (section 5), so the relevant figure is the **guided wavelength** $\lambda_g = \lambda_0 / \sqrt{\epsilon_{\text{eff}}}$, with an effective permittivity between 1 and ϵ_r . Confusing λ_0 with λ_g is the classic error in switched line design: on the selected beamformer board the analytical estimate is $\lambda_g \approx 68.8$ mm, barely more than half of λ_0 .

1.3 Free space impedance

In a plane wave the ratio of the electric to the magnetic field strength is fixed by the medium. In vacuum:

$$\eta_0 = \frac{|\mathbf{E}|}{|\mathbf{H}|} = \sqrt{\frac{\mu_0}{\epsilon_0}} \approx 376.7 \text{ } \Omega$$

μ_0 and ϵ_0 are the permeability and permittivity of vacuum, in H/m and F/m. η_0 reappears in antenna formulas and in the closed form microstrip impedance of section 5.

1.4 Phasors, and why complex numbers are used everywhere

A field oscillating at one frequency is completely described by two numbers: its amplitude and its phase. Writing it as the real part of a rotating complex number,

$$E(z, t) = E_0 \cos(\omega t - \beta z + \varphi_0) = \text{Re}\{\underline{E}(z) e^{j\omega t}\}, \quad \underline{E}(z) = E_0 e^{j(\varphi_0 - \beta z)}$$

Symbol	Meaning	Unit
$\omega = 2\pi f$	angular frequency	rad/s
β	phase constant, the phase accumulated per metre of travel	rad/m
$\underline{E}(z)$	the phasor : a complex number holding amplitude E_0 and phase	V/m
j	the imaginary unit, $j^2 = -1$ (engineering notation)	

This document uses the engineering convention $e^{j\omega t}$, in which a wave travelling towards $+z$ carries the factor $e^{-j\beta z}$. Three things make phasors indispensable.

1. **Adding waves becomes adding vectors.** Two waves of the same frequency arriving at a point add as two arrows in the complex plane. Aligned, they reinforce; opposed, they cancel. A phased array is nothing more than this addition, done deliberately (section 7).
2. **A delay becomes a multiplication.** Travelling a distance l multiplies the phasor by $e^{-j\beta l}$: the magnitude is unchanged and the phase decreases by βl .
3. **A time derivative becomes multiplication by $j\omega$.** Differential equations in time become algebra, which is why every RF quantity in this project, S-parameters, the array state, channel gains, is complex.

1.5 Phase and distance

The phase accumulated over a distance z is

$$\varphi(z) = \beta z, \quad \beta = \frac{2\pi}{\lambda}$$

so one wavelength of travel turns the phase by 2π radians, 360 degrees. In free space at 2.44 GHz, $\beta_0 = 2\pi/\lambda_0 = 51.1$ rad/m, which is 2.93 degrees per millimetre.

1.6 From here to a switched line phase shifter

If a signal can be routed through either of two lines whose lengths differ by Δl , the two routes deliver it with a phase difference

$$\Delta\varphi = \beta \Delta l = \frac{2\pi}{\lambda_g} \Delta l = \frac{2\pi f \sqrt{\epsilon_{\text{eff}}}}{c} \Delta l$$

Symbol	Meaning	Unit
$\Delta\varphi$	phase difference between the long and the short route	rad
Δl	extra physical length of the long route	m
λ_g	guided wavelength in the printed line, not λ_0	m
ϵ_{eff}	effective permittivity of the microstrip, section 5	dimensionless

That is the whole principle of the AetherArray phase shifter (section 15). Three consequences follow from the formula alone, and each one returns later.

- **The relevant wavelength is λ_g .** With the analytical seed $\lambda_g = 68.8$ mm, a 45 degree bit needs $\Delta l = \lambda_g/8 \approx 8.6$ mm, and one millimetre of printed line is about 5.2 degrees. Using λ_0 instead would give lengths 1.8 times too long.
- **The phase is proportional to frequency.** A fixed Δl is a true time delay, not a fixed phase, so a 45 degree bit at 2.44 GHz is a slightly different angle at the band edges (section 8.5 and Figure 5b).
- **The phase is proportional to $\sqrt{\epsilon_{\text{eff}}}$.** An error in the board's permittivity scales every switched length by the same fraction, so the longest state carries the largest error (section 59).

```

short route : in ---[ l_ref ]----- out      phase -beta*l_ref
long route  : in ---[ l_ref + delta_l ]----- out  phase -beta*(l_ref + delta_l)

difference  : delta_phi = beta * delta_l      (only the DIFFERENCE is designed)

one guided wavelength lambda_g (about 69 mm on the beamformer board)
|<-----|-----|-----|-----|-----|>|
0 deg      90 deg      180 deg      270 deg      360 deg
|-----|-----|-----|-----|-----|
          45          135          225          315

```

Figure 2: the phase line intuition. Phase grows linearly along a line; a switched line bit selects between two lengths, and only their difference is the designed phase.

1.7 What is known and what is not

The free space quantities are exact consequences of f_0 **[decided]**. Every guided length in this document is an **[analytical]** estimate that waits on SIM-001 for a full wave value, and on board coupons for a measured one (sections 25 and 59).

2. Transmission lines

2.1 Why circuit intuition breaks down

Ordinary circuit theory assumes that a voltage applied at one end of a wire appears at the other end instantly. That holds while the wire is short compared with a wavelength. A common rule of thumb says lumped reasoning fails once a structure exceeds about a tenth of a wavelength. On the beamformer board a tenth of λ_g is about 7 mm, shorter than many of its traces, so at 2.44 GHz every trace, connector, switch package and antenna feed has to be treated as a structure in which the voltage and the current vary along its length.

2.2 The distributed model

A uniform line is modelled as a cascade of infinitesimal sections, each with series resistance R' and inductance L' and shunt conductance G' and capacitance C' , all per unit length. Solving the resulting telegrapher's equations at one frequency gives voltage and current waves travelling both ways:

$$V(z) = V^+ e^{-\gamma z} + V^- e^{\gamma z}, \quad I(z) = \frac{V^+ e^{-\gamma z} - V^- e^{\gamma z}}{Z_0}$$

$$\gamma = \alpha + j\beta = \sqrt{(R' + j\omega L')(G' + j\omega C')}, \quad Z_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}$$

Symbol	Meaning	Unit
V^+, V^-	complex amplitudes of the forward and backward waves	V
γ	propagation constant	1/m
α	attenuation constant: how fast the amplitude decays	Np/m; multiply by 8.686 for dB/m
β	phase constant, as in section 1	rad/m
Z_0	characteristic impedance: the ratio V/I for a single travelling wave	ohm
R', L', G', C'	line constants per unit length	ohm/m, H/m, S/m, F/m

For a low loss line, $Z_0 \approx \sqrt{L'/C'}$ and $\beta \approx \omega \sqrt{L'C'}$. Every RF part of AetherArray is designed for $Z_0 = 50$ ohm, the impedance of the analyser, the SMA connectors and the switches, except the Wilkinson arms at 70.7 ohm (section 15.6).

Loss on the selected boards. For the beamformer microstrip, the analytical model of decision 0009 gives a conductor loss of about 4.5 dB/m and a dielectric loss of about 5.3 dB/m at 2.44 GHz, so about 9.8 dB/m in total **[analytical]**. The dielectric part follows the standard quasi-static expression [B1, chapter 3]:

$$\alpha_d \approx \frac{k_0 \epsilon_r (\epsilon_{\text{eff}} - 1) \tan \delta}{2 \sqrt{\epsilon_{\text{eff}}} (\epsilon_r - 1)}$$

$k_0 = 2\pi/\lambda_0$, $\tan \delta$ is the loss tangent of the dielectric, and the result is in Np/m. With $\epsilon_r = 4.4$, $\epsilon_{\text{eff}} = 3.19$ and $\tan \delta = 0.015$ it gives 0.61 Np/m, 5.3 dB/m, matching the generated table in section 59. This matters because a longer phase state is also a lossier one: the 315 degree state carries about 60 mm more line than the 0 degree state, so about 0.6 dB more loss (section 59).

2.3 The reflection coefficient

When a line of impedance Z_0 ends in a load Z_L , the load forces its own ratio of voltage to current. At the load, at $z = 0$, with $\Gamma = V^-/V^+$:

$$V(0) = V^+(1 + \Gamma), \quad I(0) = \frac{V^+}{Z_0}(1 - \Gamma), \quad Z_L = \frac{V(0)}{I(0)} = Z_0 \frac{1 + \Gamma}{1 - \Gamma}$$

and solving for Γ :

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Γ is complex and dimensionless. It is the fraction of the incident wave amplitude that comes back, with the phase it comes back with. Four cases are worth knowing by heart.

Load	Γ	What happens	Where it appears in AetherArray
matched, $Z_L = Z_0$	0	all power absorbed, nothing reflected	the 50 ohm terminations of the disabled channels; the analyser ports after calibration
short circuit, $Z_L = 0$	-1	total reflection, inverted	a calibration standard; a via to ground
open circuit, $Z_L \rightarrow \infty$	+1	total reflection, same sign	a calibration standard; the far end of a de-selected switched line arm
mismatched, for example $Z_L = 75$ ohm	0.2	partial reflection	a patch that resonates off frequency; a line etched too narrow

Two derived quantities are used everywhere:

$$RL = -20 \log_{10} |\Gamma| \text{ dB}, \quad VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \quad L_{\text{mismatch}} = -10 \log_{10} (1 - |\Gamma|^2) \text{ dB}$$

Symbol	Meaning	Unit
RL	return loss: how far below the incident wave the reflection is	dB, positive for a passive load
VSWR	voltage standing wave ratio: ratio of the maximum to the minimum of the standing wave on the line	dimensionless, 1 for a match
L_{mismatch}	power lost to reflection at that interface	dB

Example: $\Gamma = 0.2$ gives a return loss of 14 dB, a VSWR of 1.5 and a mismatch loss of 0.18 dB, a figure `python -m rfkit.cli` budget also prints. “VSWR 2” in the patch bandwidth estimates of section 22 means $|\Gamma| = 1/3$, a return loss of 9.5 dB.

2.4 Why every RF part is a transmission line structure

Part	Why it is a transmission line problem
SMA connectors and the four jumpers	coaxial lines with their own length and impedance; one centimetre of typical coaxial cable is about 44 degrees at 2.44 GHz (README worked example, velocity factor 0.66)
PCB traces	microstrip lines, section 5; their width sets Z_0 and their length sets phase
PE4259-63 switches	short lines with parasitic inductance and capacitance; their package is a discontinuity, and their off state is a finite isolation, not an open circuit
switched line arms	the designed phase elements themselves
antenna feeds	lines that transform the patch edge impedance to 50 ohm

Every discontinuity reflects a little. Two discontinuities a distance apart create a small standing wave between them, whose effect on the transmitted phase depends on frequency and, in a switched line, on which arm is selected. That is one route by which a phase error becomes **state dependent**, the kind calibration cannot absorb (section 11.4).

3. S-parameters

3.1 Waves rather than voltages

At microwave frequencies voltage and current are hard to measure directly, but incident and reflected waves can be separated with directional couplers. S-parameters describe a network by how it scatters waves. At each port i , referred to a reference impedance Z_0 :

$$a_i = \frac{V_i^+}{\sqrt{Z_0}}, \quad b_i = \frac{V_i^-}{\sqrt{Z_0}}$$

a_i is the wave going **into** port i and b_i the wave coming **out** of it, both in $\sqrt{W}$. With RMS phasors, $|a_i|^2$ is the incident power; with peak phasors it is twice it. Only ratios are used below, so the convention cancels.

3.2 The two port

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

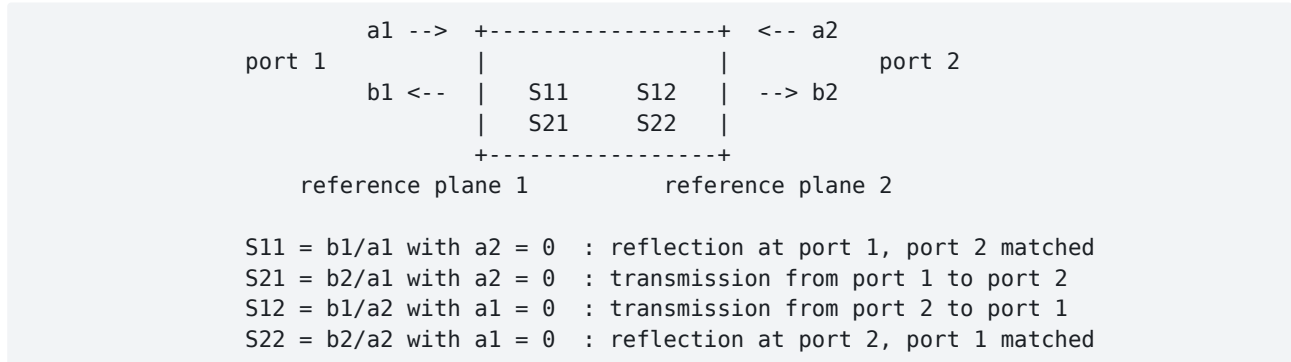


Figure 3: a two port and its scattering parameters. “Matched” means terminated in the reference impedance, so that no wave returns into that port.

Each S_{ij} is a complex number at each frequency. For a **reciprocal** network, one with no amplifier, ferrite or nonreciprocal material, $S_{21} = S_{12}$. For a **lossless** network the matrix is unitary. The RF signal path of Rev A, switches, lines and Wilkinson network, contains no amplifier, so it is expected to be reciprocal at the small signal levels used [**assumed**]; reciprocity is checkable in measurement, and decision 0008 already uses the measured asymmetry $S_{ij} - S_{ji}$ as a floor on measurement uncertainty.

3.3 Magnitude, decibels, phase

$$|S_{21}|_{\text{dB}} = 20 \log_{10} |S_{21}|, \quad \text{IL} = -20 \log_{10} |S_{21}|, \quad \varphi_{21} = \arg S_{21}$$

Quantity	Meaning	Unit
$ S_{21} _{\text{dB}}$	transmission magnitude in decibels, negative for a passive network	dB
IL	insertion loss, positive for a passive network	dB
φ_{21}	transmission phase	deg or rad

The factor is 20, not 10, because S is an amplitude ratio and power goes as its square. **Conventions matter.** The equality of insertion loss with $-20 \log_{10} |S_{21}|$ holds when the source and load are both the reference impedance; with a mismatched source or load the power actually delivered differs. Return loss is defined as $-20 \log_{10} |S_{11}|$, positive; some instruments and papers instead plot S_{11} in dB, negative, and call it return loss. This document always states which.

For a matched, lossless line of length l , $S_{21} = e^{-j\beta l}$: unit magnitude and a phase of $-\beta l$. The phase decreases with length and with frequency, and is reported **wrapped** into an interval of 360 degrees by instruments. Comparing phases therefore needs care: 359 degrees and 1 degree differ by 2 degrees, not 358, which is why `rfkit` takes every phase difference on the circle (section 31). Unwrapping along frequency recovers a continuous curve, whose slope is the group delay $\tau_g = -d\varphi_{21}/d\omega$.

3.4 What AetherArray needs from S-parameters

Need	Quantity	Where
validating the microstrip model	S_{21} of two line lengths, giving γ and so ε_{eff} and α	SIM-001; coupons C1 and C2
the phase of each switched line bit	$\arg S_{21}$ in the long state minus the short state, between the switch reference planes	SIM-003 and SIM-004 proposed; SCH-006
state dependent loss	$ S_{21} $ of each of the eight states	decision 0007 imbalance limit; decision 0009 loss estimate
splitter and combiner behaviour	balance of the four arms, isolation between outputs, input match	SIM-005 proposed; SCH-006
antenna matching	S_{11} of each element port	SIM-006 proposed; I27
coupling	the full 4×4 matrix $\mathbf{S}_A$ of the antenna board	EXP-011, decision 0008
the per channel array state	S_{21} from the common port through one enabled channel	rfkit.state, section 11

Why the phase of S_{21} is the heart of the phase bits. A bit is correct when the difference of transmission phase between its two states is the designed 45, 90 or 180 degrees at f_0 , measured between the same two planes. The absolute phase of either arm does not matter; the difference does (hardware/rev-a/layout-constraints.md section 2). A layout that makes the reference arm short and the delay arm “45 degrees long” is wrong by whatever the reference arm actually measures.

4. The vector network analyser

4.1 What it measures

A vector network analyser (VNA) contains a swept source and two or more receivers. Directional couplers or bridges at each port separate the incident wave from the reflected one, and the receivers measure **ratios** of waves, for example b_2/a_1 , in both magnitude and phase. That is what “vector” means: the instrument measures complex ratios, not only powers. A scalar analyser or a power detector measures magnitude only; it cannot tell a 45 degree bit from a 315 degree one.

4.2 Why a raw reading is not yet a measurement

The instrument’s internal paths are not perfect. Its couplers leak (finite directivity), its ports are not exactly 50 ohm (source and load match), its receivers have frequency dependent gain and phase (tracking), and the cables and adapters between the instrument and the device add their own length, loss and reflections. A raw reading of “ S_{21} ” is the device plus all of that. **Calibration** removes it: known standards are measured, an error model is solved from them, and later readings are corrected. For a two port, the familiar procedures connect a short, an open and a matched load at each port and a through between them (called SOLT, or TOSM by Rohde and Schwarz), or use lines of known relation instead (TRL). The planes at which the standards were connected become the **reference planes**: corrected S-parameters describe the device between those planes and nothing else.

De-embedding goes one step further. If the device sits inside a fixture, for example SMA launches on a board, the fixture can be characterised separately and mathematically removed, so that the result refers to planes inside the board. Coupon C3 of decision 0009, two launches back to back, exists for exactly this (section 25).

Uncertainty never reaches zero. Residual calibration errors, connector repeatability, cable movement after calibration, instrument drift and noise all remain. They are combined into an expanded uncertainty, written U in decisions 0007 and 0008, for each S-parameter at each frequency.

4.3 Why “the VNA says $S_{21} = X$ ” means nothing on its own

Consider a jumper 10 cm longer than intended. In coaxial cable with a velocity factor of 0.66, the guided wavelength at 2.44 GHz is about 81 mm, so 10 cm adds about 440 degrees of phase. An uncalibrated reading, or a reading calibrated at the wrong plane, can therefore be wrong by more than a full turn while looking perfectly plausible. A reading is meaningful only together with: the reference planes, the calibration method and kit used, the date and the conditions of that calibration, the intermediate frequency bandwidth and averaging, the source power, and an uncertainty estimate. The `rfkit` provenance record exists to carry those facts with every file (section 31).

4.4 The AetherArray analyser, today

The analyser is the subject of experiment EXP-004. Three different capabilities have to be kept apart.

Capability	Status	Evidence
The instrument exists and can reach the band	[observed] 2026-09-20: a Rohde and Schwarz ZVL, 9 kHz to 3 GHz, 50 ohm, N female ports, S21 available, complex and phase formats available, source up to 0 dBm, USB present	results/EXP-004/README.md result 2
A calibrated measurement at the board’s SMA plane is possible	unknown. No calibration kit and no adapter has been confirmed (observation O7). The kits named in the laboratory inventory are 3.5 mm kits belonging to a different instrument whose own presence is unconfirmed	docs/hardware/measurement-bench.md section 6; uncertainty I19
Measurements can be automated	unknown. USB is present on the instrument, but nothing has ever enumerated on the project computer (O9); no driver has been identified, and <code>rfkit.instrument</code> is deliberately empty of drivers	results/EXP-004/README.md result 1; docs/architecture/rf-data-layer.md section 5

The exact model and serial number, observation O1, have not been read; the identification as a ZVL3 is inferred from behaviour. Which options are installed, observation O8, is unknown, including whether the time domain option is present. Option designations are known from manufacturer sources: K1 spectrum analysis, K2 distance to fault, K3 time domain analysis [T1a]. The family is described in distributor listings as supporting one port (OSM), full two port (TOSM) and one path two port calibration [T1], a **[listing]** claim that the bench has not confirmed. Runbook SCH-001, status READY, is the 30 minute bench visit that finishes O1 and O6 to O9 (docs/runbooks/SCH-001-exp004-analyser-audit.md).

The practical blocker is therefore not frequency coverage but the connector chain: the analyser has N female ports, the boards have SMA connectors, and nothing confirmed joins them at a calibrated plane. A chain that would work, if its

parts exist, is an N male to 3.5 mm adapter followed by a 3.5 mm calibration at the adapter output, which is mechanically compatible with SMA. It is a proposal in the repository, not a plan.

5. Microstrip

5.1 Geometry

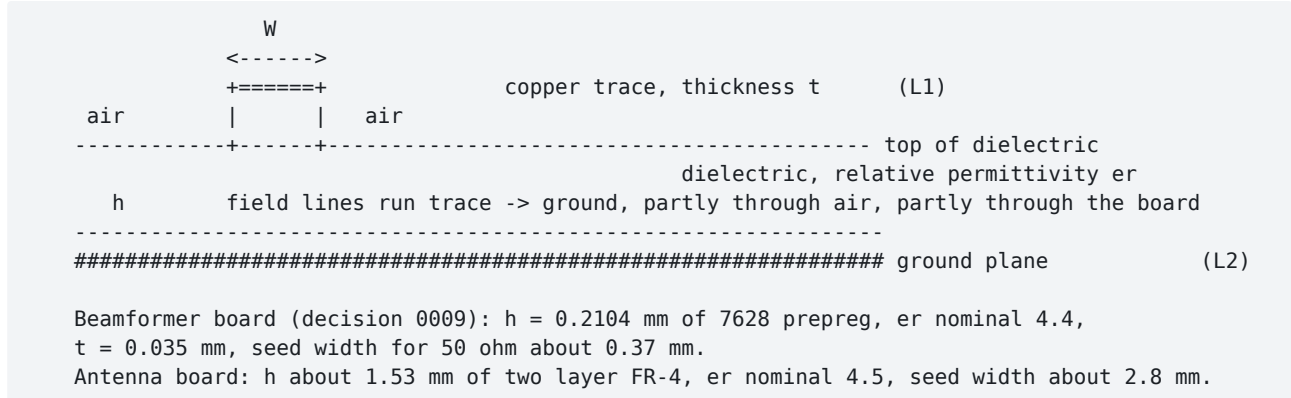


Figure 4: microstrip cross-section. The dimensions are the canonical values of `reva-stackup-r1`; the widths are analytical seeds, not layout values.

A microstrip is a copper strip of width W and thickness t on a dielectric layer of height h and relative permittivity ϵ_r , above a continuous ground plane. The signal travels as a wave guided between the strip and the ground.

5.2 Why the effective permittivity lies between 1 and ϵ_r

Below the strip the electric field runs through the dielectric; at the edges it fringes up into the air and comes back down. The wave therefore sees a mixture, and behaves as if it were in a uniform medium of **effective permittivity**

$$1 < \epsilon_{\text{eff}} < \epsilon_r, \quad \epsilon_{\text{eff}} \approx \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} F, \quad F = \left(1 + \frac{12h}{W}\right)^{-1/2}$$

Symbol	Meaning	Unit
ϵ_{eff}	effective permittivity of the quasi-TEM wave	dimensionless
F	filling factor: the share of the field beyond the uniform half that sits in the dielectric	dimensionless
h, W	dielectric height and strip width	m

This is the quasi-static closed form used in Pozar [B1, equation 3.195] and quoted in decision

9. Because some field is always in air, the wave is not purely TEM; it is called

quasi-TEM, and its ϵ_{eff} rises slowly with frequency (dispersion), which the Kirschning and Jansen model captures [B9].

5.3 How W/h sets the impedance

A wider strip has more capacitance to ground per unit length and so a lower Z_0 . For $W/h \geq 1$ the zero thickness closed form is [B1]

$$Z_0 \approx \frac{\eta_0}{\sqrt{\epsilon_{\text{eff}}} [W/h + 1.393 + 0.667 \ln(W/h + 1.444)]}$$

The essential point is that Z_0 depends on the **ratio** W/h and on ϵ_r , not on W alone. For FR-4 near $\epsilon_r = 4.4$, 50 ohm needs W/h of roughly 1.8 to 2. Therefore:

- on the beamformer board, $h = 0.2104$ mm, so a 50 ohm line is about 0.37 mm wide;
- on a two layer 1.6 mm board, $h \approx 1.53$ mm, so a 50 ohm line is about 2.8 mm wide.

The SC-70-6 package of the PE4259-63 has a lead pitch of 0.65 mm. A 2.8 mm line is more than four times that, so every one of the 87 switch RF pins (29 switches, three RF pins each) would need a taper; a 0.37 mm line matches the pads. This single ratio is why the beamformer is not built on the antenna board's laminate (decision 0009, option A2).

A worked check. With $W = 0.372$ mm and $h = 0.2104$ mm, $W/h = 1.77$, $F = 0.358$ and the closed form gives $\epsilon_{\text{eff}} \approx 3.31$ and $Z_0 \approx 52.6$ ohm at zero thickness. The `scikit-rf` `MLine` model, which includes the 35 um copper thickness and

dispersion, gives $\varepsilon_{\text{eff}} = 3.19$ at f_0 and 50 ohm at that width; its zero thickness synthesis gives 0.402 mm. Two respectable analytical models differ by about four per cent in ε_{eff} and eight per cent in width. That disagreement is precisely why the analytical width is called a seed.

5.4 Why W_{seed} is only an initialisation

`rfkit.stackup.seed` computes the width for which the scikit-rf line model gives 50 ohm on the nominal stack-up. Decision 0009 labels it **INITIALISATION ONLY**: $W_{\text{seed}} \neq W_{50}$. SIM-001 sweeps the width at 0.9, 1.0 and 1.1 times the seed in HFSS and interpolates the width at which the port impedance is 50 ohm; it never extrapolates (experiments/SIM-001-microstrip-50-ohm.md, decision criterion 2). And even W_{50} from HFSS is a model of nominal materials: the board as fabricated is characterised later on its coupons.

5.5 Material uncertainty, one effect at a time

Effect	Mechanism	What it does here	Status
nominal ε_r	the fabricator's impedance calculator uses 4.4 for 7628 prepreg, with no frequency or test method stated	sets ε_{eff} , hence every guided length	[vendor] nominal, V8
uncertainty of ε_r	published FR-4 values span roughly 4.2 to 4.6; the laminate brand is not guaranteed per order	a bound of ± 0.2 is adopted; it moves the 315 degree state by about 6.3 degrees	[assumed] bound, decision 0009
manufacturing variation	prepreg thickness, etched width, copper thickness	impedance from about 41 to 61 ohm at the worst corner of all bounds; up to 11.6 degrees on the 315 degree state	[analytical], generated table in section 59
glass weave	7628 is a coarse glass cloth; a 0.37 mm line can sit over glass bundles or resin gaps	different channels may see different local permittivity; the one material effect a coupon cannot calibrate away	unquantified, uncertainty I25
copper roughness	the foil's surface roughness, a few micrometres, exceeds the 1.34 μm skin depth at 2.44 GHz	raises conductor loss, up to about double in the Hammerstad and Jensen model, and raises apparent permittivity	bounded, not modelled; section 24
solder mask	a thin dielectric coating of uncertain thickness over the line	would add channel dependent phase	removed: mask opened over RF copper; section 24
conductor loss	finite copper conductivity, skin effect, ENIG nickel on exposed copper	about 4.5 dB/m on the beamformer line, smooth copper	[analytical]
dielectric loss	$\tan \delta = 0.015$, a laminate vendor typical value at 1 GHz	about 5.3 dB/m; does not depend on width	[vendor] typical, V12

The skin depth is $\delta_s = \sqrt{2/(\omega\mu_0\sigma)} = 1.34 \mu\text{m}$ for copper at 2.44 GHz, with $\sigma = 5.8 \times 10^7 \text{ S/m}$; the 15.2 μm inner copper of the ground plane is about eleven skin depths thick, so it behaves as a good conductor.

6. Antennas

6.1 Radiation in a few lines

An antenna turns a guided wave into a free wave. Time varying currents on a conductor radiate: far from the antenna, at distances large compared with both the wavelength and the antenna, the field falls as $1/r$, is transverse to the direction of travel, and its angular shape no longer depends on distance. That angular shape is the **radiation pattern**. The boundary of that **far field** region is conventionally taken at $R = 2D^2/\lambda$ for an antenna of largest dimension D ; for the four element array, $D \approx 0.18 \text{ m}$ and $R \approx 0.55 \text{ m}$ (decision 0004).

6.2 The figures used to describe an antenna

Quantity	Definition	Unit
directivity $D(\theta, \phi)$	radiation intensity in a direction divided by its average over all directions	dimensionless, or dBi
radiation efficiency e_r	radiated power divided by accepted power; the rest is lost in conductor and dielectric	dimensionless
gain G	$G = e_r D$	dBi
realised gain	$G(1 - \Gamma ^2)$: includes the mismatch at the feed	dBi
polarisation	the direction in which E oscillates; a simple patch is linearly polarised	
E-plane, H-plane	the two principal cuts of the pattern: containing E , and containing H	

Quantity	Definition	Unit
half power beamwidth	angle between the two directions where the pattern is 3 dB below its maximum	deg
sidelobes	secondary maxima of the pattern; their level is quoted relative to the main lobe	dB
impedance match	how close the feed impedance is to 50 ohm, through Γ	

By reciprocity, an antenna has the same pattern, gain and impedance when receiving as when transmitting. That is why a receiving array can be characterised by transmitting through it with a passive network, and why the AP-S demonstrator, which must receive (section 45), can be designed with transmit language.

6.3 The microstrip patch

A patch is a rectangle of copper on a grounded dielectric. Between the patch and the ground it behaves like a leaky resonant cavity: the fundamental mode has a standing wave along the patch length L , and the fields fringing at the two radiating edges add in phase broadside to the board. Resonance occurs approximately when L is half a wavelength in the dielectric, slightly shortened by fringing. The transmission line model in Balanis [B2] gives seeds:

$$W_p = \frac{c}{2f_0} \sqrt{\frac{2}{\epsilon_r + 1}}, \quad L_p \approx \frac{c}{2f_0 \sqrt{\epsilon_{\text{eff},p}}} - 2\Delta L$$

Symbol	Meaning	Unit
W_p, L_p	patch width and resonant length	m
$\epsilon_{\text{eff},p}$	effective permittivity of a microstrip as wide as the patch	dimensionless
ΔL	fringing extension at each radiating edge, a fraction of h	m

On the antenna construction this gives $W_p \approx 37.0$ mm and $L_p \approx 28.6$ mm [**analytical, sanity check only**] (generated table, section 22).

What controls the rest:

- **Substrate thickness h .** Bandwidth grows roughly in proportion to h/λ , and so does the share of power radiated rather than dissipated. A thin substrate stores a lot of energy for little radiation: a high quality factor, a narrow bandwidth and a low efficiency.
- **Permittivity ϵ_r .** A higher ϵ_r makes the patch smaller but its bandwidth narrower, and makes the resonance sensitive to permittivity error.
- **Feed position.** The input impedance varies from a few hundred ohm at the edge to near zero at the centre; an inset feed or a quarter wave transformer brings it to 50 ohm.
- **Loss.** On FR-4 with $\tan \delta \approx 0.015$, dielectric loss takes a large share of the accepted power.

6.4 Why the antenna board and the beamformer board use different stack-ups

Decision 0009 applied the same patch sanity check to every candidate construction.

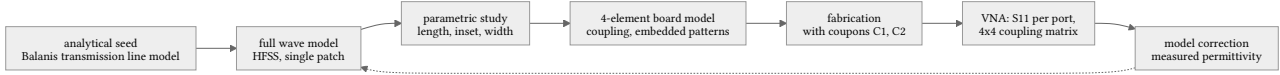
Construction	h	Patch bandwidth, VSWR 2	Radiation efficiency
beamformer, 4 layer FR-4, patch over L2	0.21 mm	about 0.14 per cent	about 9 per cent
antenna, 2 layer FR-4	1.53 mm	about 1.05 per cent	about 48 per cent

[**analytical**] estimates, generated in section 22. The thin prepreg that makes the beamformer lines narrow enough for the switch pads would make a patch with almost no bandwidth and an efficiency near ten per cent; the thick core that suits a patch makes 2.8 mm lines that do not fit the switches and leaves no inner plane for control routing. The two boards need substrates an order of magnitude apart, so they are two boards with two constructions from one fabricator and one laminate class.

Two consequences of the antenna estimate deserve attention, both [**analytical**] and both recorded in decision 0009 or derivable from it.

- **The bandwidth is narrower than the band.** About 1.05 per cent of 2.44 GHz is about 26 MHz, while band 57a is 83.5 MHz wide. If the estimate holds, a single patch is matched to VSWR 2 over roughly a third of the band. This does not contradict any decision, but every judgement made “at every point in band 57a” (decisions 0007 and 0008) will see the antenna mismatch change across the band. SIM-006 must report it.
- **The first board may miss the band.** The permittivity bound alone moves the resonance by -1.06 to $+3.38$ per cent, which is more than the bandwidth. Decision 0009 says a second antenna board order should be expected.

6.5 The antenna design workflow



Nothing in the repository designs the patch yet; its length, width, feed and spacing remain free parameters for HFSS (decision 0009). The analytical equations above say only that a patch is physically sensible on the chosen construction.

7. Phased arrays

7.1 Intuition

Several antennas fed with the same signal radiate waves that add at every point in space. In some directions the waves arrive in step and reinforce; in others they arrive out of step and cancel. Delaying the signal to each element by a controlled amount moves the direction in which they reinforce. That is a phased array: a beam steered by phase, with nothing moving. The standard references for what follows are Balanis [B2], Mailloux [B3] and Hansen [B4].

7.2 Geometry and the array factor

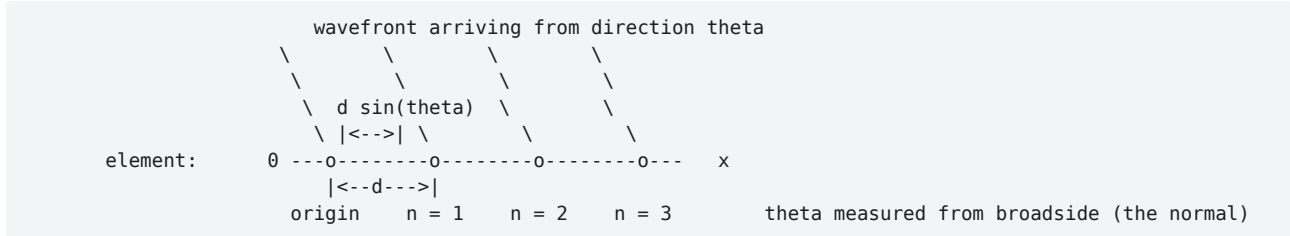


Figure 6: four element uniform linear array, element 0 at the origin, spacing d .

Take N identical elements on a line, spaced by d , element 0 at the origin, and a distant point in the direction θ measured from the array normal (broadside). The path from element n is shorter than the path from element 0 by $nd \sin \theta$, which is a phase advance of $nkd \sin \theta$ with $k = 2\pi/\lambda_0$. If element n is fed with amplitude a_n and phase ϕ_n , the field in direction θ is proportional to

$$AF(\theta) = \sum_{n=0}^{N-1} a_n e^{j(nkd \sin \theta + \phi_n)}$$

Symbol	Meaning	Unit
$AF(\theta)$	array factor: the coherent sum of the element contributions	dimensionless, complex
a_n, ϕ_n	amplitude and phase applied to element n : the command	dimensionless, rad
$k = 2\pi/\lambda_0$	free space wavenumber, 51.1 rad/m at 2.44 GHz	rad/m
d	element spacing, $\lambda_0/2 = 61.4$ mm for Rev A	m
θ	observation angle from broadside	rad

This is the convention of README.md, docs/mathematics/formulation.md and rfkit.budget.array_factor, and every formula below uses it. By reciprocity the same expression describes reception: a plane wave arriving from θ and combined with weights $a_n e^{j\phi_n}$ produces an output proportional to $AF(\theta)$.

Element pattern and total pattern. Real elements are not isotropic. If all elements had the same pattern $f(\theta)$, the total field would factor as $E(\theta) = f(\theta) AF(\theta)$, the principle of pattern multiplication. With coupling, each element radiates differently inside the array, and the correct description uses one **embedded element pattern** $g_n(\theta)$ per element: $E(\theta) = \sum_n g_n(\theta) a_n e^{j\phi_n}$. Decision 0008 uses exactly this form for the simulation stage of gate G4 (section 10).

7.3 Steering

With uniform amplitudes, AF is largest when every term has the same phase in the wanted direction θ_0 . That requires $nkd \sin \theta_0 + \phi_n$ to be the same for all n , so

$$\phi_n = -nkd \sin \theta_0, \quad \Delta\phi = \phi_{n+1} - \phi_n = -kd \sin \theta_0$$

$\Delta\phi$ is the progressive phase between neighbours, in rad. With $d = \lambda_0/2$, $kd = \pi$, so $\Delta\phi = -180^\circ \sin \theta_0$: steering to 30 degrees needs -90 degrees per element, steering to 15 degrees needs -46.6 degrees. Substituting into the array factor with $a_n = 1$ and summing the geometric series gives the classical closed form

$$|AF(\theta)| = \left| \frac{\sin(N\psi/2)}{\sin(\psi/2)} \right|, \quad \psi = kd(\sin \theta - \sin \theta_0)$$

which is exact. Its maximum, N , occurs at $\psi = 0$, that is $\theta = \theta_0$: the **coherent addition** of N unit contributions. The array output power in that direction is N^2 times one element's, while the noise from independent sources adds only as N , which is why the array gain over one element is $10 \log_{10} N$, 6.0 dB for four isotropic elements at half wavelength spacing.

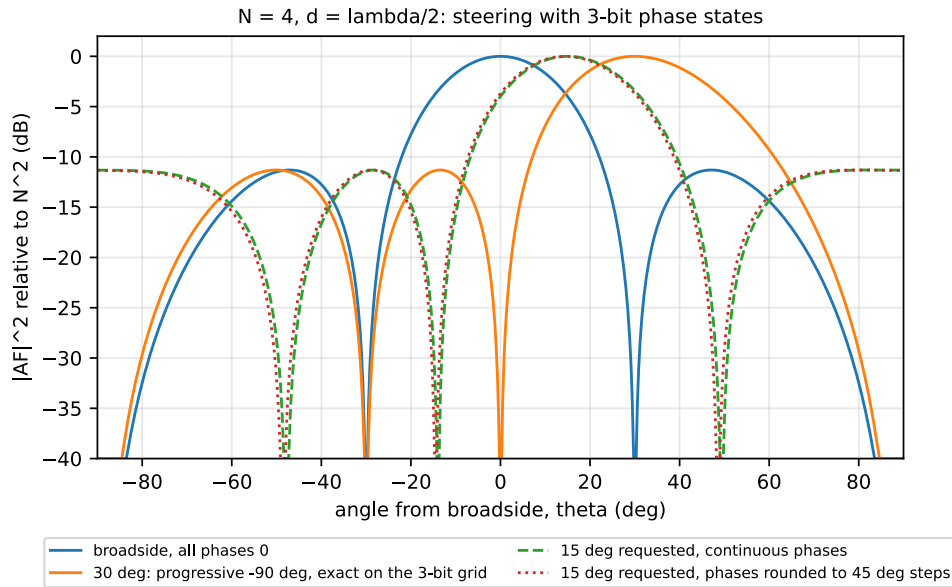
7.4 Nulls, beamwidth, sidelobes and grating lobes

- **Nulls.** $|AF| = 0$ when $N\psi/2$ is a nonzero multiple of π but $\psi/2$ is not: $\psi = 2\pi m/N$. At broadside, $N = 4$ and $d = \lambda_0/2$, the nulls fall at $\sin \theta = \pm 0.5$ and ± 1 : ± 30 and ± 90 degrees. Section 9 uses the null at 30 degrees.
- **Beamwidth.** The half power beamwidth at broadside is approximately $0.886 \lambda_0/(Nd)$ radians, 25.4 degrees for Rev A, and it widens roughly as $1/\cos \theta_0$ when the beam is steered.
- **Sidelobes.** With uniform amplitudes the first sidelobe is -11.3 dB for $N = 4$; the often quoted -13.2 dB is the limit for large N (-12.8 dB at $N = 8$, -13.15 dB at $N = 16$) **[analytical]**. Lowering sidelobes needs an amplitude taper, which Rev A cannot apply (decision 0003, section 4 of the architecture document).
- **Grating lobes.** AF repeats whenever ψ advances by 2π . A second full strength lobe, a grating lobe, appears in visible space unless

$$\frac{d}{\lambda} < \frac{1}{1 + |\sin \theta_0|}$$

which for steering anywhere up to 90 degrees gives $d \leq \lambda/2$. That is why half a wavelength is the usual spacing. Rev A's spacing is half a wavelength at 2.44 GHz; at the upper band edge it is 0.509 wavelengths, still far from a grating lobe for the benchmark steering angles up to 45 degrees.

- **Aperture.** The array spans $3d = 1.5\lambda_0 = 184$ mm between element centres.



analytical illustration: ideal isotropic elements, array factor only, no coupling

Figure 7a: array factor of four isotropic elements at half wavelength spacing, broadside, steered to 30 degrees with an exact progressive phase of -90 degrees, and steered to 15 degrees with continuous phases and with phases rounded to the 45 degree grid. The rounded 15 degree command points at 14.48 degrees. **[analytical]** illustration, generated by `tools/docs/master_reference_figures.py`.

7.5 What four elements can and cannot do

Can	Cannot
steer a 25 degree wide beam over roughly ± 45 degrees	form a narrow beam or resolve sources closer than about a beamwidth
place up to $N - 1 = 3$ nulls with full complex weights; fewer, and less precisely, with phase only quantised control	reject many interferers at once
show, clearly, every effect of phase error, quantisation, coupling and drift on pointing, gain and null depth	demonstrate the gain, sidelobe performance or scaling behaviour of a large array
be enumerated exhaustively: every reachable beam state can be evaluated	serve as a production antenna

Four elements is a small array, and this document does not present it otherwise. Its value is that it is an **instrumented** array: every element has its own connector, every channel can be isolated electronically, every state is an exact code word, temperature is logged, and the number of states is small enough to evaluate completely. Those are properties of a research platform, not of a product.

8. Quantised phase shifting

8.1 The eight states

Each Rev A channel has three switched line bits of 45, 90 and 180 degrees (decision 0003). With bit values $b_{45}, b_{90}, b_{180} \in \{0, 1\}$, the nominal phase of the channel is

$$\phi = 45^\circ(b_{45} + 2b_{90} + 4b_{180}) \in \{0^\circ, 45^\circ, 90^\circ, \dots, 315^\circ\}$$

so eight states, 45 degrees apart. State 7, all three bits set, is the 315 degree state with the most extra line. The phase is “nominal” because the realised phase of each state differs from it; the forward model of docs/mathematics/inverse-calibration.md section 2 uses nominal phases on purpose and lets the array state absorb what it can of the difference.

8.2 Quantisation error and its floor

To command an arbitrary phase, the controller picks the nearest state; the effect of this rounding on array patterns is a classical subject [B7]. If the wanted phases are spread uniformly, the rounding error is uniform on ± 22.5 degrees, and its standard deviation is

$$\sigma_q = \frac{\Delta}{\sqrt{12}} = \frac{45^\circ}{\sqrt{12}} = 12.99^\circ$$

Δ is the step, 45 degrees; $\sqrt{12}$ is the standard deviation of a uniform distribution of unit width. Decision 0007 propagates this floor to the array, with the finite array formulas implemented in `rfkit.budget` and checked by Monte Carlo [analytical]:

Consequence of the 3-bit floor	Value	Formula
pointing standard deviation at broadside	1.85 degrees	$\sigma_\theta = \sigma_q / (kd \cos \theta_0 \sqrt{\sum_i (i - \bar{i})^2})$, with $\sum_i (i - \bar{i})^2 = 5$ for four elements
coherent gain loss	0.167 dB	$G/G_0 = e^{-\sigma^2} + (1 - e^{-\sigma^2})/N$
error sidelobe floor	-18.9 dB relative to the peak	$10 \log_{10}(\sigma^2/N)$
worst case error on one channel	22.5 degrees	half a step; beyond it states come out of order

The large array approximation $e^{-\sigma^2}$, the classical tolerance result [B6], overstates the gain loss at four elements by about a third, 0.223 against 0.167 dB; decision 0007 corrected an earlier draft that used it. **These numbers are the floor no calibration can beat**, because they come from the bit count, not from any error. Decision 0007 anchors every acceptance threshold to them (section 57).

The rounding is deterministic for a given steering angle; σ_q is its average over angles. Some angles need no rounding at all. With $d = \lambda_0/2$ the progressive phase $\Delta\phi = -m \times 45^\circ$ is exactly reachable when $\sin \theta_0 = m/4$: at 0, 14.48, 30, 48.59 and 90 degrees. A request for 15 degrees rounds to the 14.48 degree command, an error of 0.52 degrees (Figure 7a).

8.3 Only relative phase matters

Adding the same phase c to every channel multiplies AF by e^{jc} , which changes neither its magnitude nor the direction of its maximum. **The far field beam shape depends only on relative phases**. Two consequences:

1. The common phase is unobservable from the beam and can be fixed by convention, for example by taking channel 0 as the reference (requirement R7).
2. The number of distinct beam shapes is the number of relative phase configurations. With one channel as reference and eight states on each of the other three,

$$|\mathcal{X}| = 8^{N-1} = 8^3 = 512$$

The 16 bit beam state word has 65 536 values, but most of them are redundant: they differ by a common phase, or by the phase bits of a channel that is disabled. Counting the enable bits as a binary amplitude, the distinct relative configurations are $\sum_{k=1}^4 \binom{4}{k} 8^{k-1} = 512 + 256 + 48 + 4 = 820$ **[analytical]**: all four channels on, three on, two on, one on.

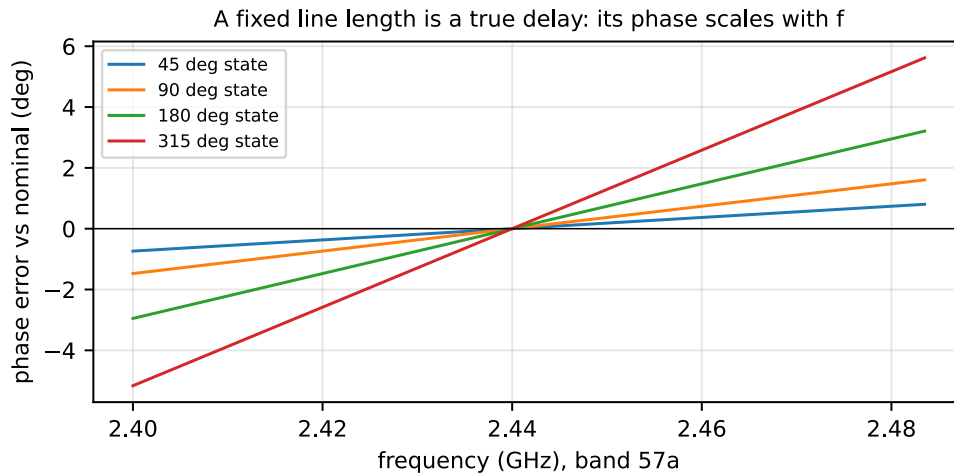
This small number is central to Part VIII. With an estimate of the array state, every one of the 512 (or 820) configurations can be evaluated in software in a fraction of a second, so the best beam for any criterion is found exactly, by enumeration. There is nothing for a learned beamformer to do (section 38).

8.4 Why the common phase is not quite free off f_0

The eight command “origins”, adding 0, 45, ..., 315 degrees to every channel, give the same beam at f_0 . Away from f_0 they do not, because a switched line’s phase scales with frequency (section 8.5): adding a common offset changes which bits are set and therefore how much line each channel carries. Decision 0007 found that a steering table which picks the origin per angle keeps band edge operation within the pointing budget, while the worst origin exceeds it at 22 of 91 angles between 0 and 45 degrees, by up to 22 per cent **[analytical]**. This constrains whoever writes the steering table; no rffit metric enforces it.

8.5 Dispersion of a switched line

A fixed length of line is a time delay, so its phase grows in proportion to frequency: $\phi(f) = \phi_{\text{nominal}} f / f_0$ for an ideal non dispersive line. Band 57a spans -1.64 to $+1.78$ per cent about 2.44 GHz, so the 315 degree state is -5.2 degrees off at 2.400 GHz and $+5.6$ degrees off at 2.4835 GHz, before any fabrication error **[analytical]**. Both simulators must reproduce this term; it is intrinsic, and only their disagreement about it is judged (decision 0007).



analytical illustration: ideal non-dispersive line, phase = nominal $\times f / f_0$

Figure 5b: phase error against frequency of the 45, 90, 180 and 315 degree states of an ideal switched line, relative to their nominal value at 2.44 GHz. **[analytical]**.

9. Null steering and interference rejection

9.1 Three different goals

Goal	What is optimised	When it is the right goal
steer the main beam	response in one direction θ_D	one source, no interference
place a null	response in one direction θ_I driven to zero	one interferer to suppress, the wanted direction loosely constrained

Goal	What is optimised	When it is the right goal
maximise a signal to interference metric	the ratio of wanted to unwanted response	both matter at once: the AP-S communication mode

9.2 Formulation

Write the steering vector for direction θ as $\mathbf{a}(\theta)$ with entries $a_n(\theta) = e^{-jnk d \sin \theta}$, and the realised weights as $\mathbf{w}$, whose entry w_n is the complex amplitude applied to element n (written $a_n e^{j\phi_n}$ in section 7, where a_n was an amplitude, not a steering vector entry), so that $\mathbf{a}(\theta)^H \mathbf{w} = AF(\theta)$ in the convention of section 7. For a wanted source at θ_D and an interferer at θ_I :

$$P_D(\mathbf{w}) = |\mathbf{a}_D^H \mathbf{w}|^2, \quad P_I(\mathbf{w}) = |\mathbf{a}_I^H \mathbf{w}|^2, \quad J(\mathbf{w}) = P_D(\mathbf{w}) - \lambda P_I(\mathbf{w})$$

or, with source powers S_D and S_I received by an isotropic element and receiver noise power σ_n^2 referred to one channel,

$$\text{SIR}(\mathbf{w}) = \frac{S_D P_D(\mathbf{w})}{S_I P_I(\mathbf{w})}, \quad \text{SINR}(\mathbf{w}) = \frac{S_D P_D(\mathbf{w})}{S_I P_I(\mathbf{w}) + \sigma_n^2 \|\mathbf{w}\|^2}$$

Symbol	Meaning	Unit
$\mathbf{a}_D, \mathbf{a}_I$	steering vectors towards the wanted and the interfering source	dimensionless
P_D, P_I	array power response towards each	dimensionless
λ	trade-off weight of the scalar objective, not a wavelength here	dimensionless
S_D, S_I	received powers of the two sources at one element	W
σ_n^2	noise power per channel	W

The SINR form is the textbook one for a receiver with independent channel noise [B5]. In Rev A the channels are combined passively before any receiver, so the noise is added after the combiner and the simpler SIR is the meaningful figure; which of the two applies depends on the receiver chosen for the demonstrator, which is not chosen (section 47).

With unconstrained complex weights the best null keeping the wanted response is the projection of $\mathbf{a}_D$ orthogonal to $\mathbf{a}_I$, $\mathbf{w} \propto (\mathbf{I} - \mathbf{a}_I \mathbf{a}_I^H / N) \mathbf{a}_D$, which puts an exact null on θ_I at the cost of some gain towards θ_D . Rev A cannot apply it: it has no amplitude control and only eight phases per channel. Its reachable set is the 512 relative configurations (820 with the enable bits), so the problem becomes

$$\mathbf{x}^* = \arg \max_{\mathbf{x} \in \mathcal{X}} J(\hat{\mathbf{H}}_t \mathbf{w}_{\text{nominal}(\mathbf{x})})$$

an exact enumeration once the array state estimate $\hat{\mathbf{H}}_t$ exists (docs/mathematics/inverse-calibration.md section 5).

9.3 Degrees of freedom at $N = 4$

Four channels give three relative complex weights. With full complex control they can place three nulls, or one null and keep a well formed main beam. With phase only control, the amplitudes are fixed and only the three relative phases remain; with three bits each of those takes eight values. Some pairs of wanted and interfering directions will be well served by some state, others will not, and an interferer close to the wanted direction, within about a beamwidth, cannot be rejected without losing the wanted signal too. Which angle pairs are achievable, and how robustly, is exactly the question of the $N = 4$ feasibility gate proposed in section 50; it has deliberately not been computed for this document, because its acceptance rule has to be fixed first.

9.4 Why nulls are a sensitive meter of calibration error

A main beam is forgiving: the 13 degree quantisation floor costs only 0.17 dB of gain. A null is the opposite. It exists because four contributions cancel, and any residual error leaves an uncanceled remainder. For small independent errors of standard deviation σ , in radians of phase or in relative amplitude, the expected power left in a null, relative to the beam peak, is

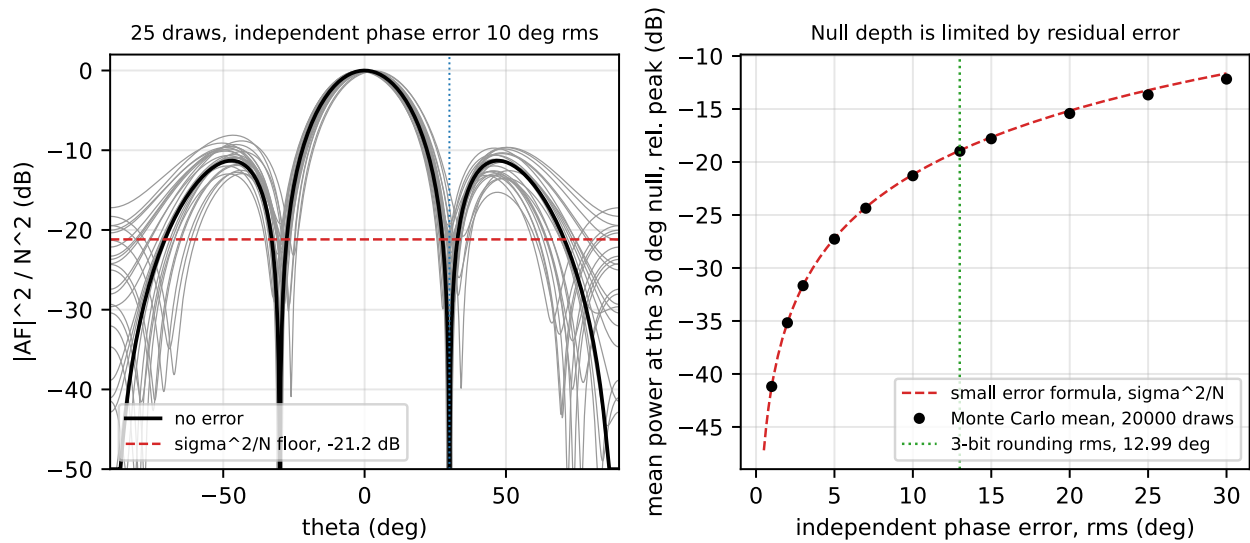
$$\frac{\mathbb{E}[|AF(\theta_{\text{null}})|^2]}{N^2} \approx \frac{\sigma^2}{N}$$

an approximation valid for $\sigma \ll 1$ rad, which the Monte Carlo of Figure 7b confirms. Its consequences, at $N = 4$ [analytical]:

Residual error, rms	Expected null power relative to the beam peak
1 degree	−41 dB

Residual error, rms	Expected null power relative to the beam peak
2 degrees	−35 dB
5 degrees	−27 dB
10 degrees	−21 dB
13 degrees, the 3-bit floor	−19 dB

The same mechanism is the “error sidelobe floor” of decision 0007. Phase quantisation, amplitude imbalance, coupling and drift all add to σ^2 , and any of them fills the null. A drift of a few degrees, almost invisible in the main beam, moves a null by several decibels. **That is why the demonstrator proposed in Part IX shows calibration through nulls rather than through the main beam.** Two cautions apply: the formula describes the average over random errors, not a particular deterministic null chosen by enumeration, which can be deeper or shallower; and a measured null depth is also limited by the measurement’s own noise floor and by multipath in the room.



analytical illustration: ideal isotropic elements, array factor only, no coupling; random draws from seed 20261004

Figure 7b: left, the broadside pattern with its null at 30 degrees, and 25 patterns with independent phase errors of 10 degrees rms; right, the mean power at the 30 degree null against the rms phase error, Monte Carlo and the small error formula. **[analytical]** illustration with random draws from a fixed seed; it is not data.

10. Mutual coupling

10.1 Two mechanisms

Antenna mutual coupling is radiative: part of the wave radiated or received by one patch is picked up by its neighbours, so each element’s current depends on what the others are doing. It is described by the antenna board’s 4×4 scattering matrix S_A , measured at the four element connectors; its off diagonal terms are the coupling. Its practical signatures are an **active reflection coefficient** that changes with the steering state and **embedded element patterns** that differ from the isolated element pattern and from one another.

RF path coupling happens in the beamformer: imperfect isolation between channels and reflections at the beamformer outputs. Decision 0008 splits it into a reverse part, which acts together with S_A through waves reflected by the antennas, and forward crosstalk, a channel’s transfer changing with a neighbour’s state, which it treats as a state dependent diagonal error judged under decision 0007.

Rev A’s two board split makes these separable: each board can be characterised on its own at the element connector plane (decision 0008, context).

10.2 Why a diagonal array state is an approximation

$$\mathbf{H} = \mathbf{D} + \mathbf{C}$$

$\mathbf{D}$ is diagonal: one complex gain per channel. $\mathbf{C}$ is the cross channel part: coupling. The Rev A calibration model keeps only $\mathbf{D}$. That model is exact only if coupling is negligible or if it does not depend on the commanded state. In general it does, because the contribution of element m to element n carries the relative phase of their commands.

Decision 0008 writes the coupled forward model explicitly. For the waves $\mathbf{t}(s)$ the beamformer delivers into matched loads in state s :

$$\mathbf{a}(s) = (\mathbf{I} - \mathbf{S}_{Bo}(s) \mathbf{S}_A)^{-1} \mathbf{t}(s), \quad \mathbf{e}(s) = (\mathbf{I} - \mathbf{S}_A) \mathbf{a}(s), \quad E(u, s) = \mathbf{v}(u)^T \mathbf{e}(s)$$

Symbol	Meaning
$\mathbf{S}_{Bo}(s)$	beamformer reverse path: output reflections on the diagonal, output to output isolation off it
$\mathbf{a}(s)$	waves actually incident on the antennas
$\mathbf{e}(s)$	radiating currents, in the canonical minimum scattering approximation [A23, A24]
$v_m(u) = e^{jmkdu}, u = \sin \theta$	far field phase of element m

The middle equation is an approximation exact for no real patch, so the simulation stage of G4 also uses the solver's embedded element patterns, with no approximation.

10.3 Gate G4: a test of the model, not a limit on coupling

Decision 0008 records four findings, obtained from the model on synthetic matrices before any data, which shaped the test.

1. **A calibration residual cannot detect coupling.** When coupling does not depend on the state, a probe calibration at one direction fits the diagonal model exactly. The error appears only when the beam is steered away from where the calibration looked.
2. **A single raw coupling number cannot decide.** On a synthetic uniform array with nearest neighbour coupling of -25 dB, the pointing error of the broadside calibration ranges from 0.38 to 1.27 of the budget as the **phase** of the coupling varies. Same $|S_{ij}|$, opposite verdicts.
3. **A single number can guarantee something narrower.** A screen, a bound on the relative size of the coupled currents, guarantees a pass when it is below 0.040 (about -28 dB aggregate, -34 dB per neighbour), but it rejects arrays that full propagation passes by several dB. It is reported and never decides.
4. **Coupling is structured.** It is reciprocal, close to Toeplitz and deterministic; treating it as independent random terms would be the wrong model.

The test that follows judges what matters, the calibrated beam. At every frequency in band 57a, for steering angles from -45 to $+45$ degrees in one degree steps and all eight command origins, the beam of the calibrated diagonal model is compared with the coupled beam, against a budget of $\eta_c = 0.10$ of the quantisation floor's error variance, allocated separately from decision 0007's:

$$\delta\theta^2 \leq \eta_c \sigma_{\theta,q}^2(\theta_0), \quad L(1 + \epsilon) \leq \eta_c L_q$$

$\delta\theta$ is the difference of the two beam maxima, $\sigma_{\theta,q}(\theta_0)$ the quantisation pointing spread at that steering angle, L the coherence loss caused by the relative current error ϵ , and $L_q = 0.167$ dB the floor's. The outcome is PASS, FAIL, INTERMEDIATE or UNRESOLVED by rules fixed before any data (section 58). On the synthetic nearest neighbour chart printed by `python -m rfkit.cli g4-chart`, full propagation passes at every coupling phase down to -27 dB and fails at some phases at -25 dB [**synthetic**]; this is an idealisation and not a prediction of EXP-011.

The full coupling model is an extension, not the default. If G4 fails, the first promotion freezes a measured coupling matrix in the forward model, which keeps the unknowns at $2N - 2$ provided the coupling does not drift. Only if that also fails are coupling terms estimated, raising the unknowns to order $2N^2$, 32 at $N = 4$, and reopening every measurement count in the repository (decision 0008, "What follows from each outcome"). No coupling has been simulated or measured; the antenna geometry it needs is not designed.

10bis. Integrated sensing and communication (ISAC)

This section is numbered 10bis so that the numbering of everything after it is unchanged. It teaches the concept once; Part VIII applies it to the learning track and Part IX to the contest.

10bis.1 What ISAC means

ISAC stands for integrated sensing and communication. It names the family of radio systems in which communication and sensing share resources that used to be separate: spectrum, waveform, transmitter, antenna aperture, RF front end, processing, or simply the same electromagnetic field [L32]. Communication tries to deliver information from a transmitter to a receiver through the propagation channel. Sensing tries to infer something about the propagation channel itself: where reflectors are, whether something moved, whether a person is present. Both functions observe

the same field; they ask opposite questions of it. For communication the environment is a nuisance to be equalised or rejected; for sensing it is the signal.

ISAC is therefore not “Wi-Fi used as radar”. That is one example among several, and not the one most of the literature is about.

10bis.2 Three architectural families

Family	What is shared	Typical form	AetherArray
A. Joint waveform, joint transmitter	the transmitted waveform is designed for both data and sensing	OFDM signals whose echoes are processed for range and Doppler while carrying data [L42]; automotive and 6G joint waveform design [L32]	not implemented ; Rev A has no transmitter of its own
B. Shared RF hardware and spatial aperture	the same antenna array, RF chain and spatial processing serve both functions	an array that alternates or combines communication beams and sensing beams	yes, proposed : one four element reconfigurable receiving array for both
C. Opportunistic, passive sensing	the field of an existing transmitter, whose waveform the sensing receiver did not design	passive bistatic radar with broadcast or Wi-Fi illuminators [L33, L43]; Wi-Fi sensing from channel measurements [L34]	yes, proposed : commercial Wi-Fi or Bluetooth sources as illuminators

The proposed AetherArray demonstrator belongs to **B and C together**: a receive only array whose aperture and RF chain are shared between communication oriented interference rejection and opportunistic sensing, lit by transmitters it does not control. It does nothing in family A.

A reconfigurable array is valuable in family B for a simple reason: spatial selectivity helps both functions. Communication wants gain towards the wanted source and nulls towards interferers; sensing wants to look at the environment from several spatial viewpoints. An array that can be switched between patterns on one clock edge serves both with the same hardware, and the same calibration determines how good both are.

10bis.3 Illuminators of opportunity

An **illuminator of opportunity** is a transmitter that exists for its own purpose, a Wi-Fi access point, a Bluetooth beacon, a phone in hotspot mode, whose field a separate receiver exploits for sensing. Nothing is transmitted for the sensing function.

```

COTS transmitter (Wi-Fi AP, BLE beacon, hotspot phone)
|
+----- direct path -----\
|                             \
+--> person or object --> scattered / reflected path --> AetherArray (receive only)
|                             /
+--> walls, floor, furniture --> other multipath ----/
                        person in a path: shadowing; edges: diffraction; motion: time variation

```

Figure 20: an illuminator of opportunity. The array receives the direct path and every scattered path; a person changes some of them.

The field at the array is a sum over propagation paths. For a narrowband signal at one frequency, the complex signal at element n is approximately

$$s_n(t) = \sum_{\ell} \alpha_{\ell}(t) e^{jnk d \sin \theta_{\ell}(t)} c(t) + \nu_n(t)$$

Symbol	Meaning
ℓ	propagation path: direct, single reflection, multiple reflection, diffraction
$\alpha_{\ell}(t)$	complex amplitude of path ℓ : path loss, reflection coefficient, delay phase
$\theta_{\ell}(t)$	arrival direction of path ℓ
$c(t)$	the transmitter’s own signal, unknown and bursty for Wi-Fi or Bluetooth
$\nu_n(t)$	noise and other sources

The physical mechanisms a person or an object changes:

Mechanism	Effect on the paths
reflection and scattering	adds a path or changes its amplitude and direction

Mechanism	Effect on the paths
shadowing	attenuates a path that crosses the body, often the direct path
diffraction	redistributes field around edges, including the body's
multipath change	changes the relative phases of paths, so their sum fluctuates
motion	makes $\alpha_\ell(t)$ and $\theta_\ell(t)$ vary in time; a moving reflector shifts phase by 2π per wavelength of path length change, 123 mm at 2.44 GHz

The narrowband model ignores delay spread across a Wi-Fi channel's bandwidth; that is acceptable for a power based receiver and is an approximation, stated as such.

Advantages	Limitations, and what they imply for experiments
low cost: no transmitter to build	the waveform is not controlled: power, timing and channel are the transmitter's; experiments must record them or use ratio features
spectrum reuse: no extra emission	traffic and power vary: a Wi-Fi AP's activity depends on its users; idle periods give few readings
commercial compatibility, as the AP-S rules require	synchronisation is limited: the receiver does not know when bursts arrive unless it decodes them
receive only hardware	multipath depends on geometry: results are specific to a room and a placement; the transmitter and array positions must be recorded and repeated
educational clarity	repeatability is hard: people near the setup, including the operator, change the field; EXP-005 Phase B measures exactly this
	sensing performance depends on where the transmitter is: a person who blocks no strong path is nearly invisible

10bis.4 One field, two problems

Write the combiner output for commanded state $\mathbf{x}$ as $r(t) = \mathbf{u}^\top \mathbf{s}(t)$, with realised weights $\mathbf{u} = \mathbf{H}_t \mathbf{w}(\mathbf{x})$: the nominal weights of the state, passed through the array state of section 0.3. Its mean power is

$$P(\mathbf{x}, t) = \mathbb{E}[|r(t)|^2] = \mathbf{u}^\top \mathbf{R}(t) \mathbf{u}^*, \quad \mathbf{R}(t) = \mathbb{E}[\mathbf{s}(t) \mathbf{s}(t)^H]$$

Symbol	Meaning	Unit
$\mathbf{s}(t)$	element signals, the field sampled by the four antennas	$\sqrt{W}$, complex
$\mathbf{R}(t)$	spatial covariance of the received field: 4×4 , Hermitian; it is set by the sources and the environment	W
$\mathbf{u}$	realised complex weights of the commanded state	dimensionless

The expectation is over the transmitter's signal and the noise, over an averaging time short compared with environmental change.

Communication asks which state best serves a wanted transmitter against an interferer. With both sources present, $\mathbf{R} = \mathbf{R}_D + \mathbf{R}_I + \mathbf{R}_v$, and the desired and interfering powers are $P_D(\mathbf{x}) = \mathbf{u}^\top \mathbf{R}_D \mathbf{u}^*$ and $P_I(\mathbf{x}) = \mathbf{u}^\top \mathbf{R}_I \mathbf{u}^*$. In free space with one path each, these reduce to the P_D and P_I of section 9.2; indoors each source contributes several paths, and a null placed on an interferer's direct path does not null its reflections. The objective $J(\mathbf{x}) = P_D(\mathbf{x}) - \lambda P_I(\mathbf{x})$, or a signal to interference ratio, is meaningful only when the receiver can separate the two powers, by channel, time or transmitter identity (section 47). Beam steering, interference suppression and null placement are then a choice among the reachable states, $8^{N-1} = 512$ relative configurations, or 820 with the enable bits, and the choice is an exact enumeration against the estimated array state (section 8.3). **ISAC creates no reason to use machine learning for beam synthesis**; section 38 explains why once, and nothing in this section changes it.

Sensing asks whether, and how, the environment changed. Cycling through K receive states gives a vector

$$\mathbf{p}(t) = [P(\mathbf{x}_1, t), \dots, P(\mathbf{x}_K, t)]^\top, \quad P(\mathbf{x}_k, t) = \mathbf{u}_k^\top \mathbf{R}(t) \mathbf{u}_k^*$$

Each pattern weights the paths differently: a beam towards the door sees a reflection from the door strongly, a pattern with a null towards the transmitter suppresses the direct path and leaves the scattered field visible. A person who adds, removes or moves a path changes $\mathbf{R}(t)$, and the change appears in the components of $\mathbf{p}$ whose patterns look that way.

Why several patterns carry more information than one received signal strength. Each power is a linear function of the entries of $\mathbf{R}(t)$, because $\mathbf{u}^\top \mathbf{R} \mathbf{u}^* = \sum_{m,n} u_m R_{mn} u_n^*$. A Hermitian 4×4 matrix has 16 real parameters. A single omnidirectional element measures one of them, a diagonal entry: the total power, which says almost nothing about where the field comes from. K well chosen patterns measure up to 16 independent combinations, including the cross terms R_{mn} that hold the relative phases between elements and therefore the arrival directions. The pattern diverse

power vector is a sketch of the spatial covariance obtained without coherent receivers per channel. This is a statement about what can in principle be observed; which patterns are informative in a given room, and whether 16 are needed, is an experimental question.

10bis.5 The calibration problem both functions share

The measurement does not depend on the environment alone. In the notation of this document,

$$y_t = G(\mathcal{E}_t, \mathbf{H}_t, \mathbf{x}_t) + \epsilon_t$$

where $\mathcal{E}_t$ is the state of the environment and the sources, $\mathbf{H}_t$ the array state, $\mathbf{x}_t$ the commanded state and ϵ_t the measurement noise. For power readings, section 10bis.4 makes G explicit, and with the diagonal array state of Rev A something sharper follows. Substituting $\mathbf{u}_k = \mathbf{H}_t \mathbf{w}_k$:

$$P(\mathbf{x}_k, t) = \mathbf{w}_k^T \tilde{\mathbf{R}}(t) \mathbf{w}_k^*, \quad \tilde{\mathbf{R}}(t) = \mathbf{H}_t \mathbf{R}(t) \mathbf{H}_t^H, \quad \tilde{R}_{mn} = h_m(t) R_{mn}(t) h_n^*(t)$$

The measurements depend on the environment and on the hardware only through their product $\tilde{\mathbf{R}}$. A phase drift $\delta\phi_m$ on channel m rotates every cross term $\tilde{R}_{mn}$ by $\delta\phi_m$, which is exactly what a change of arrival direction does. A gain drift on channel m scales its row and column, which is what a change in a path's strength does. From the sensing data alone, hardware drift and environmental change are not separable:

A measured change is not necessarily an environmental change.

Source of a change in $\mathbf{p}(t)$	Examples
environment, the signal	a person enters, an object moves, a reflector shifts, shadowing or multipath changes
hardware, the confound	amplitude drift, phase drift, temperature, switch path variation, detector and acquisition drift

The two can be told apart only with information from outside the sensing data: an independent measurement of $\mathbf{H}_t$ (a calibration), or prior knowledge of how each evolves. Their time scales help: people move in seconds, thermal drift takes minutes to hours. They do not settle it, since a moved piece of furniture is a slow environmental change. First order, the confound is visible in the differential of the same expression:

$$\delta P_k \approx \underbrace{\mathbf{u}_k^T \delta \mathbf{R} \mathbf{u}_k^*}_{\text{environment}} + 2 \underbrace{\text{Re}\{(\delta \mathbf{H}_t \mathbf{w}_k)^T \mathbf{R} \mathbf{u}_k^*\}}_{\text{hardware}}$$

Both terms land in the same K numbers.

The consequence for each function:

```

COMM :  H_t drifts --> realised weights wrong --> beam off, null filled --> desired / interferer
                                              discrimination worse
SENSE:  H_t drifts --> spatial signature p(t) changes --> a change detector reports an event
                                              that is hardware drift, not the room

```

This is the bridge between the three threads of the project. Calibration research estimates $\mathbf{H}_t$. The learning track tries to estimate it with fewer new measurements by using its history. ISAC is where an error in $\mathbf{H}_t$ becomes a wrong communication beam and a false sensing event.

A consequence for calibration in a room. An over the air calibration against an ambient source estimates, per channel, the product of the hardware term and the incident field at that element, $h_n s_n$, not h_n alone. In free space with a single path from a known direction, the field term is known and can be divided out; indoors it contains multipath that the calibration absorbs, and that changes when the room changes. A calibration done that way is a channel calibration, useful for the communication mode in that room, but it is not the hardware state the drift prior is about. Hardware labels for the learning track therefore come from the conducted route or a controlled probe (sections 11.3 and 16), and any calibration done at a demonstration has to say which of the two it is.

10bis.6 Where the learning contribution sits

Part VIII defines the method; this paragraph only places it. The diagonal state $\mathbf{H}_t = \text{diag}(h_i(t))$ with $h_i(t) = g_i(t) e^{j\delta\phi_i(t)}$ is unchanged, and so is the Bayesian structure:

$$p(\mathbf{H}_t | y, \mathbf{x}, \mathcal{D}) \propto p(y | \mathbf{H}_t, \mathbf{x}) p_\theta(\mathbf{H}_t | \mathcal{D})$$

with the learned temporal prior p_θ , the physical likelihood $p(y | \mathbf{H}_t, \mathbf{x})$, and the posterior as the corrected estimate after sparse new measurements. Its purpose in an ISAC system is not “learning produces a beam command”. It is:

```
drift history --> better prior on H_t now --> fewer new calibration measurements
--> calibrated spatial response restored sooner --> COMM and SENSE more reliable
```

M_{required} , the number of new physical measurements needed to recover a target (section 0.2), has two system level forms here. **Communication:** at equal recovered desired to interferer performance, $M_{\text{sparse}} < M_{\text{full}}$ would support the research claim. **Sensing:** at equal stability or detection performance, the same inequality could support it too, but only once the sensing metric is defined before the experiment, for example the false alarm rate of a change detector during periods with no environmental change. No such result exists.

10bis.7 Why this is ISAC, and in what narrow sense

The proposed system is, precisely:

a reconfigurable receiving array sharing the same spatial RF aperture and hardware between communication oriented interference rejection and opportunistic environmental sensing.

The two functions share the antenna array, the phase reconfigurable RF chain, the calibration, the measurement infrastructure, the deterministic control and the spatial patterns. The illuminator stays external and commercial. That is a legitimate member of families B and C, and a deliberately narrow one. AetherArray is **not** a 5G or 6G ISAC base station, not a joint waveform design system, not a monostatic radar, not a range and Doppler radar (a power detector measures neither delay nor frequency shift), and not a centimetre level localisation system. Calling it any of those would be semantic inflation.

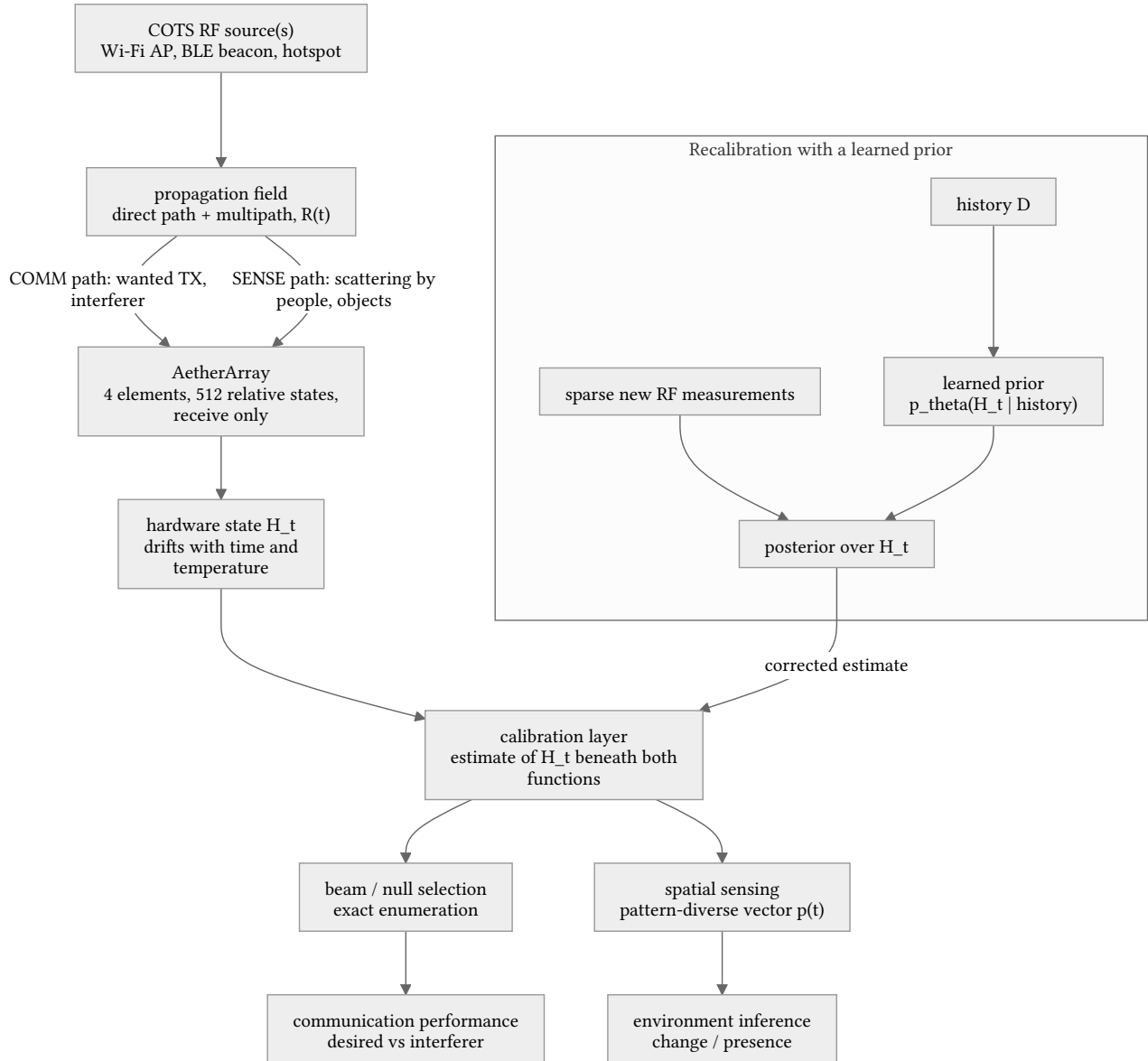


Figure 19: the signature figure. One RF aperture, two ISAC functions, one calibration layer beneath both, and the learned prior as a way to restore that layer with fewer new measurements.

10bis.8 ISAC claims and their evidence

Claim	Type	Current status	Evidence today	What would establish it
the array can steer a receive response	theoretical, simulation	analytical only	array factor, sections 7 and 8; no simulation of the real geometry, no hardware	an HFSS model of the array (EXP-011, SIM-008) and a measured pattern change between states (EXP-007 to EXP-009)
the array can suppress an interferer while keeping the wanted link	system level	not established	the ideal theory of section 9; the $N = 4$ feasibility gate is not written or run	the gate (section 50), then a physical desired and interferer experiment with a source separating receiver
pattern diverse power vectors contain sensing information	plausible, literature supported in other forms	not demonstrated on AetherArray	the observability argument of 10bis.4; Wi-Fi sensing literature with different receivers [L34]	a repeated, controlled sensing experiment with a pre-registered task and error rate
hardware drift degrades sensing stability	physical hypothesis	unmeasured	the confound $\tilde{\mathbf{R}} = \mathbf{H}\mathbf{R}\mathbf{H}^H$ of 10bis.5; no drift has been measured	repeated sensing with a static room, calibrated against uncalibrated, over a drift period
drift degrades null depth	physical hypothesis, analytical	unmeasured	σ^2/N , section 9.4	a measured null over time, with temperature logged

Claim	Type	Current status	Evidence today	What would establish it
a learned prior reduces new recalibration measurements	research hypothesis	unproven; gated by G2	none; nothing built	a held out temporal experiment against baselines A to D, section 42
the system qualifies as ISAC in families B and C	definitional	true by design, narrow	10bis.7	not an empirical claim; it holds if both modes run on the shared aperture

11. Calibration

11.1 What calibration means here

The controller commands channel i with a nominal complex weight x_i , set by the enable bit and the three phase bits. The hardware applies something else:

$$x_i \longrightarrow h_i x_i, \quad h_i = g_i e^{j\delta\phi_i}, \quad \mathbf{H}_t = \text{diag}(h_1, \dots, h_N)$$

Symbol	Meaning	Unit
x_i	nominal weight: $e_i e^{j\phi_i}$, enable bit e_i times the nominal state phase	dimensionless
h_i	complex error of channel i	dimensionless
g_i	gain error of channel i , realised amplitude relative to nominal	dimensionless
$\delta\phi_i$	phase error of channel i	rad
$\mathbf{H}_t$	the diagonal array state at time t	

Calibration estimates $\mathbf{H}_t$ and uses it. With phase only control, the correction cannot be $\mathbf{H}^{-1}\mathbf{x}$ as in a textbook; it is the selection, among the reachable states, of the one whose realised weights $\hat{\mathbf{H}}_t\mathbf{x}$ best serve the criterion. Gain errors are estimated and not corrected (decision 0003, section 4).

11.2 Kinds of error

Error	Example	Absorbed by a diagonal state?
common gain and phase, the same on every channel and every state	probe antenna gain, cable to the analyser, distance	yes, and unobservable anyway: it does not change the beam shape
per channel, independent of the commanded state	different jumper lengths, connector differences, switch to switch insertion loss variation	yes : this is exactly what one complex number per channel holds
per channel, dependent on the commanded state	a 45 degree bit that is really 47 degrees; a long state lossier than a short one; reflections between discontinuities that change with the selected arm	no
common to all channels, proportional to the state phase	a permittivity error or a frequency offset scaling every line length by the same fraction	no : it depends on the state, so it steers the beam (decision 0007)
coupling between channels	section 10	only if it does not depend on the state

Why state dependent errors are the dangerous ones. A state independent error is a constant per channel; one calibration measures it and every beam benefits. A state dependent error is different for each of the eight states of each channel, so it reaches every calibrated beam differently, and the diagonal model cannot represent it. It can only be bounded by design, which is why decision 0007 derives a requirement on the hardware's own state dependent phase error, 2.29 degrees at f_0 , and an amplitude imbalance allowance of 0.82 dB peak to peak, or modelled explicitly per state, at the cost of 32 complex unknowns instead of 4.

11.3 What a full calibration is

A **full calibration** estimates all identifiable parameters of the chosen model from scratch, without using any previous estimate. On Rev A it can be done two ways.

Route	Procedure	Measurements	Needs
complex, per channel	enable one channel, terminate the other three, measure S_{21} from the common port, repeat for each channel	$N = 4$ complex readings	the analyser; per channel isolation, which Rev A provides electronically

Route	Procedure	Measurements	Needs
power only, REV	for each channel, step its phase through at least three states while the others stay fixed, and read the total power	$3N = 12$ power readings at the minimum	a power reading at the sum port and one fixed probe (section 11.5)

The complex route is the “expensive label” of the learning track: it gives every channel’s complex transfer directly, and Rev A obtains it without touching a cable (docs/architecture/rev-a-rf-architecture.md section 5.3).

11.4 Identifiability: what can be known at all

Of the $2N$ real numbers in $\mathbf{H}_t$, two cannot be determined from the array output. A common phase rotation of every h_i is confounded with the phase of the probe path, and a common scaling of every g_i is confounded with the gain of the probe path, the probe antenna, the distance and the cables, which are not known to the required precision. Neither changes the beam shape. With one channel taken as reference, the identifiable set is

$$d = 2N - 2$$

real parameters, $N - 1$ relative gains and $N - 1$ relative phases: **six at $N = 4$** . Since a scalar reading provides one real number, six readings is the absolute floor for any method that starts with no information about the state, whatever the algorithm.

11.5 Why power only measurement is harder

A power reading is quadratic in the unknowns:

$$y_k = \left| \sum_{n=1}^N w_n(\mathbf{x}_k) h_n \right|^2 + \epsilon_k$$

so recovering $\mathbf{h}$ means recovering a complex vector from magnitudes alone, up to a global phase: **phase retrieval**. The classical rotating element field vector method (REV) of Mano and Katagi [A6] sidesteps it channel by channel. Stepping the phase φ of channel n while the others, summing to S , stay fixed gives

$$P(\varphi) = |S|^2 + |y_n|^2 + 2|S||y_n|\cos(\psi_n + \varphi - \arg S)$$

a sinusoid with three unknowns, hence at least three phase states per channel and $3N$ readings in total (docs/mathematics/formulation.md section 6; [A20] for the count). Note a direct consequence of the formula: the offset and the amplitude of the sinusoid are symmetric in $|S|$ and $|y_n|$, so the two magnitudes can be swapped without changing $P(\varphi)$. The method needs a further assumption, such as $|y_n| < |S|$, or a further measurement, to choose between them.

What the phase retrieval literature actually establishes. This point needs care, because the repository states it more strongly than the literature supports.

Result	Status	Source
$4N - 4$ generic intensity measurements suffice to determine any vector in $\mathbb{C}^N$ up to global phase	proved: a sufficient count for generic measurement vectors	Conca, Edidin, Hering, Vinzant 2015 [A12]
$4N - 4$ is also necessary	proved only for dimensions $N = 2^k + 1$; conjectured in general	[A12]; [L3]
at $N = 4$, eleven measurements can be injective	an explicit frame of 11 vectors in $\mathbb{C}^4$ with a certificate	Vinzant 2015 [L4]
lower bound at $N = 4$	10 or 11 from a general bound; a 2026 preprint claims exactly 11	[L5]; [L6], unreviewed

All four are verified at index level only (Part XV). Three corrections follow for the repository’s argument, which docs/architecture/ml-calibration.md section 3 and decision 0002 state as “recovering a complex vector from intensity measurements alone generically requires at least $4N - 4$ ”:

- $4N - 4$ is a **sufficient** count for generic vectors, not a proven lower bound at $N = 4$, where 11 can suffice.
- These results concern **generic** measurement vectors. The Rev A probe vectors are structured, unit modulus and restricted to 45 degree phases; whether a given set of them is injective has to be checked for that set. Uncertainty I16 already records this as open.
- Injectivity is uniqueness for **every** possible state. A method that uses prior information, for example that the errors are small, or that one magnitude exceeds another, can need fewer readings. REV itself relies on such an assumption, and the fast amplitude only method of Long and co-authors reports about $2N$ readings with three phase states [L7], although an experimental comparison found it less accurate than REV [A22].

11.6 What remains of the “no headroom” argument

Count at $N = 4$	Readings	Nature
identifiable parameters, the absolute floor	6	necessary for any method starting from nothing
complex route, one channel at a time	4 complex, 8 real numbers	what the analyser path needs
fast amplitude only method [L7]	about 8, to verify	structured, with prior assumptions
generic injective power only set	11 to 12	uniqueness for every state
REV at its minimum of three states	12	structured, with one assumption

The honest conclusion is narrower than the repository’s wording and survives intact in substance. On first calibration at $N = 4$, the best classical routes already sit within a few readings of the floor of six, and going below the injective count requires prior information about the state. A learned first calibration estimator trained on simulated arrays is a way of supplying population prior information; at this scale it has little room to save measurements, its labels exist only in simulation, and any saving it showed would be hard to separate from the choice of baseline. **So $N = 4$ provides no honest argument that machine learning reduces first calibration measurement counts, and the project does not make that claim.** The prior information worth using is the array’s own history, which exists only for recalibration. That is decision 0002, and this document recommends restating its counting argument in these terms (Appendix J).

The complex route strengthens the conclusion: since gate G1 passed (decision 0004), the analyser can measure each channel’s complex transfer in N readings, which is already close to the floor. Decision 0002’s second reopening condition, “if EXP-004 finds that phase can be measured directly, ... the comparison in section 3 of the architecture document is redone”, was triggered by that result; decision 0004 records that decision 0002 does not reopen, which is correct in substance, but the comparison itself has not been redone in writing.

12. Drift

12.1 What drift is

An array calibrated at time t is described by $\mathbf{H}_t$. Later it is described by a different state:

$$\mathbf{H}_{t+\Delta t} \neq \mathbf{H}_t$$

The calibration made at t then describes hardware that no longer exists. How long a calibration stays valid, the question of EXP-010, is unmeasured in this project and, by the repository’s reading of the literature, rarely measured outside climate chambers (research/state-of-the-art.md section 4).

12.2 Possible sources, and what is known about each

None of the sources below has been measured on AetherArray, because no AetherArray hardware exists. The table classifies each by the evidence for its relevance here.

Source	Mechanism	Classification
temperature of the beamformer board	permittivity, copper dimensions and switch characteristics change with temperature	plausible ; no temperature coefficient for the selected laminate is recorded
temperature of the detector	the AD8318 output drifts with temperature, ± 0.5 dB over its full range [V6]	documented by the vendor as a range figure; no per degree slope is given, so the slope near room temperature is unresolved (I13)
switch state repeatability	a switch returning to a state may not return to exactly the same transfer	unresolved and important: if not below the measurement floor, the drift experiment measures the switches (decision 0003; E6)
connector and cable movement	mating a connector or flexing a cable changes phase	plausible , and modelled as a jump , not drift: every session records a connector handling flag (I14)
supply variation	switch and detector behaviour with supply voltage	plausible
analyser drift after calibration	the instrument’s own error terms move with time and temperature	plausible ; uncertainty not yet characterised (O1, O7)
ageing	slow material and contact changes	plausible , on time scales beyond the planned experiments
environment	people and objects near a radiated measurement	not hardware drift , but it changes radiated readings and can be mistaken for drift; EXP-005 Phase B measures it

One mechanism deserves emphasis because it connects drift to the limits of the diagonal model. If the board’s permittivity changes uniformly with temperature, every switched length changes phase in proportion to its electrical length.

That is a common proportional error: it depends on the commanded state, so a diagonal state cannot absorb it and it steers the beam (decision 0007). Temperature drift may therefore be partly state dependent. Whether it is large enough to matter is unknown **[to verify]**; the board temperature sensor of requirement R4, beside the phase network, is what will make it visible.

12.3 What drift does

Affected quantity	How drift enters	Sensitivity
beam pointing	the component of the phase drift that is linear in element index	about 0.14 degrees of pointing per degree of independent phase error at broadside (from section 8.2)
gain	the variance of the phase and amplitude drift	small: 0.1 dB for 10 degrees rms
null depth	the variance of the drift, section 9.4	large: a few degrees of drift move a null by several dB
sensing features	any change of the receive patterns used as features	a hardware change looks like an environmental one (section 48)

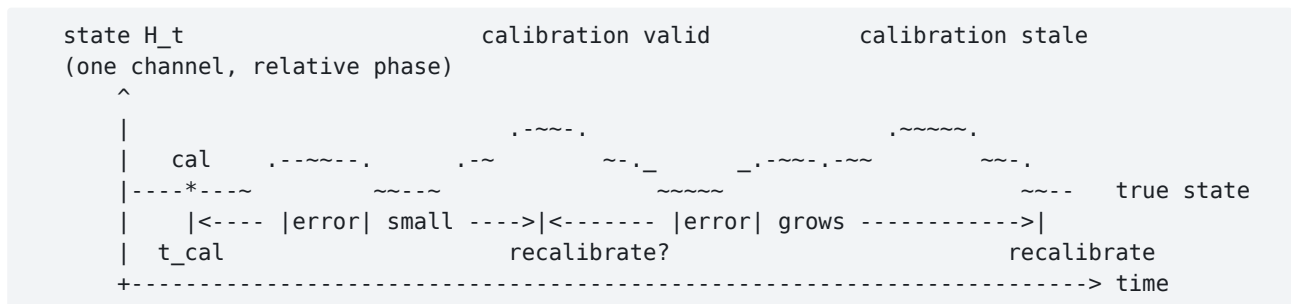


Figure 9: conceptual drift of one channel's relative phase after a calibration. Not data; no drift has been measured.

Gate G2 asks whether drift over a few hours is larger than the measurement's own repeatability. If it is not, there is nothing for a drift prior to learn, the learning track is abandoned, and decision 0002 is superseded rather than reinterpreted (decision 0002, conditions for reopening).

Part II. The engineering problem

II.1 From one array to many channels

Phased arrays are used in radar, satellite terminals, cellular base stations and test systems, and the larger ones have hundreds or thousands of channels. Every channel carries its own gain error, its own phase error, its own temperature dependence and its own share of manufacturing variation, and the elements couple to their neighbours. The array only does what its designer intended once those errors have been measured and compensated: once it has been calibrated.

Calibration costs something every time it is done:

Cost	Why
measurement time	each state must be applied, settled and read, with enough integration to beat the noise
RF hardware	probes, couplers, reference channels, switching for calibration paths
downtime	an array being calibrated is usually not doing its job
test complexity	positioners, chambers, fixtures, procedures, operators
energy	transmitting calibration signals, running instruments
computation	small compared with the rest; the measurements dominate (benchmarks/metrics.md)

A first calibration, done once at the factory or in a chamber, may be expensive and still acceptable. Repeated recalibration in the field is different: its cost recurs for the life of the array, and the array drifts in between.

II.2 Recalibration has structure

The array state does not jump arbitrarily from one calibration to the next. Between $t - 1$ and t , the same switches, lines and connectors are present, at a temperature that has changed by a few degrees, after a time that is known. Much of the change is plausibly smooth, correlated across channels, and related to measurable covariates such as temperature and elapsed time. That is the hypothesis that makes a learned prior worth testing:

$$p_{\theta}(\mathbf{H}_t \mid \mathbf{H}_{t-1}, \mathbf{H}_{t-2}, \dots, T_t, \Delta t) \text{ instead of } p(\mathbf{H}_t) \text{ uninformative}$$

Symbol	Meaning
p_{θ}	a probability model with parameters θ learned from the array's own history
$\mathbf{H}_{t-1}, \mathbf{H}_{t-2}, \dots$	earlier calibrated states
T_t	temperatures logged with the session
Δt	time elapsed since the last calibration

If the hypothesis holds, a recalibration starts from a good guess with a known uncertainty, and only the dimensions that the history cannot predict have to be measured afresh.

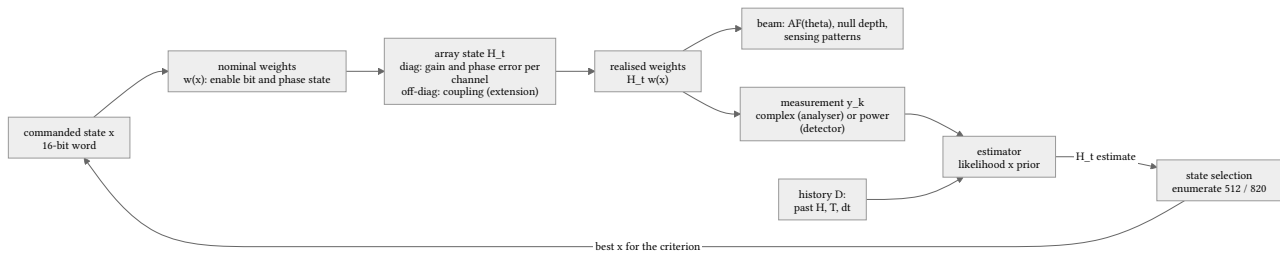


Figure 8: the calibration model. The commanded word becomes nominal weights; the array state turns them into realised weights; measurements of the realised weights, with the history, give an estimate of the state, from which the best reachable command is selected.

II.3 Initial calibration and recalibration are different problems

	Initial calibration	Recalibration
prior information about $\mathbf{H}_t$	none beyond the physical model and population statistics	the previous calibration, the elapsed time, the temperature history

	Initial calibration	Recalibration
minimum new readings	at least $2N - 2$ for any method; several more for power only uniqueness (section 11.6)	can be fewer than $2N - 2$ for a target accuracy, if the prior pins most dimensions
where learning could help	little room at $N = 4$; labels only in simulation	the subject of the project: labels from hardware, attributable savings
repository specification	ML-A, EXP-013, a control	ML-B, EXP-015, the central claim

AetherArray’s machine learning claim applies to the second column. The first column is kept as a control that shows the learning machinery works at all (decision 0002, point 2).

13. Why this matters at scale, and why a four element array can still be useful

13.1 What a four element array offers

Property	Why it matters for this research
real RF hardware	the drift, tolerances, connectors and switch nonidealities are physical, not modelled
a known number of channels	identifiability and counts are exact and small
repeatable state control	every beam state is an exact 16 bit code word applied on one clock edge
observable drift	temperature is logged beside the phase network and the detector
ground truth	the analyser measures each channel’s complex transfer, electronically isolated, with no cable handling
cheap experiments	a few tens of euro of parts; unattended runs
complete enumeration	512 relative states: every beam can be evaluated exactly

13.2 What it does not offer

A physical validation at $N = 4$ is not a proof of behaviour at $N = 128$.

A four element array cannot show how coupling behaves between the centre and the edge of a large aperture, how thermal gradients across a large board correlate channel errors, how an active transmit and receive module with amplifiers drifts, or how measurement counts behave when the identifiable dimension is 254 instead of 6.

13.3 The intended scaling study

The scaling study is **[proposed here]**; the repository does not contain it. Its logic would be:

1. measure on Rev A the statistics that a simulator needs and cannot invent: phase drift and amplitude drift distributions, their temporal correlation, their dependence on temperature, the measurement noise, and the coupling structure from EXP-011;
2. parameterise an array simulator with those distributions, labelled as measured on four channels;
3. simulate recalibration with and without a learned prior at $N = 4, 8, 16, 32, 64, 128$, with the same baselines as on hardware;
4. check the $N = 4$ simulation against the $N = 4$ hardware result before believing the larger ones.

How the counts scale with N , for a diagonal state, **[analytical]**:

N	identifiable parameters $2N - 2$	complex route, one channel at a time	REV minimum $3N$	generic injective power only count $4N - 4$, sufficient
4	6	4	12	12 (11 known to suffice)
8	14	8	24	28
16	30	16	48	60
32	62	32	96	124
64	126	64	192	252
128	254	128	384	508

For $N > 4$, REV needs fewer readings than the generic sufficient count $4N - 4$, and at large N it falls below any injective count, since known lower bounds grow as about $4N$ [L5]. That is possible only because REV relies on structure and on prior assumptions (section 11.5). The room for prior information grows with N , which is why the repository expects adaptive methods to gain more at larger N (docs/architecture/ml-calibration.md section 5, M4).

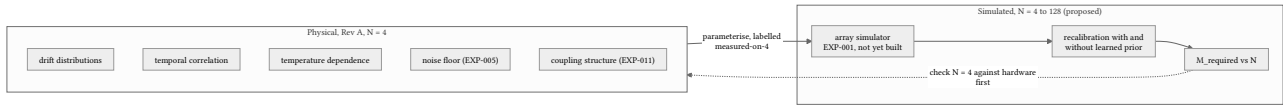


Figure 18: four physical channels and a simulated scaling study. The simulated part is a proposal; EXP-001, the array simulator it needs, has not been built.

Limits of the extrapolation, stated in advance.

- Large arrays use different hardware: integrated beamformer chips, active modules with amplifiers, digital or hybrid architectures. Their drift mechanisms differ from those of a passive switched line board.
- Thermal and coupling topology change with size: gradients, edge effects and correlated errors across subarrays have no analogue at four elements.
- Full wave simulation does not scale with the study. HFSS Student is limited to 64 000 volume mesh elements [V23], and even the full licence cannot model a 128 channel array repeatedly; a scaling study would use idealised coupling models, such as the Toeplitz model already in `rfkit.coupling`, or periodic unit cell analysis, and say so.
- Measured statistics from one board are a sample of one board.

Reason	Explanation
per element access, requirements R1 and R2	every element has its own connector, so each element can be measured alone, the coupling between any two can be measured, and the mutual coupling calibration method B6 [A7] remains possible. An integrated splitter would close all three permanently (docs/hardware/rev-a-requirements.md section 3)
substrate physics	the switched lines need a thin dielectric and an inner ground; the patches need a thick one. A factor of about seven in patch bandwidth and five in efficiency separates them (section 6.4)
control routing	the beamformer needs an inner plane so the beam state lines of decision 0005 can cross under RF lines without cutting the RF ground
replaceable antennas	the beamformer board survives a change of antenna geometry, of frequency or of element type; a missed patch resonance costs an antenna board, not the whole system
fabrication risk	decision 0009 expects a second antenna board order may be needed; the expensive, dense board is not affected
native experiments	the four jumpers make the known cable error experiment EXP-007 and the reconnection sensitivity metric natural rather than contrived

The price is real: eight connectors and four cables add loss and add drift sources. That drift is inside what EXP-010 measures; it is at least observable rather than hidden (docs/architecture/rev-a-rf-architecture.md section 5.2).

15. Beamformer channels

15.1 What each channel contains

Stage	Parts	Function
enable	one PE4259-63: throw 1 to the phase chain, throw 2 to a 50 ohm termination	pass the channel, or terminate it so it can be excluded electronically
45 degree bit	two PE4259-63 and two printed arms	select the reference or the delay arm
90 degree bit	two PE4259-63 and two printed arms	as above
180 degree bit	two PE4259-63 and two printed arms	as above

Seven switches per channel, 28 for four channels, plus $U900$, the path selector at the common node, added during capture: 29 PE4259-63 in all (hardware/rev-a/README.md). One control line drives both switches of a bit, so the channel takes four control lines and the array sixteen.

15.2 The switched line principle

```

                reference arm (electrical length theta_ref)
                +-----+
in -- RFC [SPDT A]                                [SPDT B] RFC -- out
                +-----+
                delay arm -----+
                (electrical length theta_ref + theta_bit)

control = 0 : both SPDTs select the reference arm -> phase -theta_ref
control = 1 : both SPDTs select the delay arm    -> phase -theta_ref - theta_bit

designed quantity: theta_delay - theta_ref = theta_bit = 45, 90 or 180 degrees at f0,
measured between the RF ports of SPDT A and SPDT B (layout-constraints.md section 2)

```

Figure 5: one switched line bit. Two single pole double throw (SPDT) switches route the signal through one of two printed arms; the difference of their electrical lengths is the bit.

An SPDT switch connects its common port (RFC) to one of two throws. Two of them, facing each other, select one of two paths. The phase difference between the paths is $\beta \Delta l$ (section 1.6). Three bits in cascade give the eight states of section 8.

15.3 What the realised phase depends on

The analytical difference lengths of decision 0009 are 8.6, 17.2 and 34.4 mm for the 45, 90 and 180 degree bits on the beamformer construction [**analytical, INITIALISATION ONLY**]. They are seeds, not layout values, because the realised phase difference depends on more than a straight line length:

Factor	Effect
effective permittivity	sets λ_g ; known a priori only to an assumed ± 0.2 on ϵ_r (section 59)
frequency	a fixed length is a time delay; 5.6 degrees on the 315 degree state at the upper band edge (section 8.5)
bends and meanders	a 34 mm arm will not be straight; corners and closely spaced meander turns change the effective length and couple to themselves
discontinuities	pads, width steps and the transition into each switch add reactance that differs between arms if their routing differs
switch package and parasitics	each SPDT adds its own insertion phase and loss; if its two throws differ, that difference enters the bit
launch and reference planes	the bit is defined between the switch RF ports, not between connectors
line width	changes Z_0 and ϵ_{eff} together; a width tolerance of ± 20 per cent is guaranteed by the fabricator [V11]
loss	the longer arm is lossier, so each bit also changes amplitude: the state dependent loss of section 59

Two further hazards are specific to switched lines with finite isolation. They are standard engineering concerns, not repository findings, and both must be checked in simulation [**to verify, SIM-003 proposed**].

- **Reflections between discontinuities.** Small mismatches at both ends of an arm create a standing wave whose effect on the transmitted phase depends on the arm's length, so it differs between states: a state dependent error by construction.
- **Resonance of the de-selected arm.** The arm that is not selected is a line held between two switch throws in their off state, close to an open circuit at both ends and weakly coupled to the through path by the switch's off state isolation. A line open at both ends resonates when its electrical length approaches a multiple of half a guided wavelength. The delay arm of the 180 degree bit is at least half a guided wavelength long, so depending on the reference arm length it may sit near such a resonance, producing a sharp, state dependent loss and phase excursion in the band. The usual remedies are choosing the reference length to keep both arms away from resonance, or splitting the bit.

15.4 The PE4259-63

Property	Value	Evidence
function	SPDT RF switch with integrated CMOS control logic	[vendor] V1
frequency range	10 MHz to 3000 MHz	[vendor] V1
insertion loss	0.35 dB typical at 1000 MHz, 0.5 dB at 2000 MHz	[vendor], typical spot values; nothing at 2.44 GHz
isolation	30 dB typical at 1000 MHz, 20 dB at 2000 MHz	[vendor], typical; no guaranteed minimum at any frequency (I18)
supply	1.8 V to 3.3 V	[vendor]
package	SC-70-6, 0.65 mm lead pitch	[vendor]
availability	active, more than 300 000 units in distributor stock, 0.84 USD at one unit, read 2026-09-18	[vendor] listing
control thresholds	not read : the datasheet is a scanned image	open item H1
control polarity	the single pin truth table, which level selects which throw, not read	firmware inversion constant, decision 0005

The datasheet is a scanned image that could not be read by machine; isolation at 2.44 GHz, the logic input thresholds and the truth table need a person with the file open (experiments/EXP-004-instrument-audit.md, follow-up).

15.5 Open hardware items H1 to H5

Item	Question	Why it matters	Closes when
H1 logic level compatibility	does the switch read the buffer's output levels correctly?	blocks board release. A common rail buffer prevents overvoltage but does not prove the switch sees a valid high and low	the PE4259 input thresholds at the board supply and the selected buffer's guaranteed output levels are recorded with margin; fallback: a level translator specified against a named standard plus a bench measurement of the threshold on a sample (SCH-011)
H2 expansion header supply	can the DE1-SoC header source about 70 mA at 5 V?	the detector is 68 of the 69 mA the board draws	the header current limit is read; otherwise a separate supply, which enlarges H5
H3 converter part	which converter digitises the detector on the RF board, if any?	decision 0005 placed it there as a precaution	EXP-005 Phase A decides whether it stays, then a part is chosen

Item	Question	Why it matters	Closes when
H4 buffer part	which registered buffer latches the sixteen lines?	needed for H1 and for setup and hold times	a part is selected
H5 ground strategy	how are the two boards' grounds joined?	a ribbon between a digital board and a receive chain is a loop	a decision before layout; EXP-005 condition C5 informs it

Source: docs/architecture/control-architecture.md section 8. All five are open on 2026-10-04, and gate F4 of decision 0006 requires H1 to H4 closed and H5 decided before fabrication.

15.6 The Wilkinson network and the path selector

The four channels meet in a four way Wilkinson network built from three two way stages, each with two quarter wave arms of 70.7 ohm and a 100 ohm isolation resistor (hardware/rev-a/layout-constraints.md section 3). An ideal Wilkinson stage is matched at all ports and isolates its two outputs from each other, which keeps the channels from loading one another. Its seed arm width on the beamformer construction is 0.182 mm **[analytical]**, about twice the 0.09 mm process minimum; an optional coupon C4 checks the etch of that narrowest line. The common node then passes through $U900$, which connects it either to the analyser connector or to the detector, so that neither loads the other; it adds about 0.5 dB to the common arm **[estimate]**.

16. Enable and terminate

Each channel's enable switch either passes the signal into the phase chain or connects the element port to a 50 ohm termination. That single switch is what makes several experiments possible without touching a cable.

Use	How the enable switch provides it
isolate one channel	enable one, terminate three, measure; the per channel label of the learning track is obtained electronically
labels for supervised learning	the expensive full calibration that supervises the cheap one can run unattended
element by element baseline B2	the classical method of measuring each channel alone
fault finding	a dead or misbehaving channel shows up in a single measurement
coupling measurements	terminated neighbours present a defined 50 ohm load, as EXP-011 requires
binary amplitude	subsets of channels can be switched on, which adds 308 relative configurations to the 512 (section 8.3)

Two limits must be stated. First, switching channels off is not amplitude control: it cannot taper the aperture or lower sidelobes in any designed way, and Rev A makes no claim to amplitude beamforming (decision 0003). Second, a terminated channel is not perfectly silent. Its signal leaks into the chain through the enable switch's isolation, about 20 dB typical at 2 GHz [V1], with no value obtained at 2.44 GHz. When one channel is measured at the common port with the other three terminated, their leakage adds to it; EXP-004 gives the residual as $\sqrt{3} \cdot 10^{-I/20}$ for random relative phases and $3 \cdot 10^{-I/20}$ when the three add coherently, for isolation I in dB **[analytical]**:

Isolation I	Residual, random phases	Amplitude error	Phase error	Residual, coherent worst case
20 dB	−15.2 dB	17 per cent	10 degrees	−10.5 dB
25 dB	−20.2 dB	10 per cent	6 degrees	−15.5 dB
30 dB	−25.2 dB	5 per cent	3 degrees	−20.5 dB

This degrades per channel measurements **taken at the common port**, the B2 baseline. It does not degrade a conducted measurement between the common port and the enabled channel's own element port, because the other channels' signals do not reach that port. It does not affect REV either, which never switches channels off. Which reference plane the learning labels use is therefore a real design choice; the repository leans on the conducted route but does not fix it in one place (Appendix J).

17. The RF detector

17.1 What a logarithmic detector does

The AD8318 converts RF power at its input into a DC voltage proportional to the power in decibels. Over its accurate range:

$$V_{\text{out}} \approx s(P_{\text{dBm}} - P_{\text{icpt}}), \quad s \approx -25 \text{ mV/dB}$$

Symbol	Meaning	Unit
V_{out}	detector output voltage, about 0.4 V to 2.2 V over the usable range	V
s	logarithmic slope, nominally -25 mV/dB , negative: more power gives less voltage	V/dB
P_{dBm}	input power	dBm
P_{icpt}	intercept, a constant of the part and of the frequency	dBm

The vendor specifies 1 MHz to 8 GHz, $\pm 1 \text{ dB}$ conformance over a 55 dB range below 5.8 GHz, $\pm 0.5 \text{ dB}$ stability over temperature across its full range, and a single 5 V supply at about 68 mA [V6] [vendor]. A 0.1 dB change of received power is a 2.5 mV change of output.

17.2 What it cannot do

- **It measures power only.** It returns no phase, so through the detector the array state must be inferred from power readings: the phase retrieval problem of section 11.5.
- **It is broadband.** It responds to all power in its range of frequencies. It cannot by itself separate a wanted transmitter from an interfering one in the same band, which matters for the AP-S communication mode (section 47).
- **Its constants are not known to the needed accuracy.** The slope and intercept vary from unit to unit and with frequency and temperature, so the detector must be characterised against the analyser used as a stepped source (docs/hardware/measurement-bench.md section 4). Its temperature drift is corrected empirically against the MCP9808 beside it, which is why that sensor is mandatory (requirement R4).

17.3 Why a converter is needed, and where it goes

The FPGA fabric is digital; it cannot sample an analogue voltage. Every analogue quantity reaches it through an analogue to digital converter (ADC).

Option	Description	Status
local converter on the RF board	a serial ADC beside the detector, read by the fabric over four lines	decided as a precaution , decision 0005 point 2; part not selected (H3); requirement stated as at least 14 effective bits over a 2 V span, so one step is below 0.01 dB
DE1-SoC on board converter, LTC2308	eight channels, 12 bit, up to 500 ksp/s, 0 V to 4.096 V input range at the header [T6]	kept as an independent path ; it samples a different chain from a different ground reference, so a disagreement between the two paths is informative

The case for the local converter is cost asymmetry, not a demonstrated failure: carrying a signal where 0.1 dB is 2.5 mV along a ribbon beside sixteen switching lines could pick up interference **correlated with the commanded state**, which would imitate a calibration coefficient (section 36). Fitting the converter costs a few euro; omitting it and being wrong costs a board revision. EXP-005 Phase A turns the argument into a measurement.

An unresolved detail: the LTC2308 has a 2.5 V internal reference, yet the header range is quoted as 0 V to 4.096 V, which implies some circuit in front of the converter, a divider or an external reference. Which it is matters for the source impedance the converter sees, and is item B1 of EXP-005, not yet closed (results/EXP-005/README.md, preparation P3).

17.4 Resolution, averaging and dither

A 12 bit converter over 4.096 V has a step of

$$\text{LSB} = \frac{4.096 \text{ V}}{2^{12}} = 1.0 \text{ mV}, \quad \frac{1.0 \text{ mV}}{25 \text{ mV/dB}} = 0.04 \text{ dB}$$

One step is larger than the 0.02 dB effect EXP-005 looks for. Averaging m readings reduces random noise by $\sqrt{m}$, but only if the noise moves the reading across at least one step. If the input sits still between two codes, every reading returns the same code, and averaging recovers nothing: the error is a fixed rounding, not noise. That is why EXP-005 has a validity precondition, V1, that the raw codes of a burst show at least two adjacent distinct values, and why dither, deliberately added noise, would be required otherwise.

18. Temperature sensors

Two MCP9808 digital sensors sit on the beamformer board: one at I2C address 0x18 beside the phase network, one at 0x19 beside the detector. The vendor gives about 0.25 degrees Celsius typical accuracy and 0.0625 degree resolution

[V5] **[vendor]**. The AD8318 also provides an analogue die temperature output, read by a converter channel, as a third, independent temperature.

Nothing in the repository says temperature causes a known correction to the array state; no coefficient has been measured. Temperature is logged because it is useful in four ways:

Use	Why
covariate	a drift prior can condition on it: $p(\mathbf{H}_t \mid \mathbf{H}_{t-1}, T_t, \Delta t, \dots)$
experimental context	a session is interpretable only with its conditions
potential predictor	if drift tracks temperature, the prior will learn it; if not, that is a finding
diagnostic	separates array drift, near 0x18, from detector drift, near 0x19

A temperature reading is taken with every measurement record, not on a separate schedule, and outside the quiet window of section 19, because an I2C transaction is switching activity on two more lines of the same cable (`docs/architecture/control-architecture.md` sections 4.2 and 5.1). Whether temperature enters the first prior model, or only later, is a modelling choice to be made on data, not now.

19. The DE1-SoC

19.1 What it is

The Terasic DE1-SoC is a development board built around an Intel Cyclone V system on chip. One chip contains two very different computers:

Part	What it is	What it is good at
FPGA fabric	a field programmable gate array: a large array of logic cells and wiring configured into custom digital circuits, which run in parallel on a clock	deterministic timing: an operation takes exactly the same number of clock cycles every time
HPS	the hard processor system: an ARM processor running ordinary software	storage, networking, files, floating point computation, everything without a hard deadline
GPIO	two 40 pin expansion headers, 36 user pins each, connected directly to the FPGA at 3.3 V with protection diodes [T6]	sixteen beam state lines and the rest of the interface
ADC	the LTC2308, reached from the fabric over a four wire serial interface [T6]	the independent detector path of section 17

Three DE1-SoC boards are owned (`docs/hardware/inventory-and-needs.md`); one is enough for Rev A, and EXP-005 Phase A uses two.

19.2 Why not a microcontroller alone

Decision 0003 originally specified an STM32G0 microcontroller. Decision 0005 replaced it, and the reason is the measurement, not the beamforming: an analogue array needs no programmable logic to form a beam.

Requirement of the drift experiment	Microcontroller	FPGA fabric
identical delay between applying a state and sampling, every time	subject to interrupts and host traffic: it jitters	an exact number of clock cycles
all sixteen lines change at one instant	ports written in sequence	one clock edge
trigger and timestamp from one clock	usually two clocks	one free running counter
control lines provably static while sampling	by convention	by construction, in the sequencer
unattended sequencing without the host	limited	native

Each microcontroller weakness enters the data as scatter correlated with the measurement sequence, the one kind of error this project cannot tolerate (section 36). The costs are also recorded: gateway is new work, the board is external so the interface crosses a cable, and the schematic must be re-captured. If EXP-005 shows the floor is set by the environment rather than by control timing, decision 0005 allows a fallback to the simpler route for the control path alone, without touching the RF design.

19.3 The partition

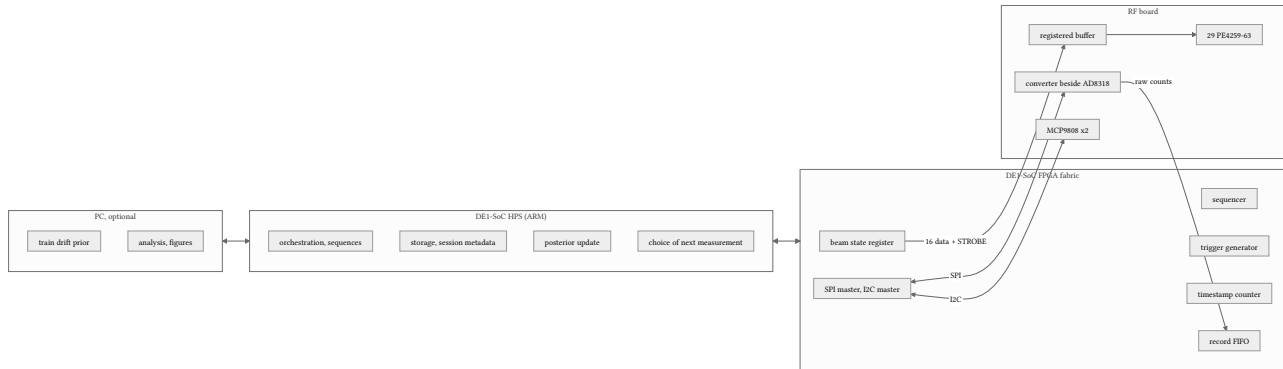


Figure 11: hardware and software partition. Everything with a deadline is in the fabric; everything probabilistic is above it (docs/architecture/control-architecture.md section 2).

Why learning does not run in the FPGA. Nothing about inference has a hard deadline: a posterior update over six parameters, or an enumeration of 512 states, takes milliseconds on the processor, between measurements that take far longer. Putting floating point inference in fabric would add design effort and risk for no timing benefit. The fabric owns no floating point, no inference and no storage, by design.

19.4 One measurement, step by step

```
time --->
HPS / sequencer  load word k+1 into register
16 data lines    =====X===== (static)
STROBE           _|^_               rising edge latches
RF board buffer  outputs change on STROBE edge, all 16 at once
settling interval |<----- N_settle clock cycles, counted at the controller ----->|
TRIGGER          _|^_               [c1][c2]..[cm]
conversion(s)    taken at STROBE edge ..... and at TRIGGER edge
timestamp        |<===== no control line change, no I2C traffic =====>|
quiet window
I2C temperature read [read]
record           word, read-back word, both timestamps, settling, raw counts, T, versions
```

Timing of one sequencer entry (docs/architecture/control-architecture.md section 5.1). The settling interval is counted from the strobe at the controller, so cable and buffer delay lie inside it on purpose. The value of N_{settle} is set by the settling sweep of EXP-005.

The quiet window, from the strobe edge to the end of the last conversion, forbids any control line change and any I2C traffic. The I2C clause matters as much as the control line clause, because an I2C transaction is switching on two more lines of the same cable.

19.5 The 16 bit beam state word

$$b = 4c + f, \quad c \in \{0, 1, 2, 3\}, \quad f \in \{0, 1, 2, 3\}$$

Bit b of the word carries field f of channel c , with f meaning enable, 45, 90 and 180 in that order. Bits 0 to 3 are channel 0, bits 12 to 15 channel 3. Logical polarity is defined: a one enables the channel or selects the delay arm. Electrical polarity is not: it depends on the unread PE4259 truth table and becomes one inversion constant per field in gateway. The monitor path select MON_SEL is a seventeenth line, deliberately outside the word.

19.6 What the read-back proves, and walk one bit bring-up

Each record carries a read-back word. With an ordinary registered buffer, the only word that can be read is the controller's own output register. That detects a sequencer or software fault. It does **not** detect a broken conductor, a bad connector contact or an unpowered or failed buffer: the controller would read back exactly what it intended while the switches never received it. Detecting those would need sixteen return lines, which the interface does not carry. Instead, bring-up walks a single bit through all sixteen positions and watches the RF response change as expected: each bit should toggle exactly one channel's enable or one bit's phase, and nothing else (docs/architecture/control-architecture.md section 5.2).

19.7 What exists

No gateway exists. Quartus 17.1 is installed on the project computer, and no programmer has ever been attached to it (results/EXP-005/README.md, P2). The beam state register, sequencer, trigger, timestamp counter, serial masters and record FIFO are specified only.

20. Grounding and the digital to RF interface

Connecting an FPGA board by ribbon cable to a board carrying a 2.44 GHz receive chain creates several coupled problems.

Problem	Mechanism	Why it matters here
return currents	each switching line's current returns through the cable's ground conductors	with too few or badly placed grounds, return currents share paths with the analogue signal
common impedance	a shared ground conductor has impedance, so one circuit's current shifts another circuit's reference	the detector's 2.5 mV per 0.1 dB is easily disturbed
digital noise	fast edges contain energy far above the clock frequency	it can couple into the detector input or the analogue lines
ground loops	the DE1-SoC, the RF board, the analyser and a PC each have a ground; joined at several points they form loops	loop currents from other equipment add slowly varying offsets
EMI	the ribbon is an antenna for both the digital edges and the RF	radiated pickup in the receive chain

The specific danger is not noise in general, which averaging reduces, but noise that depends on the commanded state, which averaging does not remove and which a calibration absorbs as if it were a property of the array. The mitigations decided so far are partial: interleaved grounds in the 2 by 20 connector, a registered buffer at the board edge that restores edges and removes skew, the quiet window, an unbroken L2 ground beneath every RF trace with digital routing on L3 and L4 (decision 0009), and the converter beside the detector.

H5, the ground strategy between the two boards, is open. If the header cannot supply the board (H2), the board takes its own supply and the question grows. EXP-005 condition C5, a ground strap between the boards, is designed to show whether the grounding arrangement is a live variable; if C5 differs from C3 by more than 0.5 mV, H5 becomes a measured requirement rather than a layout question (experiments/EXP-005-repeatability-floor.md section 7.6). No final ground strategy is claimed here.

Part IV. PCB stack-up and RF physical design

A **stack-up** is the layer construction of a printed circuit board: which copper and dielectric layers, in which order, of which thickness and material. For an RF board it decides the impedance of every line, the phase of every switched length and much of the loss. Decision 0009 selected the Rev A stack-ups on 2026-10-03, recorded as reva-stackup-r1 in hardware/rev-a/stackup/reva-stackup.json. Every value in that file carries a status, never mixed: guaranteed, a published specification limit; typical, a laminate vendor's typical value and never a tolerance; nominal, a value a design tool or a fabricator's calculator uses; assumed, an engineering assumption made in decision 0009; and, for tolerances, unbounded, with no number. The tables below between generation markers come from that file.

21. The beamformer stack-up

21.1 The construction

beamformer, RF beamformer and control interface board: JLCPCB, 4 layer FR-4, 1.6 mm, controlled impedance stack-up JLC04161H-7628.

Layer	Material	Role	Thickness	Status	Source
L1	copper	RF microstrip, RF components, local escapes outside RF keep-out	0.035 mm	nominal	V8
PP1	fr4_prepreg_7628	RF substrate	0.2104 mm	nominal	V8
L2	copper	continuous RF reference plane, no routing, no splits	0.0152 mm	nominal	V8
CORE	fr4_core_np155f	core	1.065 mm	nominal	V8
L3	copper	power rails and slow digital: beam state lines, strobe, I2C, converter	0.0152 mm	nominal	V8
PP2	fr4_prepreg_7628	lower prepreg	0.2104 mm	nominal	V8
L4	copper	digital and connector routing, ground pour stitched to L2	0.035 mm	nominal	V8

antenna, four element antenna array board: JLCPCB, 2 layer FR-4, 1.6 mm, 1 oz finished copper.

Layer	Material	Role	Thickness	Status	Source
L1	copper	patches, feed lines, SMA launches	0.035 mm	nominal	V11
CORE	fr4_two_layer	RF substrate	1.53 mm	assumed	D0009
L2	copper	continuous ground under every patch and feed, no routing	0.035 mm	nominal	V11

Material	Designation
copper	electrodeposited copper, foil type not published by the fabricator
fr4_prepreg_7628	7628 glass prepreg, resin content 49 per cent, in the fabricator's NP-155F based stack-up; brand not guaranteed per order
fr4_core_np155f	NP-155F core assumed by the fabricator's calculator
fr4_two_layer	two layer FR-4 core; brand not fixed by the fabricator, one of NP-140F, KB-6164, S1141 or S1000H (V14)
lpi_soldermask	liquid photoimageable solder mask

Material	Property	Value	Status	Source	Frequency and method
copper	conductivity	5.8e+07 S/m	assumed	D0009	not applicable
fr4_prepreg_7628	permittivity	4.4	nominal	V8	not stated; fabricator impedance calculator value, no frequency stated
fr4_prepreg_7628	loss tangent	0.015	typical	V12	1 GHz; IPC-TM-650 2.5.5.9
fr4_core_np155f	permittivity	4.6	nominal	V8	not stated; fabricator impedance calculator value, no frequency stated
fr4_core_np155f	loss tangent	0.015	typical	V12	1 GHz; IPC-TM-650 2.5.5.9
fr4_two_layer	permittivity	4.5	nominal	V11	not stated; fabricator capability page value for two layer boards, no frequency stated
fr4_two_layer	loss tangent	0.015	assumed	D0009	not stated
lpi_soldermask	permittivity	3.8	nominal	V8	not stated; fabricator impedance calculator value

The RF microstrip is on L1 over a continuous ground on L2, through one 0.2104 mm layer of 7628 glass prepreg. L3 carries the supply rails and the slow digital lines of decision 0005, and L4 carries digital and connector routing under a ground pour stitched to L2. The keep-out rule keeps L2 unbroken beneath every RF trace, divider and switch, with

at least three prepreg thicknesses of plane on either side [**assumed**]; a line on L3 or L4 that passes beneath RF copper may not change reference layer there.

21.2 Why this construction

Advantage	Explanation
line width matches the switches	a 50 ohm seed width of about 0.37 mm against the 0.65 mm pitch of the SC-70-6 package
the widest 50 ohm line of the published single prepreg options	about four times the 0.09 mm process minimum; finer glass styles give 0.21 mm or less, with 1.8 to 2.8 times the conductor loss
one homogeneous dielectric under the RF	a single material to model and to calibrate
an inner plane for control routing	beam state lines cross under RF lines without cutting the RF ground
controlled impedance at no extra charge	the fabricator commits to this published construction when impedance control is ordered [V8]
cost and lead time	a low cost class and a few days of production, against one to two project budgets for an RF laminate

Disadvantage	Explanation
loss	the highest of the compared constructions: dielectric loss alone about 5.3 dB/m
state dependent loss	the 315 degree state loses 0.59 to 0.86 dB more than the 0 degree state, against an amplitude imbalance allowance of 0.82 dB for everything together (section 59)
coarse glass weave	7628 cloth has large glass bundles; a 0.37 mm line can see a different local permittivity on different channels
weak permittivity provenance	a calculator nominal with no frequency and no test method

21.3 Uncertainties, as recorded

Construction	Tolerance	Input	Bound	Status	Source
beamformer	rf dielectric thickness	h	minus 10 to plus 10 per cent	assumed	V11
beamformer	rf dielectric permittivity	ϵ_r	minus 0.2 to plus 0.2	assumed	D0009
beamformer	rf copper thickness	t	minus 0 to plus 0.0056 mm	assumed	V9
beamformer	etched width	w	minus 20 to plus 20 per cent	guaranteed	V11
antenna	rf dielectric thickness	h	minus 10 to plus 10 per cent	assumed	V11
antenna	rf dielectric permittivity	ϵ_r	minus 0.3 to plus 0.1	assumed	D0009
antenna	rf copper thickness	t	unbounded, no number	unbounded	V11
antenna	etched width	w	minus 20 to plus 20 per cent	guaranteed	V11

Uncertainty	What the sources say	Consequence
permittivity provenance and frequency	4.4 for 7628 prepreg in the fabricator's impedance calculator, no frequency or method stated [V8]; the fabricator says its values are deduced, not the supplier's raw data [V10]; the laminate vendor gives 4.2 to 4.4 at 1 GHz for one laminate thickness and 3.9 to 4.1 for another [V12]	a bound of ± 0.2 is adopted [assumed]; it is an envelope, not a guarantee
core permittivity contradiction	4.6 on the stack-up page [V8], 4.43 in the calculator guide [V9]	affects no RF line; recorded, not resolved
copper thickness contradiction	0.035 mm on the stack-up page [V8], 1.6 mil, about 0.041 mm, in the calculator [V9]	bounded as +0 to +0.0056 mm; confirmed at order time
impedance tolerance contradiction	10 per cent on the capabilities page [V11], 20 per cent elsewhere [V10]	confirmed at order time
laminate brand	not guaranteed per order; multilayer boards may use one of several laminates [V14]	a coupon order does not characterise the production order
glass weave	no source quantifies the spatial variation	uncertainty I25; the one material effect a coupon cannot remove

Why a manufacturer's nominal permittivity is not a measured RF truth. A permittivity value is meaningful only with its frequency and its test method. FR-4 permittivity falls slowly with frequency, and different methods measure different things: Rogers itself publishes a process value of 3.48 for RO4350B, measured in a clamped stripline at 10 GHz, and a design value of 3.66, measured by a differential phase length method from 8 to 40 GHz, and its own chart reads

higher still near 2.5 GHz [V17]. A calculator value with neither frequency nor method is a starting point. The board's own coupons, measured on the analyser, become the authority once a board exists (section 25).

Order time requirements, from decision 0009: request impedance control with stack-up JLC04161H-7628 so that the published construction is used; declare the RF nets uncoated and ask that their widths are not adjusted for a coated model; specify ENIG; record the laminate the fabricator reports.

22. The antenna stack-up

The antenna board is two layer FR-4, 1.6 mm, with 1 oz finished copper: patches, feed lines and launches on L1, one unbroken ground on L2. Its core thickness, 1.53 mm, is **[assumed]** from the finished thickness; its permittivity, 4.5, is the fabricator's capability page value with no frequency **[vendor nominal]**; its loss tangent, 0.015, is **[assumed]**; and its laminate brand is one of four [V14]. It is the least well defined material in the project.

The trade-off is the one of section 6.3: a thicker substrate gives a patch more bandwidth and more radiation efficiency, at the cost of wide feed lines, which on an antenna board with short feeds and no switches is harmless.

SANITY CHECK ONLY. The patch length, width, feed and spacing stay free parameters for HFSS. These figures only say whether a patch is physically sensible on each construction.

Construction	W_p (mm)	L_p (mm)	Bandwidth, VSWR 2	Radiation efficiency	Resonance shift over the permittivity bound
beamformer	37.4	29.3	0.14 per cent	9 per cent	-2.19 to +2.34 per cent
antenna	37.0	28.6	1.05 per cent	48 per cent	-1.06 to +3.38 per cent

Four elements at half a free space wavelength need an antenna board about 260 mm long, sanity check only.

These figures are estimates from the transmission line patch model of Balanis [B2], computed by `rfkit.stackup.patch_sanity`, and say only that a patch is physically sensible on this construction. They are not a design: the patch length, width, feed and spacing are free parameters for HFSS. The resonance shift over the permittivity bound is larger than the estimated bandwidth, so the first antenna board may not resonate in band 57a; decision 0009 expects that a second antenna board may be ordered, and moves the antenna board to RO4350B (option B2) if two boards in a row miss the band.

23. Why not a Rogers laminate?

RF laminates such as Rogers RO4350B are designed for this frequency range. Their advantages are real:

- lower loss: a loss tangent of about 0.0031 at 2.5 GHz against about 0.015 for FR-4 [V17];
- a guaranteed process permittivity tolerance of ± 0.05 and guaranteed thickness tolerances [V17], where FR-4 offers calculator nominals;
- more uniform glass, so less channel to channel variation.

Decision 0009 compared them, with the same computations applied to every construction:

Construction	Process	h (mm)	ϵ_r , status	$\tan \delta$	W_{seed} (mm)	λ_g (mm)	315 degree loss spread, smooth to doubled conductor loss (dB)	315 degree error from the ϵ_r bound (deg)	Patch bandwidth, efficiency	Width flags
beamformer, selected	JLCPCB, 4 layer FR-4, 1.6 mm, controlled impedance stack-up JLC04161H-7628	0.210	4.4, nominal	0.015	0.372	68.8	0.59 to 0.86	6.3, assumed	0.14 per cent, 9 per cent	none
antenna, selected	JLCPCB, 2 layer FR-4, 1.6 mm, 1 oz finished copper	1.530	4.5, nominal	0.015	2.836	66.5	0.36 to 0.39	9.8, assumed	1.05 per cent, 48 per cent	none
ro4350b_thin_2l, candidate	JLCPCB, 2 layer RO4350B, 0.51 mm core, finished 0.6 mm, 1 oz, ENIG	0.508	3.66, typical	0.0031	1.073	73.2	0.17 to 0.27	1.9, guaranteed	0.39 per cent, 49 per cent	wide: over the SC-70-6 lead pitch, a taper at every switch pin
ro4350b_thick_2l, candidate	JLCPCB, 2 layer RO4350B, 1.52 mm core, finished 1.65 mm, 1 oz, ENIG	1.524	3.66, typical	0.0031	3.291	72.5	0.10 to 0.13	1.9, guaranteed	1.18 per cent, 81 per cent	none
fr408hr_4l, candidate	OSH Park, 4 layer FR408HR, 1.6 mm, ENIG	0.200	3.61, typical	0.009	0.405	74.7	0.46 to 0.73	3.0, assumed	0.15 per cent, 12 per cent	none

And it found three reasons not to choose them for Rev A:

1. **Cost.** At the accessible fabricators, the RF laminate options cost between half and twice the whole project budget of 50 to 70 EUR for one board, against a few euros for FR-4.
2. **Fabrication constraints.** At the fabricator that publishes both, RO4350B is two layer only, with no stated impedance control; the thin version is a 0.6 mm board with no inner layer for control routing, and needs a taper at every switch pin if made thick.
3. **A measurement is needed anyway.** Even Rogers's nominal permittivity is ambiguous at 2.44 GHz, between a process value, a design value and a chart (section 21.3). Keeping the 315 degree state within 2.29 degrees needs the beamformer permittivity known to about ± 0.073 (section 59), which no candidate guarantees as a nominal value. On every candidate the effective permittivity must therefore be measured on the board. What differs is how much the material varies around the measured value, and how much loss it adds.

Rogers is a reopening path, not a rejection. Decision 0009 names the triggers: coupon measurements outside the assumed bounds, channels differing by more than the coupon uncertainty (the glass weave effect), a state dependent loss that makes the amplitude imbalance exceed 0.82 dB, two antenna boards missing the band, or a budget change. The first places to look are FR408HR with spread glass (B3), the Eurocircuits RO4350B and FR-4 hybrid (B4), and RO4350B for the antenna board (B2).

24. Solder mask and copper roughness

Solder mask is opened over RF copper. The fabricator publishes a nominal mask thickness, a guaranteed minimum of 10 μm , no maximum and a permittivity of 3.8 with no frequency [V8, V11]. A mask over the RF lines could therefore be bounded on one side only, and its thickness varies across a board, which would become channel to channel phase error that no common correction removes. Opening the mask over RF microstrip and patches removes the parameter: the nominal HFSS model has no mask on RF copper, matching the fabrication drawing. The price is ENIG on the exposed lines, whose nickel adds conductor loss; that enters the loss bound and the coupon attenuation, not the nominal model. The mask stays elsewhere, including the dams between switch pads.

Copper is smooth in the nominal model, and its roughness is bounded, not invented. No fabricator publishes its foil type or roughness. Rough copper raises conductor loss, because the current at 2.44 GHz flows within about 1.3 μm of the surface and has to follow the roughness, and it raises the apparent permittivity. A Huray model, which represents the surface as stacked spheres, needs nodule size and density data that nobody has published for these boards, so decision 0009 asserts none. Instead, the loss study bounds conductor loss between smooth copper and twice smooth, the asymptote of the Hammerstad and Jensen roughness correction [B8]. The apparent permittivity rise, which that correction does not capture, is absorbed into the effective permittivity the coupons measure. The only roughness data found for any candidate was for Rogers foils, about 3 μm for standard 1 oz foil [V18].

25. Coupons

A **coupon** is a small test structure fabricated on the same panel as the real circuit, so that it shares the circuit's laminate lot, thickness and etch, and can be measured when the circuit itself cannot be measured in pieces. Decision 0009 defines four, plus an optional calibration set, on each board; they are defined but not yet laid out.

Coupon	Structure	What it identifies
C1	a 50 ohm thru line between two SMA launches, of the SIM-001 short length	the impedance of the line as built; with C2, the launches
C2	the same line made a quarter guided wavelength longer at f_0	with C1, the propagation constant γ by the two line method: ϵ_{eff} and attenuation of the board as built, which absorb permittivity, thickness, roughness and ENIG together
C3	two SMA launches back to back with the shortest line	the launch, so switched line measurements can be de-embedded to the switch reference planes
C4, optional	a 70.7 ohm line of the C1 length	the etch on the narrowest line, the Wilkinson arms
TRL set, optional	thru, reflect and line standards	only if observation O7 cannot provide a calibration at the SMA plane

The two line method. Write the transmission matrix of each measured line as the launch at one end, the uniform line, and the launch at the other end, $\mathbf{T}_i = \mathbf{X} \mathbf{L}(l_i) \mathbf{Y}$, with $\mathbf{L}(l) = \text{diag}(e^{-\gamma l}, e^{\gamma l})$. If the launches are identical, the product

$$\mathbf{M} = \mathbf{T}_{\text{long}} \mathbf{T}_{\text{short}}^{-1} = \mathbf{X} \text{diag}(e^{-\gamma \Delta l}, e^{\gamma \Delta l}) \mathbf{X}^{-1}$$

has eigenvalues $e^{\mp \gamma \Delta l}$ that do not depend on the launches at all. From them:

$$\gamma = \alpha + j\beta, \quad \varepsilon_{\text{eff}} = \left(\frac{c \beta}{2\pi f} \right)^2, \quad \alpha_{\text{dB/m}} = 8.686 \alpha$$

Symbol	Meaning	Unit
$\mathbf{T}_i$	measured transmission (cascade) matrix of line i , from its S-parameters	dimensionless
$\mathbf{X}, \mathbf{Y}$	the launch at each end, unknown and cancelled	dimensionless
Δl	length difference, a quarter guided wavelength at f_0 : 17.2 mm on the seed	m

The quarter wavelength choice matters: at a difference of 0 or 180 degrees the two eigenvalues coincide and the extraction becomes ill conditioned, which is the same reason line standards in TRL calibration are chosen near 90 degrees. `rfkit.lineparams` implements this extraction, and the same code serves SIM-001, whose two simulated line lengths, 10 mm and 27.2 mm, are the coupon lengths, so simulation and measurement go through the same extraction.

What coupons do and do not do. They calibrate the model: a measured ε_{eff} or attenuation updates the material values in a new revision of the canonical file and the models are rerun; that is model calibration and does not reopen decision 0009. They do not replace knowledge of the stack-up, they are measured only after fabrication, and they cannot detect channel to channel variation, because they sit in one place on the panel.

SCH-012, the stack-up validation at the bench, is NOT READY: the coupons are not laid out, the extraction procedure and its uncertainty have not been written before data, and the analyser uncertainty waits on EXP-004 observations O1 and O7 (`docs/runbooks/register.md`).

Part V. The simulation and analysis stack

Why several tools? Because each one answers a different question, and because agreement between independent routes is the only evidence a model can offer before hardware exists.

Tool	Question it answers	Strength	Blind spot
analytical models	roughly what should happen, and how sensitive is it?	instant, transparent, good for scaling and seeds	geometry details, discontinuities, coupling
HFSS	what do Maxwell's equations give for this exact geometry?	geometry aware, includes fringing, coupling and radiation	only as good as the model's materials, ports and mesh; slow
ADS, or scikit-rf circuit models	what does an independent distributed circuit model give?	fast, independent model form	idealised discontinuities; needs component models
the analyser	what does the real board do?	includes everything real	only as good as its calibration and reference planes; needs hardware
rfkit	do these agree, by rules fixed in advance?	one tested place for comparison rules and provenance	judges only what it is given

26. Analytical calculations

Analytical models are closed form equations: the microstrip formulas of section 5, the patch formulas of section 6, the array factor and its error statistics of sections 7 to 9.

Good for	Bad for
sanity checks: is a 0.37 mm line plausible for 50 ohm on 0.21 mm of FR-4?	switches and their packages
initialisation: the width and length seeds of SIM-001	bends, meanders and their self coupling
scaling laws: how the 315 degree error grows with permittivity error	connectors and launches
quick sensitivity estimates: the tolerance tables of section 59	discontinuities and their reflections
error budgets: decision 0007's thresholds	fringing in non uniform geometry
	mutual coupling between patches
	detailed loss including roughness and plating

analytical seed $\neq$ final geometry

In this repository, every analytical length or width is printed under the label INITIALISATION ONLY or SANITY CHECK ONLY, and the stack-up validator refuses to store any geometry in the canonical file (hardware/rev-a/stackup/README.md section 1).

27. HFSS

27.1 What it is

Ansys HFSS is a full wave, three dimensional, frequency domain electromagnetic solver based on the finite element method. The model's volume is divided into small tetrahedra, the fields inside each are approximated by simple functions, and Maxwell's equations become a large linear system solved at each frequency. The result is the field everywhere in the model and the S-parameters at its ports. It is driven by four kinds of input:

Input	Meaning in SIM-001
geometry	substrate, trace, air box, ground; built by script from the canonical stack-up
materials	substrate permittivity and loss tangent, copper conductivity, from reva-stackup-r1
boundaries	outer faces perfect electric conductor; ground of finite conductivity
ports	one wave port at each end: a two dimensional eigenmode solve gives the port's mode, its impedance Z_{pi} and its propagation constant

27.2 Adaptive meshing and convergence

HFSS solves at one frequency, estimates where the solution error is largest, refines the mesh there, and solves again. The change in S-parameters between successive passes, ΔS , is the convergence measure. SIM-001 requires $\Delta S \leq 0.02$ on two consecutive passes, at most 20 passes, at a solution frequency of 2.44 GHz, then an interpolating sweep from 1 to 3 GHz in 5 MHz steps. Decision 0007 adds a requirement for any solve used in a simulator comparison: the convergence criterion must be recorded, and for the verdict to concern models rather than mesh it must satisfy $\Delta S \leq |S_{21}| T$, with T the phase threshold in radians, about $0.04 |S_{21}|$.

27.3 Why a colourful field plot is not evidence

A field plot shows that the solver produced a solution. It does not show that the solution is right. A full wave result counts as evidence in this project only with:

Requirement	What it rules out
convergence recorded: passes, final ΔS , element count	a solution still changing with the mesh
port check	a wave port too small that couples to the walls; SIM-001 reruns with ports enlarged by half
boundary check	a shield or radiation boundary too close
material provenance	values typed by hand or from the wrong board; every model reads the canonical file
frequency coverage	conclusions drawn from one frequency; verdicts are judged across band 57a
reproducibility	a result nobody can regenerate; the builder, its inputs and the exported files are versioned
an independent check	agreement with a model of different form: the closed form, the two line extraction and the port solution are three readings in SIM-001

27.4 HFSS Student and full HFSS

Work runs where it is scientifically sufficient, not where the biggest tool is (docs/runbooks/README.md rule 2). HFSS Student, installed locally as release 2025 R2, is the default. Its documented limits [V23]: 64 000 elements in a three dimensional volume mesh, 8 000 in a three dimensional surface mesh, 2 000 triangles in two dimensions; DXF and STEP import only; local solves only, on at most four cores; no SBR+, mesh assemblies, circuit model generation from S-parameters, geometry export, optiSLang, LSDSO, Workbench, beta features or Linux. No port limit is stated, and whether Touchstone export with port impedance comments is supported is not documented either way, which SIM-001 step 5 checks. A model moves to the full licence at school only if its converged mesh needs more than the volume limit, or a listed feature; a straight 50 ohm line should need far less, so exceeding the limit there would first be treated as a modelling fault.

27.5 Uses in AetherArray

Use	Status
50 ohm microstrip on the beamformer construction	SIM-001, READY; execution blocked
phase sections, meanders and the Wilkinson arms	proposed, Part VI
switch discontinuities, if a package model can be built	proposed; no PE4259 model source identified
the four way divider	proposed
single patch, then the four element antenna board: coupling S_A and embedded element patterns	EXP-011 Stage 1, criterion fixed; geometry not designed

No HFSS solve has produced a result. The first execution of the SIM-001 builder against AEDT Student, on 2026-10-03, failed inside the first PyAEDT call: the AEDT server process started but never opened its scripting connection, and no geometry was created. A second probe with no builder code failed the same way. The session notes point, without confirmation, to AEDT Student never having completed an interactive first launch on that account; the remedy recorded is to open it once by hand and rerun step 2 unchanged (results/SIM-001/notes.md).

28. ADS

Keysight ADS (Advanced Design System) is a circuit and system simulator for RF and microwave design. Instead of meshing a geometry, it connects models: ideal and physical transmission line elements, such as a microstrip line model of a given width and length on a given substrate; S-parameter blocks imported from Touchstone files, for example a component vendor's switch model; lumped elements; and network analysis around them. It is fast and its model form is independent of HFSS's: a line in ADS is a closed form distributed model, not a solved field.

That independence is what makes it useful. If ADS and HFSS agree on the state dependent phase of a switched line channel, the agreement is evidence about the design; if they disagree, the disagreement locates a modelling error, often a discontinuity or a coupling path one model omits. The intended chain is

analytical seed --> ADS or scikit-rf circuit model --> HFSS full wave --> VNA on the board
(rfkit.stackup) (independent model form) (geometry aware) (reality)

with each step compared to the previous by `rfkit`. Decision 0007 fixed the acceptance limits for the HFSS against ADS comparison before any data: 2.29 degrees on the state dependent part of the S_{21} phase difference and 0.40 dB on the state dependent part of its magnitude, judged at every grid point of band 57a and at f_0 , with the convergence condition of section 27.2 (section 57).

ADS is at school only and optional: nothing in the repository needs it to be reproduced. The portable circuit route is `scikit-rf`'s transmission line media. Two gaps are recorded: `rfkit` has no source label yet for a `scikit-rf` circuit model, which a decision 0007 comparison would need, and no source has been identified for a PE4259 model usable in ADS (SCH-009). **No ADS result exists.**

29. PyAEDT

PyAEDT, packaged as `ansys-aedt-core`, is a Python interface to Ansys Electronics Desktop, which hosts HFSS. Aether-Array uses it so that solver models are built by script rather than by hand.

Benefit	How SIM-001 realises it
reproducibility	the builder <code>tools/sim/sim001_hfss.py</code> is versioned; rerunning it rebuilds the same six designs
parameterisation	widths of 0.9, 1.0 and 1.1 times the seed, two lengths, port size as variables
no manual transcription	substrate thickness, permittivity, loss tangent and copper come from <code>rfkit.stackup.sim001_parameters</code> , which reads the canonical file
sweep automation	the solve, the sweep and the exports are scripted
export provenance	each design writes a renormalised <code>.s2p</code> , a <code>-portdata.s2p</code> referred to the port impedance, and a JSON sidecar with the stack-up fingerprint and convergence fields, checked by <code>rfkit.stackup.check_sim_export</code>
versioning	the builder, the analysis script and the canonical file are all in git

The rule that generated geometry must consume the canonical configuration, never copied constants, is enforced by construction: the builder has no numeric material value of its own, and `python sim/sim001_hfss.py --dry-run` prints the complete geometry from the canonical file without AEDT, a step the CI runs. The builder also refuses to add designs to an existing project, stops the AEDT servers it started if it fails, and offers `--student`, `--graphical`, `--port-scale` and `--seed-only` for the protocol's steps. The analysis script `tools/sim/sim001_analyse.py` applies the committed SIM-001 criterion; it has been tested on an emulated run only. `ansys-aedt-core 1.1.0` is recorded in the SIM-001 protocol but is not in `requirements.txt`, because no CI job runs AEDT.

30. Touchstone

A Touchstone file, extension `.snp` for an N port, is a plain text table of network parameters against frequency; `.s2p` is a two port, `.s4p` a four port such as the antenna board's coupling matrix. A minimal two port file looks like this:

```
! AetherArray example, synthetic: not a measurement
# GHz S RI R 50
! freq re(S11) im(S11) re(S21) im(S21) re(S12) im(S12) re(S22) im(S22)
2.400 0.010 -0.020 0.700 -0.650 0.700 -0.650 0.012 -0.018
2.440 0.011 -0.019 0.640 -0.710 0.640 -0.710 0.013 -0.017
```

The option line, starting with `#`, gives the frequency unit, the parameter type, the number format (RI for real and imaginary, MA for magnitude and angle, DB for decibels and angle) and the reference impedance. Lines starting with `!` are comments; HFSS uses them to carry the port impedance and propagation constant, which SIM-001 reads. For two port files the column order is S11, S21, S12, S22, an exception to the row order used for larger files.

Limitation	Consequence
metadata is free text in comments	calibration state, reference planes, stack-up and instrument settings are not standardised; <code>rfkit</code> carries them in a provenance record beside the file

Limitation	Consequence
one reference impedance per file in version 1	port renormalisation must be explicit
no uncertainty	the analyser's expanded uncertainty must travel separately

Touchstone is the bridge of the project: whatever produced the S-parameters, the analysis reads the same kind of file.

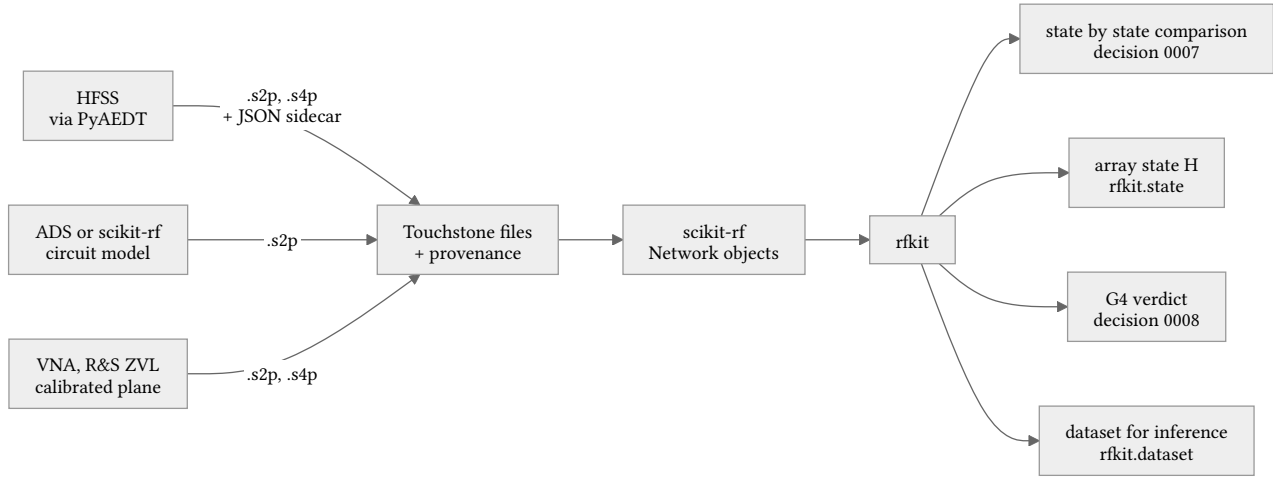


Figure 12: one data path for every source of S-parameters.

31. scikit-rf and rfkit

scikit-rf is an open source Python library for RF and microwave engineering. Its `Network` object holds S-parameters against frequency and reads and writes Touchstone; it also provides transmission line media such as the `MLine` microstrip model used for the stack-up seeds, calibration algorithms, de-embedding, time domain transforms and vector fitting. It is pinned at version 1.12.0 in `requirements.txt`.

rfkit is the project's layer on top of it, in `tools/rfkit/`. It encodes, once and with tests, the rules that would otherwise have to be remembered in every analysis notebook.

Module	What it does, as implemented on 2026-10-04
provenance, io	every trace carries its source, path, checksum, ports, reference impedance, sweep, calibration state and stack-up fingerprint; synthetic traces are labelled <code>synthetic</code>
grid	finds the band shared by all traces and the coarsest common grid; refuses to extrapolate, raising an error instead
metrics	values at a frequency and over a band; magnitude in dB; phase unwrapping; phase differences on the circle, so 359 and 1 degrees differ by 2; phase spread about the circular mean; amplitude imbalance
compare	pairwise comparison, whose S_{21} verdicts read “not applicable”, and <code>compare_states</code> , which keeps only the state dependent part and judges it against decision 0007 over band 57a
budget, thresholds	the error budget and the derivation of every threshold; values rounded down, tests fail if a recorded value is not its derivation
coupling	gate G4: the coupled forward model, the diagonal model, the rules of decision 0008, the guard matrices, the synthetic chart
state	per channel S_{21} to the diagonal array state, reference channel explicit, raw complex values kept; the full matrix form supported by the data structure
dataset	repeated measurements with session, time and temperature, and the inference record
calibration	interfaces that refuse to run until calibration standards exist; <code>apply_calibration</code> raises an error by design
instrument	the adapter boundary for automation, deliberately without drivers
stackup	loading and validating the canonical stack-up; seeds, sensitivity, loss and patch checks; SIM-001 inputs and export checks; the generated documentation tables
lineparams	the two line extraction of section 25

Command line entry points: `compare`, `compare-states`, `state`, `budget`, `g4`, `g4-chart`, `example` and `stackup` (`tools/rfkit/README.md`).

Why a project specific layer is worth its cost: the comparisons this project makes, HFSS against ADS against the analyser, are easy to make wrongly in ways that produce plausible numbers. Comparing over a band only one trace covers, calling 359 against 1 degree a 358 degree error, judging a constant offset that calibration removes as a failure, or printing PASS

against a threshold nobody derived: each is prevented by one tested function rather than by vigilance. **Status: 160 tests pass at the baseline, all on synthetic or analytically constructed data. No simulated or measured file has yet been processed.**

32. Python

Python is the working language of everything that is not gateway or schematic: experiment orchestration, analysis, plotting, simulation scripting, optimisation, the probabilistic inference to come, tests and the runbook builder.

Package	Role	Status
NumPy 2.4.6, SciPy 1.17.1	numerical arrays, root finding, statistics	pinned in requirements.txt
scikit-rf 1.12.0	the RF data layer	pinned
pytest 9.1.1	the test suite	pinned
matplotlib 3.10.9	figures, and equations in the runbook PDFs	pinned
mistune, reportlab, pillow, pymupdf	the school runbook PDFs	pinned
ansys-aedt-core 1.1.0	the SIM-001 builder	recorded in SIM-001, not pinned, not run in CI
scikit-learn, PyTorch, a Gaussian process or Kalman filter library	the learning track	not chosen and not dependencies ; the model class is to be chosen on data (section 40)

The pins date from 2026-09-25 with Python 3.12, and the CI uses Python 3.12. The repository rule is that there is one package list; a second would be deleted.

33. Git, CI and reproducibility

RF research benefits from software engineering discipline for a plain reason: the evidence chain from a material value to a verdict passes through many files and several tools, and any step done by hand is a step where a number can be retyped wrongly, a threshold moved after the data, or a synthetic trace mistaken for a measurement.

Practice	In this repository
version control	every document, decision, script and design file is in git; the commit is the timestamp
decision records	nine accepted decisions, each with options, evidence, known limitations and conditions for reopening; superseded text is kept, not deleted
canonical configuration	one stack-up file read by every tool; numbers never retyped
generated documentation	tables between markers regenerated from the canonical file; tests fail on drift
pre-registration	decision rules committed before data: EXP-005 section 7, decision 0007, decision 0008, the SIM-001 criterion
tests	rfkit test suite, 160 tests at the baseline
continuous integration	three workflows: documentation conventions (tools/check-docs.sh); RF tests, worked example, budget derivation, stack-up table check and the SIM-001 dry run; runbook build and check
runbooks	every school task gets a step by step PDF built from Markdown, with the source's checksum in the PDF; a task cannot be READY without it

The test count is quoted with its date because it changes; the command in the front matter recomputes it.

Part VI. Simulation roadmap

Only SIM-001 exists in the repository. The later stages below are a **[proposed here]** sequence, assembled from the follow-up list of SIM-001, the decision 0007 simulator comparison, EXP-011 and the school task register. Their numbers are suggestions; registering them, with criteria written before data, is future work.

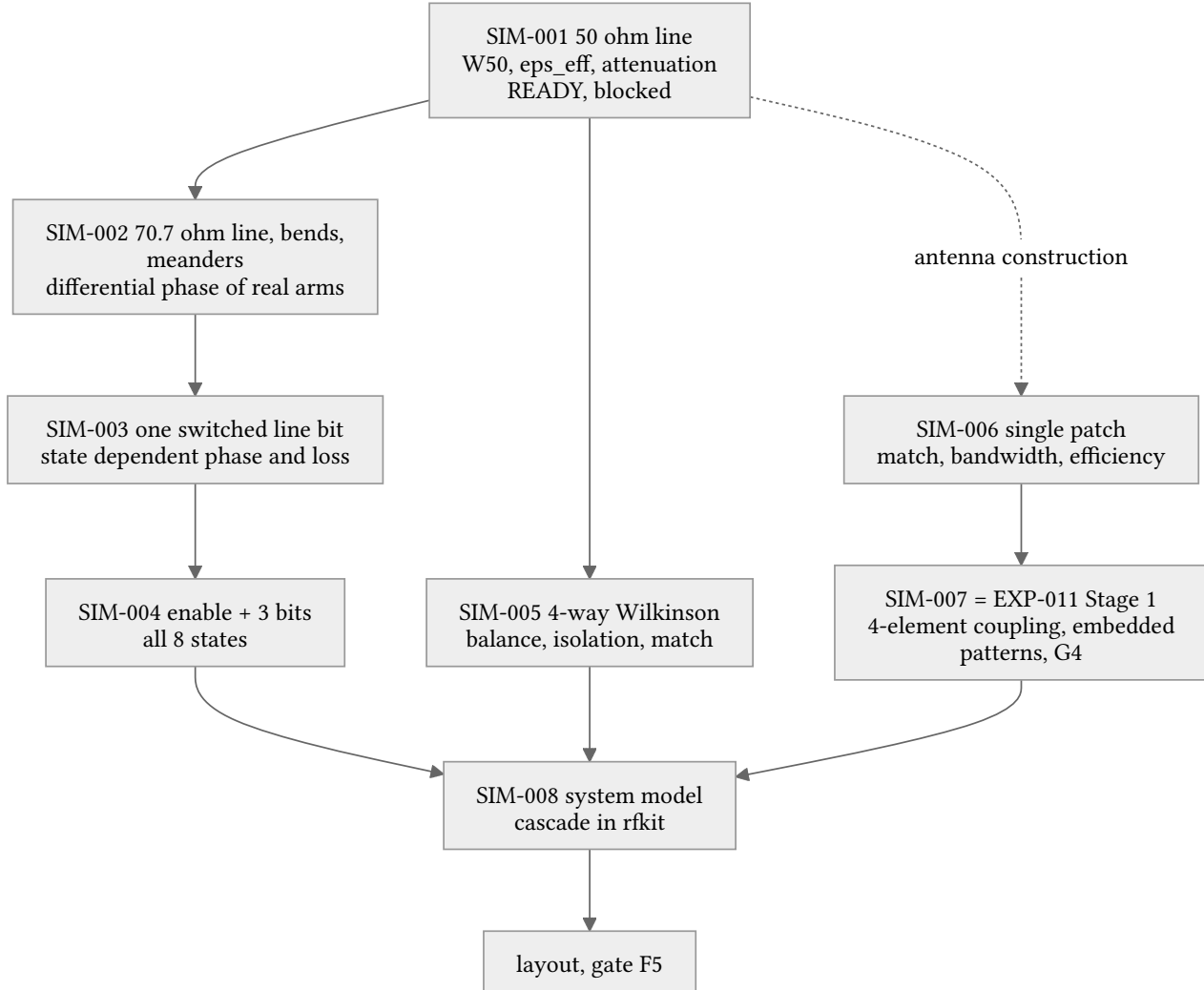


Figure 13: simulation roadmap. SIM-001 is registered; SIM-002 to SIM-008 are proposed numbering.

Why sequential? Each stage needs the previous one's output as a trusted input. It is irrational to optimise a full array before the transmission line model it is built from has been validated: an error in ε_{eff} would then be spread across every bit, every arm and every coupling result, and could not be separated from them.

Stage	Input	Model	Expected output	Pass or fail	Unlocks	Where
SIM-001 , 50 ohm microstrip, registered	canonical stack-up, seed width	straight line, two lengths, three widths, wave ports	W_{50} by interpolation; ε_{eff} and attenuation by two line extraction and by port solution	converged at $\Delta S \leq 0.02$ twice; W_{50} inside the sweep; acceptable if at least twice the process minimum; other figures reported, not judged	every printed length; gate F5	LOCAL, HFSS Student
SIM-002, Wilkinson line and real arm shapes, proposed	W_{50} , ε_{eff} from SIM-001	70.7 ohm line; meandered 45, 90 and 180 degree arms with mitred bends	differential phase of each arm as it will be routed, against the straight line value	criterion to be registered; decision 0007's 2.29 degrees bounds the state dependent error a layout may introduce	the arm geometry	LOCAL, likely within Student limits

Stage	Input	Model	Expected output	Pass or fail	Unlocks	Where
SIM-003, one switched line bit, proposed	SIM-002 arms; a switch model	two SPDTs and two arms, switch as an S-parameter block if a model exists	phase and loss of each state across band 57a; off arm resonance check	decision 0007: HFSS against circuit model within 2.29 degrees and 0.40 dB on the state dependent part	trust in the bit topology	EITHER; full HFSS if over the mesh limit (SCH-010)
SIM-004, full channel, proposed	SIM-003	enable switch and three bits in cascade, likely by cascading simulated blocks in <code>rflkit</code>	all eight states: phase error against nominal, loss spread	derived requirement: state dependent phase within 2.29 degrees of nominal at f_0 ; imbalance within 0.82 dB	the channel design; amplitude imbalance against decision 0009's loss estimate	EITHER
SIM-005, divider, proposed	W_{50} , 70.7 ohm width	three stage Wilkinson with resistors, then U_{900}	balance, output isolation, input match	criterion to be registered	the combiner	LOCAL
SIM-006, single patch, proposed	antenna construction	patch, feed, ground, radiation boundary	resonance, S_{11} , bandwidth, efficiency, pattern	criterion to be registered; resonance inside band 57a with margin for the permittivity bound	the antenna geometry, uncertainty I27	LOCAL, within Student limits [assumed]
SIM-007, coupling, registered as EXP-011 Stage 1	the four patch board	four ports at the connector plane, embedded element patterns	S_A in <code>.s4p</code> , last two passes, patterns container	decision 0008: PASS, FAIL, INTERMEDIATE or UNRESOLVED	gate G4; the coupling matrix for EXP-013	EITHER; SCH-004 on escalation
SIM-008, system model, proposed	SIM-004, SIM-005, SIM-007	cascade of simulated blocks and the coupled forward model, not a full wave model of everything	predicted per channel states, beams and nulls for all 512 states	criterion to be registered	the predictions that hardware validation (SCH-006) will test	LOCAL

Part VII. Measurement and experimental method

34. Evidence hierarchy

The repository uses several vocabularies for evidence, each introduced where it was needed. Together they form one discipline: never let a weaker kind of evidence be read as a stronger one.

Vocabulary	Classes	Where defined	What it prevents
confidence markers	[established], [assumed], [to verify]	CONVENTIONS.md section 3	an assumption hardening into a fact by repetition
instrument evidence	[observed] with a date, [inventory], [vendor], [listing]	docs/hardware/measurement-bench.md section 1	a measurement plan built on an instrument nobody has seen
material values	guaranteed, typical, nominal, assumed, unbounded	decision 0009; the canonical stack-up	a typical value used as a tolerance; a guessed tolerance given a number
thresholds	provisional-theory-derived, unresolved, not-a-limit	decision 0007; rfkkit.thresholds	a tool printing PASS against a number nobody derived
data provenance	source label on every trace, including synthetic	rfkkit.provenance	synthetic test data read later as a measurement
task readiness	NOT READY, BLOCKED, READY, DONE	docs/runbooks/README.md	a school task attempted without a procedure
experiment state	planned, running, to do; results cells to measure	experiments/, results/	an empty cell being filled with a plausible number

Read as a hierarchy of strength, for a question of fact about this project's hardware:

strongest	measured on project hardware, calibrated, repeated observed directly on the bench, dated simulated on the actual geometry, converged, checked vendor documentation for the exact part or model analytical computation from documented inputs listing or secondary source inference or engineering judgement, labelled	none yet the analyser's capabilities, 2026-09-20 none yet PE4259, AD8318, MCP9808, DE1-SoC, laminates seeds, budgets, sensitivity tables ZVL calibration methods, dynamic range local converter precaution, keep-out margin data rates of an unattended rig
weakest	unverified	

The ordering depends on the question. Decision 0004 made the point explicitly: for “can this instrument reach 2.44 GHz and measure a complex S_{21} ”, a bench observation is stronger than a datasheet, because a datasheet describes a model family and the observation describes the unit in the room. For “how accurately does it measure”, the datasheet of the identified model and a calibration are what count.

35. EXP-004: the analyser audit

35.1 What has to be known

Observation	What it is	Reading on 2026-09-20	What it decides now
O1	manufacturer, model, serial	Rohde and Schwarz ZVL; exact model and serial not read	provenance and every accuracy figure; the one reading that could still reopen the frequency, if the model were specified below 2.44 GHz
O2	maximum frequency	3 GHz, pass	the frequency; closed
O3	minimum frequency	9 kHz, pass	headroom; closed
O4	transmission measurement	S_{21} available, pass	gate G1; closed
O5	complex formats	available, pass	gate G1; closed
O6	source power	capability up to 0 dBm; level to be used not recorded	a procedure parameter
O7	calibration kit and connectors	N female ports; no kit, no adapter confirmed	calibrated validation of any board; the most consequential gap
O8	time domain option	not taken	the echo strategy for EXP-005; not a gate
O9	remote interface	USB present; enumeration not verified	automation for EXP-014; not a gate

Source: results/EXP-004/README.md, result 2: four observations complete, four partial, one not taken. Runbook SCH-001 is READY and finishes them in about 30 minutes at the bench. Result 1 is a negative result worth recording: no local evidence on the project computer identifies any instrument, because none has ever been plugged into it.

35.2 Why the frequency could be fixed before the model was identified

Decision 0004 fixed $f_0 = 2.44$ GHz on 2026-09-23 from the bench observation alone, and recorded why the earlier logic, which held the frequency open until the exact model was read, was wrong. Two questions had been merged: whether the instrument can do something, which an observation answers best, and how well it does it, which needs the datasheet and a calibration. Only the first gates design; the second gates validation.

The regulatory side was settled from the primary source [R1]: band 57a, 2400 to 2483.5 MHz, allows 10 mW equivalent isotropic radiated power with no duty cycle restriction, which is what a swept measurement needs; every sub-band between 863 and 870 MHz requires a duty cycle limit or a spectrum access technique. With a 2 dBi probe, the observed 0 dBm source ceiling keeps the radiated level about 8 dB below the limit, so compliance is guaranteed by the instrument rather than by the operator. The national table of frequency allocations, which may be more restrictive than the European instrument, has not been checked; it is a LOCAL lookup in the register.

36. EXP-005: the repeatability floor, and whether the control path disturbs it

36.1 Three things called noise

Quantity	What it is	What it does to calibration
ordinary measurement noise	random scatter from sample to sample	sets the floor; averaging reduces it
environmental drift	slow movement of the mean with temperature, time, handling	slow enough to track; the subject of EXP-010
state correlated error	a shift of the reading that depends on which beam state was commanded	does not look like noise. It looks like a property of the array, so the calibration absorbs it faithfully and reports a hardware coefficient that does not exist

The third is why the quiet window (requirement R9) and the local converter (open item H3) exist, and both were adopted in decision 0005 as precautions. EXP-005 decides them by measurement.

36.2 Phase A, executable now

Phase A needs no detector. It concerns the acquisition path, and a stable direct voltage stands in for the detector's output with the advantage that its true value is constant, so any state correlated change is unambiguously the path.

Element	Specification
apparatus	two owned DE1-SoC boards: board 1 drives a ribbon and converts the far path, board 2 converts a local path with a short lead. Same converter type on both, so the only difference is the cable
source S1 conditions	0.4 V to 2.2 V, near 1.2 V; stable to 0.2 mV over about 60 s; 100 ohm or less source impedance; not a DE1-SoC rail
beam state set interleaving	C1 static; C2 lines toggling during conversion; C3 the quiet window; C4 the temperature bus only; C5, optional, quiet window plus a ground strap
counts	16 words: eight single bit patterns, then eight from a fixed seed written into the protocol
	conditions interleaved in every cycle, never blocked, so slow drift cancels in the contrasts
	1000 samples per state per condition per cycle, 16 states, 20 cycles; a settling sweep from 1 us to 10 ms; at least 5 ribbon replugs; at least 3 sessions on different days

The state correlated error is the part of the between state spread that within state noise does not explain:

$$e_{\text{state}} = \sqrt{\max\left(0, \sigma_{\text{between}}^2 - \frac{\sigma_{\text{within}}^2}{m}\right)}$$

Symbol	Meaning	Unit
σ_{between}	standard deviation of the per state means	mV
σ_{within}	pooled standard deviation inside a state	mV
m	samples per state, 1000	count

Subtracting $\sigma_{\text{within}}^2/m$ matters: with finite samples the per state means scatter even when nothing depends on the state, and reporting that scatter as an effect would manufacture one.

36.3 The threshold, and what kind of number it is

The decision threshold is **0.5 mV, 0.02 dB at the detector slope**. It is a design judgement, not a law of physics: it is one fifth of a 0.1 dB drift signal, so an acquisition path contributing less than this cannot dominate what the drift experiment tries to see, and it is small against the detector's own ± 0.5 dB temperature figure. Its validity rests on the drift signal being of order 0.1 dB, which is itself unmeasured.

Rules, fixed before any measurement (EXP-005 section 7), after four validity preconditions V1 to V4 (codes dither; within state spread of the local path at most 2 mV; source drift within a cycle at most 0.5 mV; the two paths agree at rest within 3 mV):

Question	Outcomes
R9, the quiet window, on the far path	demote if C2 shows at most 0.5 mV; keep provisionally between 0.5 and 1.0 mV; keep, justified, if C2 exceeds 1.0 mV and C3 is at most 0.5 mV; escalate , reopening decision 0005, if C3 exceeds 0.5 mV whatever C2 shows
H3, the local converter, in C3	demote if the far path's within state spread is at most 1.2 times the local one, its state correlated error at most 0.5 mV, and its reconnection shift within 0.5 mV of the local one; keep, justified, if the spread exceeds twice the local one or the error exceeds 1.0 mV; otherwise keep provisionally
C4, the temperature bus	above 0.5 mV makes scheduling sensor reads outside the window mandatory
C5, the ground strap	a difference from C3 above 0.5 mV upgrades H5 to a measured requirement

36.4 Status

Not executed. Condition C1 has not been run; every result cell reads to `measure`. Item B1, the analogue input header and the circuit in front of the converter, is advanced but not closed; item B2, the converter example to use, is not chosen; the harness is not built (`results/EXP-005/README.md`). Phase B, the original question of whether received power can be measured repeatably in the room, needs a detector, antennas and interconnect, none owned, and its decision rules for both routes have not been written; it is gate F3 of decision 0006.

37. The future drift experiment

The learning track needs data that only a fabricated array can produce: the drift of its own switches, lines, connectors and antennas over time. Three planned experiments supply it.

Experiment	Question	State of its protocol
EXP-010	how long does a calibration stay valid?	not detailed; "little work but a lot of calendar time"; gates EXP-014 and EXP-015 (<code>experiments/plan.md</code>)
EXP-014	can the array calibrate itself repeatedly, unattended, for weeks?	method and record schema fixed; criterion: a full classical calibration completes unattended, repeatedly, with session to session spread at or below the EXP-005 floor
EXP-015	does a learned prior recalibrate with fewer measurements than from scratch?	specification ML-B; failure declared if the required count is not below the from scratch count on held out sessions, or if G2 has failed

The record schema of `docs/architecture/control-architecture.md` section 6 is the contract for that data. Every measurement writes: session identifier; sequence index; commanded and read back beam state words; strobe and conversion timestamps from one clock; the settling delay applied; raw converter counts; the converter reference and channel; the phase network, detector and die temperatures; gateway and software versions; the connector handling flag; and the analyser state when it was used. A field not captured is absent, not defaulted. Three are easy to omit and expensive to lose: raw counts, from which the noise model is estimated; the read back word; and the version fields, without which a timing change between sessions is invisible.

What the repository does not yet fix, and this document does not invent: the duration of a campaign, the schedule of full calibrations against cheap ones, whether temperature is varied deliberately or left to the room, the label's reference plane (section 16), and the session rate. The rate of about 100 labelled recalibration pairs per week is an **unverified estimate** from an assumed ten minute cycle (decision 0002).

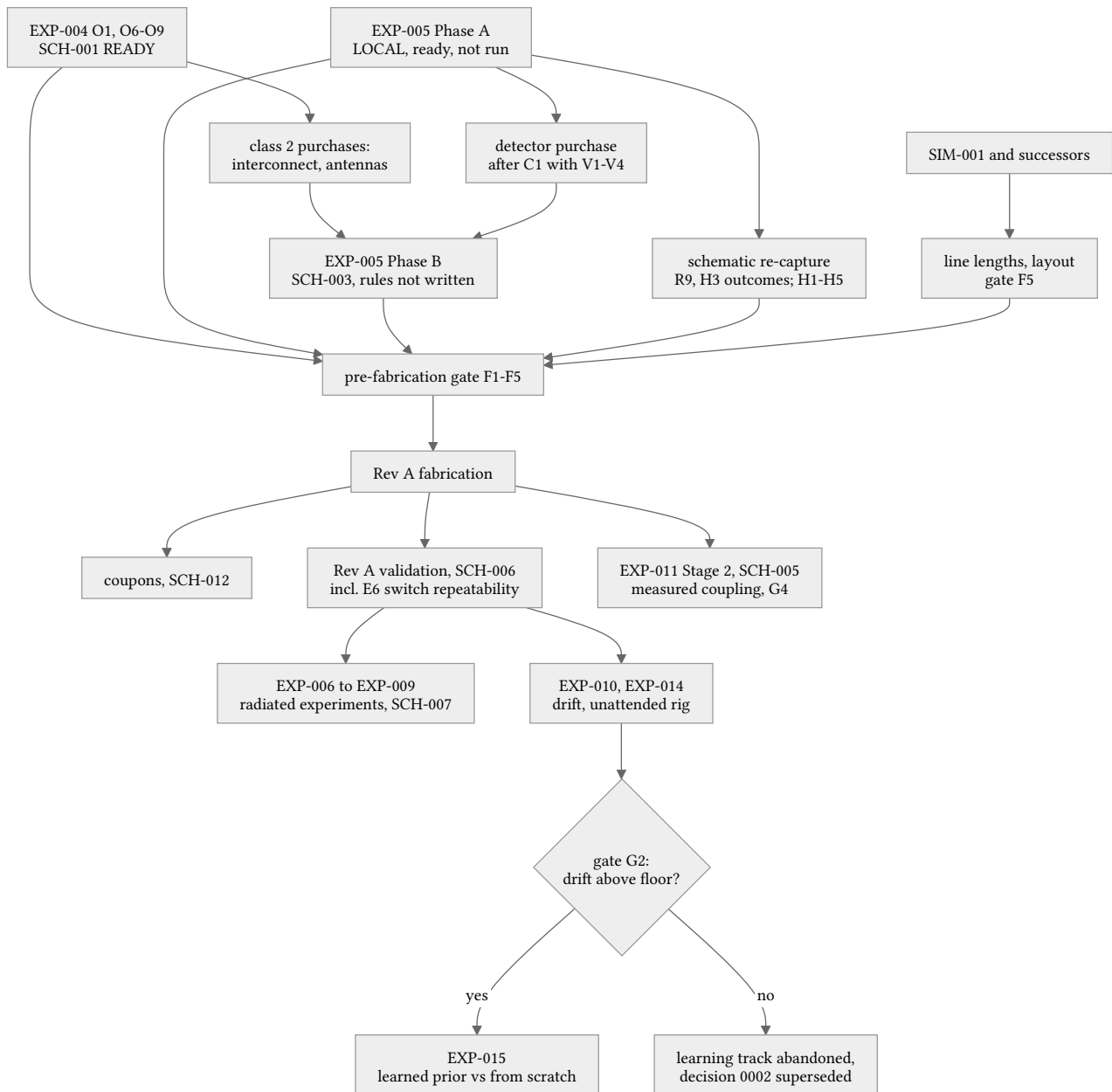


Figure 14: measurement roadmap and its gates, from docs/runbooks/register.md, decision 0006 and experiments/plan.md.

Part VIII. Machine learning

Machine learning is not used because the project needs an AI label. It is used, if gate G2 allows it at all, for one specific thing that physics cannot supply: how this particular array's state tends to move between calibrations. A physical model knows that a switched line's phase is $\beta \Delta l$; it does not know by how many degrees channel 2 of this board drifts per degree Celsius on a Tuesday afternoon, how that drift correlates with channel 3's, or how quickly it decorrelates. That structure, if it exists, has to be learned from the board's own history.

38. What machine learning does not do

Statement	Why
1 It does not replace Maxwell's equations	the likelihood of every measurement is explicit physics, the forward model of <code>docs/mathematics/inverse-calibration.md</code> section 2; it is never learned
2 It does not replace HFSS	full wave simulation designs and checks the geometry; nothing learned stands in for it
3 It does not choose the beam on Rev A	beam synthesis is an exact enumeration of 512 relative states against the estimated array state (section 8.3); decision 0005 withdrew the earlier scheduling of a surrogate pattern synthesis track
4 It does not perform a first calibration with fewer measurements than identifiability allows	no method can determine six unknowns from fewer than six readings without prior information; at $N = 4$ the classical routes already sit within a few readings of that floor (section 11.6)
5 It does not output antenna commands end to end	nothing learned ever predicts a measurement or a command directly; the learned part predicts where the state probably is
6 It is not trained on simulated labels and then claimed to solve real drift	the labels of the central experiment come from full calibrations run on the hardware; the simulation trained estimator ML-A is a control, and its mandatory negative test reports how it degrades out of its training distribution
7 It does not run in the FPGA	inference has no hard deadline; the fabric owns timing only (section 19.3)
8 It does not replace analyser ground truth	the analyser's per channel complex measurement is the label that supervises the cheap measurements

Why exhaustive enumeration is a strength, not a weakness. With 512 relative states, or 820 with the enable bits, the best reachable beam for any criterion, main beam gain, null depth, signal to interference ratio, can be found exactly in software once the array state is estimated. That gives the project a known optimum to compare against. A learned or surrogate beamformer would at best approximate this optimum and would add an error source to the experiment. Only in a model free comparison, where each of the 512 states is a physical measurement, could a surrogate save anything, and there it is retained as a control, not as a contribution (`docs/architecture/ml-calibration.md` section 6, ML-D).

39. What machine learning does

39.1 The learned object

The object of learning is the temporal structure of $\mathbf{H}_t$. For the diagonal model, represent the identifiable part as a vector of six real numbers, the three relative log gains and the three relative phases of channels 1 to 3 against channel 0:

$$\mathbf{z}_t = \left(\ln \frac{g_1}{g_0}, \ln \frac{g_2}{g_0}, \ln \frac{g_3}{g_0}, \psi_1 - \psi_0, \psi_2 - \psi_0, \psi_3 - \psi_0 \right)_t \in \mathbb{R}^6$$

The phases are angles, so they are taken relative to the previous calibration and kept small, which avoids the wrap at 360 degrees. A learned prior is then a conditional distribution

$$p_\theta(\mathbf{z}_t \mid \mathbf{z}_{t-1}, \mathbf{z}_{t-2}, \dots, T_t, \Delta t, \dots)$$

with parameters θ fitted on past sessions.

39.2 Learned prior plus physical likelihood

New measurements update the prior by Bayes' rule:

$$p(\mathbf{z}_t \mid y_{1:k}, \mathbf{x}_{1:k}, \mathcal{D}) \propto \underbrace{\prod_{i=1}^k p(y_i \mid \mathbf{z}_t, \mathbf{x}_i)}_{\text{physical likelihood: never learned}} \times \underbrace{p_\theta(\mathbf{z}_t \mid \mathcal{D})}_{\text{learned prior}}$$

Factor	Content	Origin
likelihood	the forward model, complex $\sum_n w_n h_n$ for the analyser or power $ \sum_n w_n h_n ^2$ for the detector, plus a noise model estimated from raw counts and the EXP-005 floor	explicit physics
prior	where the state probably is now, given the history	learned
posterior	what is believed after k new readings	Bayes' rule

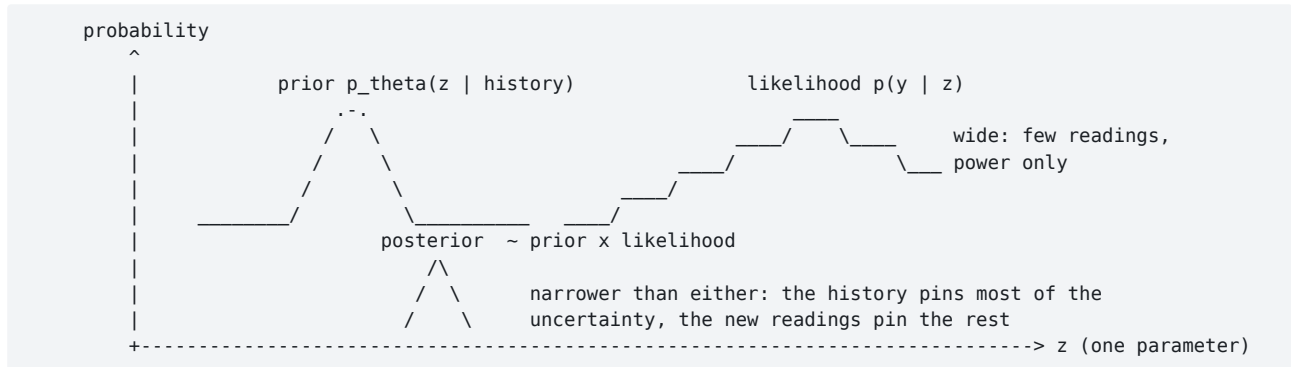


Figure 10: prior, likelihood and posterior for one parameter, conceptually. With a good prior, the posterior reaches the target width with fewer new readings.

This division is the architectural claim of the project. If the prior is good, few readings are needed to reach the target accuracy. If the prior is poor but honest about its uncertainty, the likelihood dominates and more readings are needed: the method degrades towards the from scratch cost rather than failing. **That property holds only if the prior's uncertainty is calibrated.** An overconfident prior that is wrong can produce a confident wrong posterior and stop early. The evaluation must therefore test the prior's predictive coverage on held out sessions, and a practical safeguard, [proposed here], is a residual check: if the posterior predictive residuals of the new readings exceed what the EXP-005 noise floor allows, the session falls back to a full calibration and is counted as such.

40. Candidate models

The first model should be simple, for reasons the data decide rather than taste:

Consideration	Implication		
dataset size	of order 100 labelled sessions, an unverified estimate (section 37)		
dimensionality	six outputs per session for the diagonal model		
uncertainty	the stopping rule needs a calibrated posterior, not only a point prediction		
interpretability	a learned temperature coefficient or correlation time is itself a physical finding		
temporal correlation	the data are a time series with irregular intervals		

Model	What it assumes	Fits this problem because	Weak where
linear Gaussian state space model, inferred with a Kalman filter [L30]	$\mathbf{z}_t = \mathbf{A}\mathbf{z}_{t-1} + \mathbf{B}\mathbf{u}_t + \mathbf{q}_t$, Gaussian noise $\mathbf{q}_t$ with covariance growing with Δt ; inputs $\mathbf{u}_t$ such as temperature change	few parameters; exact for complex readings, which are linear in $\mathbf{h}$; extended or unscented variants handle power readings; uncertainty native	nonlinear or regime changing drift; handling jumps must be modelled separately
autoregressive model	each parameter regressed on its own recent values and on covariates	simplest possible baseline; transparent	point predictions unless wrapped in a probabilistic form
Gaussian process over time and temperature [L31]	each parameter, or all jointly, is a smooth random function with a covariance kernel whose length scales are learned	nonparametric with few hyperparameters; uncertainty native; precedent for sparse calibration data [A19] and for temporal gain priors in radio interferometry [L23]	cost grows with data unless written in state space form, which temporal kernels allow [L25]
neural network	a flexible function learned from many examples	none at this data size	thousands of parameters against hundreds of numbers; uncalibrated

Model	What it assumes	Fits this problem because	Weak where
			uncertainty; any result would mostly reflect the random seed

A large neural network is therefore unjustified initially, and docs/architecture/ml-calibration.md section 7 reaches the same conclusion: the data budget is a consequence of the hardware, and it selects the model class before modelling taste enters. Which of the first three to use is an experiment, not a choice to make now (docs/mathematics/inverse-calibration.md section 3.2). In all three, θ , the drift dynamics, the temperature coefficients, the kernel length scales, is fitted on training sessions, for example by maximising the marginal likelihood.

41. Training dataset

Each session would contribute one record built from the schema of section 37:

Field	Role
label z_t	from a full calibration run on hardware immediately after the cheap measurements, with its own uncertainty
cheap measurements	the P power readings and the code words that produced them, the input of the sparse recalibration
temperatures	phase network, detector, die; now and at the last calibration
time	timestamp and elapsed time since the last calibration
commanded states	every code word applied
environment and handling	connector handling flag, operator presence, analyser state
uncertainty	the label's repeatability, from EXP-005 and EXP-014

The label is itself a measurement with noise. “Equal final accuracy” can only be judged to within that noise, which is why the session to session repeatability of the full calibration, EXP-014's criterion, is a prerequisite.

The split must respect time. Neighbouring sessions are correlated: they share temperature, handling history and slow drift. Shuffling sessions at random and splitting them into training and test sets would place near duplicates on both sides and leak information, producing optimistic results that would not survive deployment. The split is chronological:

```
sessions in time order: |----- train -----|--- validate ---|---- test ----|
                        early period          later period      final period, untouched until the end
```

Better still is a rolling origin evaluation, training on everything before a point and testing on the period after it, repeated for several points. Two further leaks to guard against: a connector handling event on one side of a split boundary that affects the other side, and a test period whose temperature range lies outside the training range, which must be reported as extrapolation.

42. Baselines

The claim is only as strong as its comparison. The minimum set:

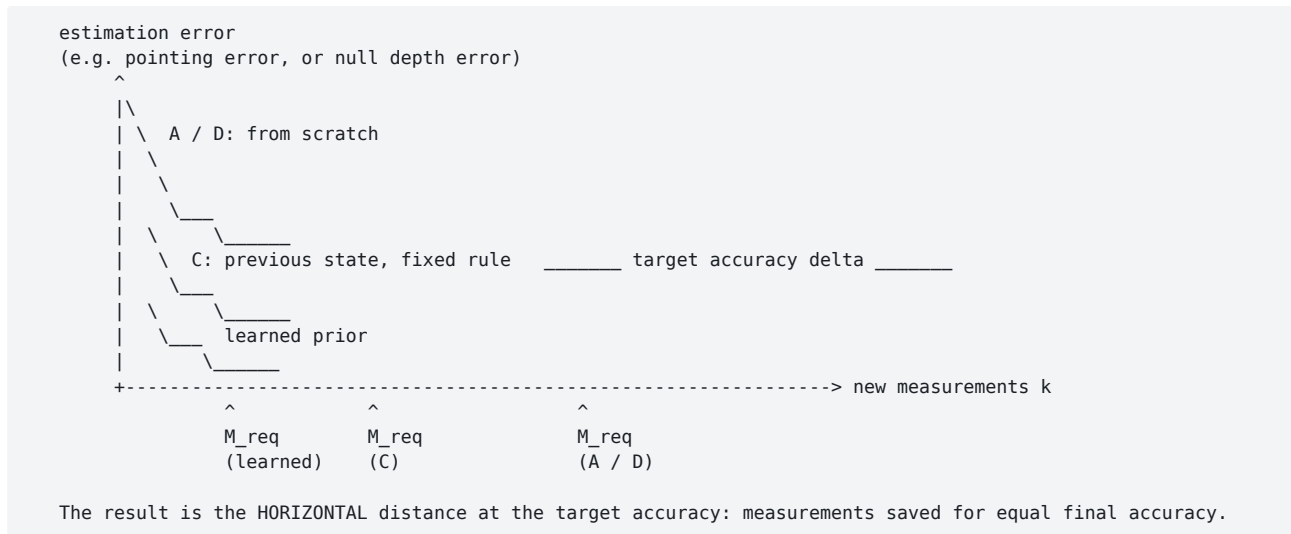
Baseline	Description	New measurements	What it shows
A, full recalibration from scratch	the best classical method at its own minimum: complex per channel readings, or REV at three states, or the fast amplitude only method if verified [L7]	4 complex, or about 8 to 12 power	the cost the method must beat
B, persistence	reuse $\mathbf{H}_{t-1}$ unchanged	0	how wrong the old calibration is now; if B already meets the target, there was nothing to recalibrate
C, simple temporal model	the same Bayesian update with a prior that is not learned from the history: the previous state with an uncertainty grown from a fixed rule, such as a random walk whose rate comes from EXP-005 and engineering judgement	as needed to reach the target	how much of any saving comes from the Bayesian update and the previous state alone
D, uninformative prior with the same update	the from scratch Bayesian baseline of the stopping rule, section 0.2	as needed	isolates the value of the history
Method	learned prior plus physical update	as needed	the claim

Baseline	Description	New measurements	What it shows
optional	learned prior plus active measurement selection	as needed	the value of choosing measurements

Baseline C deserves emphasis. A large part of any saving may come simply from starting at the last calibration with a sensible uncertainty, which needs no learning. The learned prior's contribution is the difference between the method and C, not between the method and A.

Fairness rules, consistent with `benchmarks/specification.md`: the same sessions for every method; the same measurement types; the same integration time per reading; every reading counted, including any used to detect a handling event; baselines run at their own minimum, REV at three states and not padded; results as curves with spread bands over held out sessions, never a single number.

The figure that would summarise the learning track:



Conceptual; no curve has been computed or measured.

43. Active measurement selection

Once a posterior exists, the next measurement need not follow a fixed schedule. Bayesian experimental design chooses it to be as informative as possible about the state [L26, L28]:

$$\mathbf{x}_{k+1} = \arg \max_{\mathbf{x} \in \mathcal{X}} \mathbb{I}(\mathbf{z}_t; y \mid \mathbf{x}, \mathcal{D}_k), \quad \mathbb{I} = \mathbb{H}(\mathbf{z}_t \mid \mathcal{D}_k) - \mathbb{E}_y[\mathbb{H}(\mathbf{z}_t \mid \mathcal{D}_k, \mathbf{x}, y)]$$

$\mathbb{I}$ is the mutual information between the state and the outcome of measuring in state $\mathbf{x}$, and $\mathbb{H}$ is entropy, both in nats. Intuitively: choose the beam state whose reading is expected to shrink the uncertainty about the array the most. For a measurement linear in the state, $y = \mathbf{a}^T \mathbf{z} + \epsilon$ with noise variance σ^2 and a Gaussian posterior of covariance $\mathbf{P}$, the gain is

$$\mathbb{I} = \frac{1}{2} \ln \left(1 + \frac{\mathbf{a}^T \mathbf{P} \mathbf{a}}{\sigma^2} \right)$$

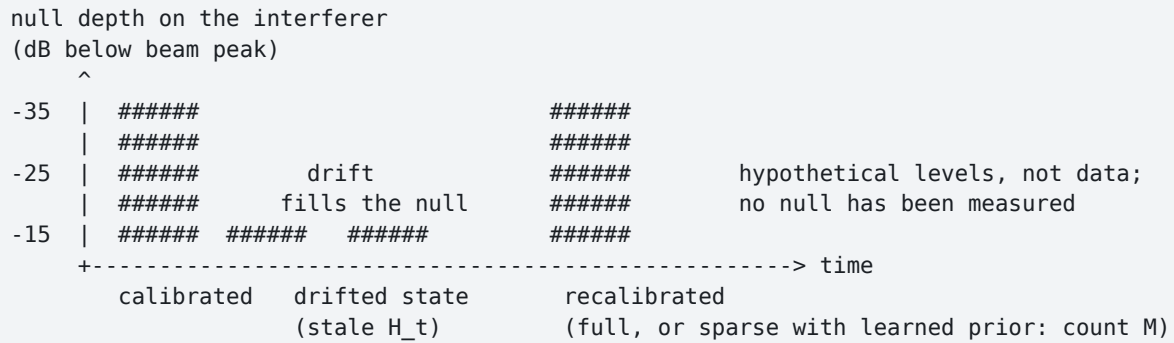
so the best measurement is the one whose outcome the current belief predicts least well relative to the noise. Power readings are nonlinear, so the gain must be approximated, by linearising around the posterior mean or by Monte Carlo. Because the candidate set is the 512 or 820 reachable states, the maximisation itself is an enumeration, not a search. For Gaussian process models, greedy mutual information selection is near optimal by a submodularity argument [L29].

This is Bayesian experimental design, which seeks a good **measurement**, not Bayesian optimisation, which seeks a good **command** [A13]; the repository corrected itself on this once (`docs/mathematics/inverse-calibration.md` section 4). Active selection is optional and later: EXP-015 names it, but **nothing is implemented**.

44. Why machine learning becomes useful in the ISAC demonstrator

Section 10bis.5 derives why both ISAC functions depend on the array state and 10bis.6 places the learning contribution; this section only says what that looks like in a demonstration. The AP-S demonstrator, if pursued (Part IX), is where the value of calibration, and therefore of cheaper recalibration, becomes visible to someone who is not an RF engineer.

Communication mode. A null placed on an interferer is deep only while the array state is accurately known (section 9.4).



Sensing mode. Hardware drift and environmental change reach the sensing features through the same product $\mathbf{H}_t \mathbf{R}(t) \mathbf{H}_t^H$, so drift can be read as a person moving (section 10bis.5).

The claim the demonstrator can carry is therefore:

Use learned temporal knowledge of RF hardware drift to restore communication and sensing performance with fewer new physical calibration measurements.

That claim is narrower and much stronger than “AI beamforming”: it is testable, it has a non-learned baseline, and its benefit is a count of measurements that a jury can watch being saved. It also carries a practical condition: drift must occur, or be induced in a controlled and declared way, within the time of a demonstration. A deliberately warmed beamformer board would be an induced drift and must be presented as such; a reconnected cable is a jump, not drift.

Part IX. The IEEE AP-S 2027 ISAC demonstrator

Status of everything in this part. The repository contains no decision, experiment or document about the AP-S Student Design Contest or about integrated sensing and communication. The demonstrator is **[proposed here]**, from the project owner’s brief. The contest facts below were researched on 2026-10-04 from official IEEE pages, but the environment this document was written in could not open them: its network policy refused connections to ieeeps.org and 2027.apsursi.org. Every contest fact is therefore **[snippet only]**: read in search engine excerpts of the official pages, consistent across many independent queries, and **not yet checked against the full call**. Quoted wording is near verbatim as indexed. Before the proposal is written, the official call [P2] must be read in full and this part corrected against it.

45. The official challenge

45.1 What the 2027 contest asks for

Item	Official requirement, as indexed	Source
edition and title	the 18th IEEE AP-S Student Design Contest; topic “Reconfigurable Receiving Antennas for Integrated Sensing and Communications (ISAC)”	[P1], [P2]
challenge	“design a dual-mode, reconfigurable receiving antenna system for ISAC that utilizes unmodified, commercial off-the-shelf (COTS) transmitters, such as Wi-Fi routers or Bluetooth beacons, as ambient signal sources, propose a setup to demonstrate its utility, and provide educational material to explain it”	[P1]
receive only	“the student-designed hardware must operate as a receiver”	[P1]
two modes	“The antenna system must dynamically switch between sensing and communication modes”	[P1]
sensing mode	ambient signals as “illuminators of opportunity”; with “a basic signal-processing backend, the antenna must optimize its characteristics to detect or track environmental changes, such as human movement or object positioning”	[P1]
communication mode	“establishing a robust data link with a specified COTS TX”; the system “must dynamically direct its main beam toward the desired TX while simultaneously placing pattern nulls or utilizing polarization mismatch to suppress strong, direct-path interference from other ambient communication sources”	[P1]
transmitters	unmodified COTS transmitters of the team’s choice, single or multiple; Wi-Fi routers or Bluetooth beacons certified for use in Japan, and smartphones in standard hotspot mode, are given as acceptable	[P1]
display	“Performance metrics/results must be displayed in real time, with easy visualization”	[P1]
education	the theory explained “in simple terms for non-engineers”, and “step-by-step instructions” to replicate the system	[P1]
cost	“The total production cost for the entire system must be less than US\$1,500”; university licensed or free software may be used; other commercial software counts in the budget	[P1]
team	2 to 5 students, at least half undergraduates by the end of May 2027; no student or mentor in more than one team	[P1]
mentor	one professional mentor who is an IEEE AP-S member; a mentor letter agreeing to supervise and to advance initial costs if necessary; the work done primarily by the students	[P1]
proposal	preliminary design due 31 December 2026 (GMT-10): a PDF of at most four pages in 12 point Times New Roman, with the demonstration setup and the quantities to be measured or post-processed, the antenna system to be built, a bill of materials up to US\$1,500, and the mentor letter	[P1]
selection and funding	six semi-finalist teams selected by 22 January 2027, each receiving US\$1,500 to build and test; stipends of up to US\$10,000 per team to attend the symposium, on submission of final materials and visa letters	[P1]
final materials	due 24 May 2027: a video of at most 10 minutes; replication instructions of at most 5 pages; a final report of at most 5 pages in the IEEE Transactions on Antennas and Propagation format, including simulation and measurement results	[P1]
judging	preliminary: likelihood of achieving the design goal and specifications, creativity, quality of the written materials. Final: achieved performance, creativity, system functionality, educational value, quality of final materials and of the on-site demonstration	[P1]
prizes	1st, 2nd and 3rd: certificates and US\$1,500, US\$750 and US\$250	[P1]
symposium	2027 IEEE International Symposium on Antennas and Propagation and JNC-USNC-URSI Radio Science Meeting, Kyoto International Conference Center, Kyoto, Japan, 20 to 25 June 2027	[P3]

Not found in any excerpt, and therefore unknown: a mandated frequency band; any rule on software defined radios, commercial receiver modules or commercial antennas; judging weights; a proposal template; whether student IEEE membership is required; a contest contact. One search summary gave “travel awards up to US\$1,500”, which conflicts with the official domain excerpts; it appears to confuse the build funds with the stipend.

45.2 Requirement and response, kept apart

Official requirement	Our design response, [proposed here]
a reconfigurable receiving antenna system	Rev A in its receive configuration: a passive switched line network, no transmitter of our own in the demonstration
COTS transmitters as ambient sources	2.4 GHz Wi-Fi routers, Bluetooth beacons or a hotspot phone; band 57a is the band they use, which the 2.44 GHz choice of decision 0004 happens to fit
communication mode with beam and nulls, or polarisation	beam towards the wanted transmitter and a null on the interferer, chosen by exact enumeration of the reachable states against the calibrated array state; polarisation is a gated option (section 51)
sensing mode	pattern diversity: a set of receive patterns cycled by the FPGA, their received powers as features
real time display	a dashboard on the HPS or a PC: per state powers, estimated pattern, null depth, sensing state, measurement count
educational material	the course of Part I is the raw material; the 16 bit word, the enumeration and the drift experiment are easy to show
cost below US\$1,500	the Rev A bill of materials was estimated at about 62 EUR before decision 0005 added a converter and buffers, and has not been re-costed; the controller and any receiver come on top; the whole system must be costed, including the DE1-SoC if the rules count owned equipment (to verify)

46. Why AetherArray fits, and what is missing

Each official requirement is traced to the physical function it asks for (section 10bis), to how AetherArray would provide it, to a metric, and to the evidence that exists and that is missing.

Official requirement	Physical function	AetherArray implementation	Measurable metric	Current evidence	Missing evidence
reconfigurable receiving antenna system	spatially selective reception, section 7	4 elements, 3-bit phase, enable per channel, 512 relative states; receive only, passive chain	pattern change between commanded states	analytical; schematic captured; not built	patch design, re-capture, fabrication, a measured pattern
communication: main beam to the wanted TX	coherent gain towards θ_D , sections 7.3 and 10bis.4	exact enumeration against the estimated array state	P_D against the ideal steered value	analytical	EXP-007 to EXP-009; a source separating receiver
communication: nulls or polarisation against interference	suppression of P_I with the link kept, sections 9 and 10bis.4	null placement by enumeration; polarisation gated (section 51)	P_I suppression; SIR or packet success, if the receiver separates sources	ideal theory only	N = 4 feasibility gate (section 50); a measured desired and interferer experiment
sensing with ambient illuminators	inference on the multipath field, sections 10bis.3 and 10bis.4	pattern diverse power vector $\mathbf{p}(t)$ over K states	detection or false alarm rate of a pre-registered task	observability argument only	protocol, classifier, controlled repeated experiment
dynamic switching between modes	shared aperture, family B	the FPGA applies any state on one clock edge	switching time, recorded	specified; no gateway	gateway and HPS software
real time metrics	a live estimate of the field and the array	record stream and inference on the HPS	dashboard latency	specified	the dashboard
(not an official requirement) robustness to drift	the shared calibration layer, section 10bis.5	Part VIII: full recalibration, and the learned prior as research extension	null depth or sensing false alarms before and after recalibration; $M_{extrequired}$	formalised only	EXP-005 to EXP-015
educational material and replication	explanation and reproducibility	public repository, runbooks, generated documentation, Part I	an outsider replicates	strong for documentation	the contest's replication guide
total cost below US\$1,500		about 62 EUR of parts estimated before decision 0005's additions, plus controller and receiver	itemised bill of materials	estimated, not re-costed, no quote	full system costing

The fit is genuine in three respects: a receiving, reconfigurable array is exactly what Rev A is; the 2.4 GHz band is where the allowed commercial transmitters operate; and the communication mode's demand for nulls is exactly where

calibration error becomes visible. It is incomplete in one structural respect, the receiver behind the array (section 47), and in schedule.

47. Communication mode

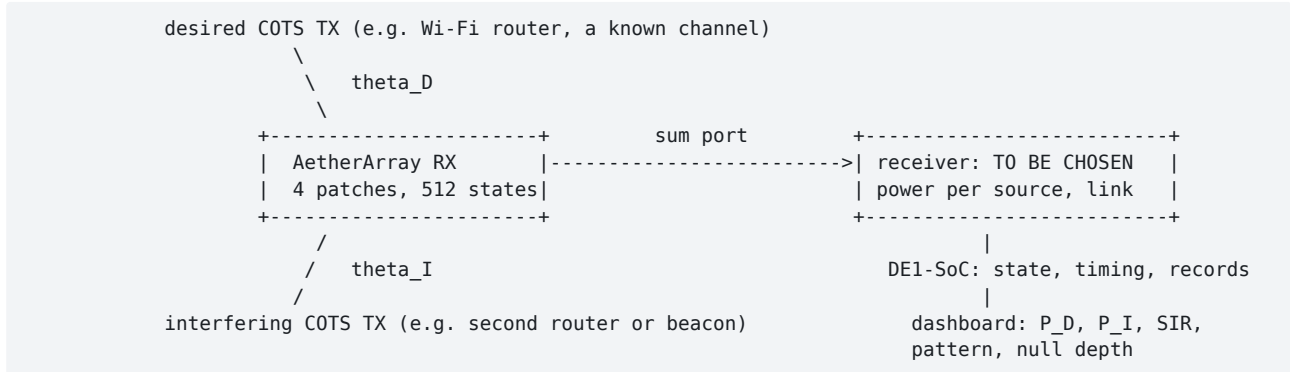


Figure 15: the proposed communication mode. The receiver behind the sum port is not chosen.

Objective. Keep a strong response towards the wanted transmitter while suppressing the interferer: maximise J or the signal to interference ratio of section 9.2 over the reachable states.

Procedure, as proposed.

1. Estimate the array state $\hat{\mathbf{H}}_t$ (Part VIII). If the wanted transmitter is used as the calibration source over the air, what is estimated is the product of hardware and incident field, $h_n s_n$, a channel calibration valid in that room (section 10bis.5); it equals the hardware state only for a dominant direct path from a known direction, or when the hardware state comes from a conducted reference.
2. Estimate or know θ_D and θ_I .
3. Enumerate all 512 relative states, or 820 with the enable bits, computing P_D and P_I from $\hat{\mathbf{H}}_t$ and an element pattern model; choose the best.
4. Apply it on one clock edge; measure the outcome; display it.

The receiver is the open architectural question. The AD8318 at the sum port measures total power across its whole frequency range. When the wanted and the interfering transmitters are both active in the 2.4 GHz band, it reports their sum and cannot attribute power to either; and ambient Wi-Fi and Bluetooth signals are bursts, not continuous carriers, so a sampled detector sees a fluctuating input. A signal to interference metric therefore needs a measurement that can tell the sources apart:

Option	How it separates the sources	Cost and risk
AD8318 alone	only if the sources occupy different times or different channels behind a filter	not controllable with unmodified commercial transmitters; filtering per channel is impractical
software defined radio at the sum port	channelisation, packet timing or identity per source	cost against the US\$1,500 cap; software effort; whether a commercial receiver is acceptable under the receive only rule is not stated in the excerpts
commercial Wi-Fi or Bluetooth receiver module fed from the sum port	reports received signal strength and packet statistics per transmitter identity	cheap and directly measures the “robust data link”; same rule question; reported RSSI is coarse and its accuracy uncharacterised

None is chosen, and this document does not choose one. Whatever is chosen also changes the measurement model of Part VIII: a per source power reading is still a power reading, so the power only likelihood applies, but its noise and its time cost differ from the detector’s. Whether ambient signals at demonstration distances reach the useful range of any of these receivers after the chain loss is **[to verify]** by a link budget.

Null visualisation. The dashboard can show the measured P_D and P_I for the chosen state, the estimated pattern with its null, and, over time, the null depth on the interferer. That last plot is what makes drift and recalibration visible (section 52).

48. Sensing mode

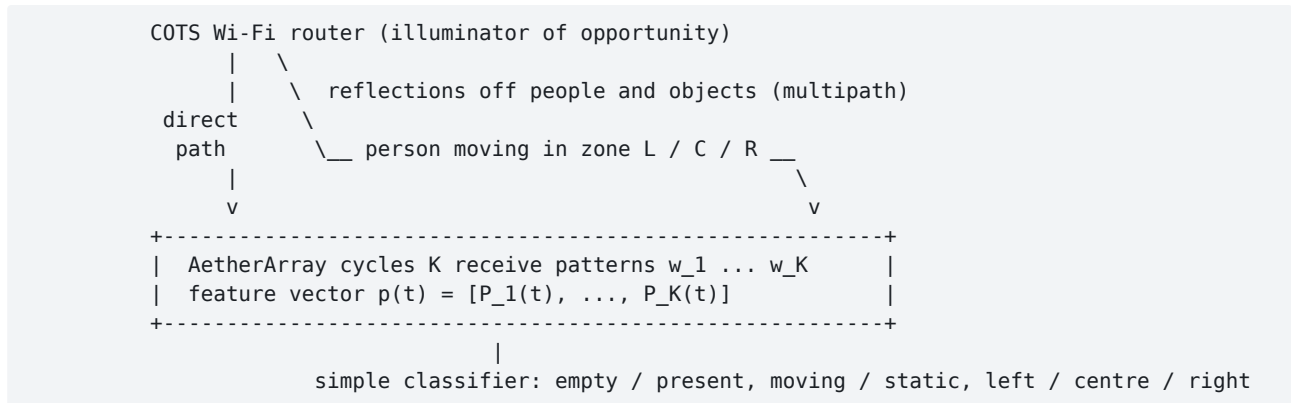


Figure 16: the proposed sensing mode.

The physics is in section 10bis: the illuminator and its paths in 10bis.3, the pattern diverse vector $\mathbf{p}(t)$ and why it carries more than one received signal strength in 10bis.4, and the confound with hardware drift in 10bis.5. What this section adds is the demonstration design. A few beams pointing in different directions and a few patterns nulled towards the transmitter, so that the scattered field is not swamped by the direct path, would form the K states. Calibration is what lets a change in $\mathbf{p}$ be attributed to the room.

Candidate tasks, in increasing difficulty: room empty or occupied; a person moving or still; a person in the left, centre or right zone. **The simplest reproducible task is preferable** to an ambitious unreliable one, for reasons specific to a contest: the venue's multipath differs from the laboratory's, visitors stand around the booth, ambient traffic varies, and a judge will believe a binary detection that works every time before a localisation that works sometimes. Ratio features such as $P_k / \sum_j P_j$ would cancel variations of the transmitter's own power, a **[proposed here]** design choice. The classifier should be as simple as the task allows, a threshold or a nearest centroid, with its error rate reported on data recorded on a different day from its training data.

No sensing protocol, classifier or data exists in the repository.

49. Why four elements may be enough

A four element array is not impressive by element count, and nothing in this document claims otherwise. Its value for the contest comes from properties a larger array would make harder to obtain:

Property	Why it serves the demonstrator
controllable states	16 bits, applied on one clock edge
exact enumeration	the best state for any criterion is known, not approximated
full characterisation	every channel measurable alone, the coupling matrix measurable pair by pair
drift experiments	temperature logged, unattended runs, labels from hardware
calibration	6 parameters: identifiable, small, explainable to non-engineers
measured null degradation and recovery	the visible demonstration of what calibration is worth

Why not move to eight elements automatically:

Grows with $N = 8$	From $N = 4$
switches	28 channel switches become 56
divider	a fourth Wilkinson stage, with its loss and area
control	32 beam state lines, against 36 user pins on one DE1-SoC header
calibration dimensionality	6 identifiable parameters become 14
measurement burden	REV at its minimum, 12 readings become 24
board size	an antenna board about twice as long
cost and schedule	more parts, more layout, more bring-up, inside a contest deadline

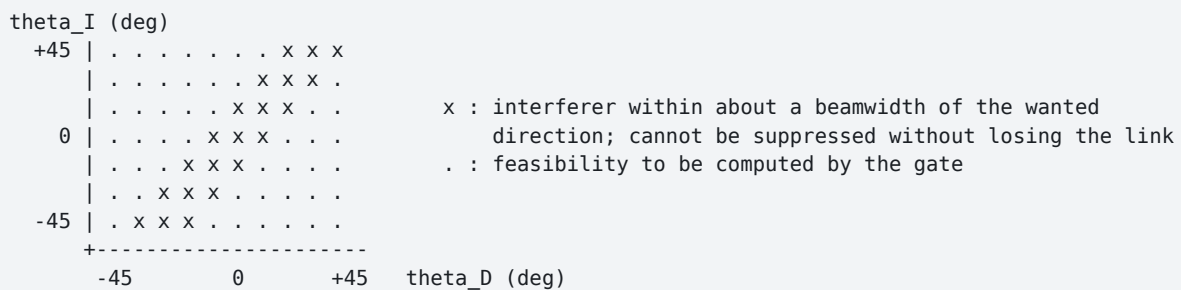
Enumeration itself would survive: 8^7 , about 2.1 million relative states, is still a few seconds of arithmetic against an estimated model, so even at $N = 8$ model based beam synthesis would not need a learned beamformer. What would not survive is the schedule. The rule this document proposes is the one the brief states: **$N = 4$ remains the architecture**

unless a deterministic feasibility study, with criteria written first, shows that a predefined AP-S requirement cannot be met.

50. The $N = 4$ feasibility gate

[proposed here], not run. Its design follows the repository’s practice of writing the rule before the data.

1. **Write the requirement first.** From the contest needs, fix what the communication mode must achieve, for example a minimum interferer suppression with at most a stated loss towards the wanted transmitter, over a stated set of angle pairs (θ_D, θ_I) with a minimum separation, and a stated robustness to phase and amplitude error. The numbers are not proposed here, because choosing them is the decision.
2. **Enumerate.** For every pair on the grid, evaluate all 512 relative states, and the 820 with enable bits, with the ideal model; record the best achievable wanted gain, interferer suppression and objective.
3. **Perturb.** Repeat with the quantisation inherent in the states, with random phase and amplitude errors at decision 0007’s levels, and at plausible drift levels; record how the best state degrades and how often the choice changes.
4. **Map.** Plot the coverage over (θ_D, θ_I) .
5. **Decide.** Retain $N = 4$ unless the predefined requirement fails over the predefined scenario set; if it fails, reopen the element count by a new decision, with the gate’s results as evidence.



Conceptual angle pair coverage map. Only the diagonal band follows from first principles; nothing else has been computed, deliberately, so that the gate’s criteria can be fixed before its results are seen.

51. The dual polarisation option

The official text allows interference to be suppressed by “polarization mismatch” as well as by nulls. A dual polarised element, two orthogonal feeds per patch with switching between them, would add:

Benefit	Explanation
polarisation reconfigurability	the receive polarisation becomes a control
interference rejection	an interferer arriving in a different polarisation from the wanted signal can be rejected without spending spatial degrees of freedom
richer sensing signatures	scattering by people changes polarisation, which adds features

And would cost:

Cost	Explanation
RF switching	a second set of feeds per element, more switches, more loss
antenna complexity	two feeds per patch, cross polar isolation to design and verify
isolation	the two polarisations must not leak into each other
routing	more lines on the antenna board, more connectors or a switch on the antenna board, which conflicts with the passive antenna board of decision 0003
calibration	a polarisation dimension in the array state
simulation	larger HFSS models, possibly beyond the Student limit

Decision 0008 already notes that one polarisation and one plane are judged, and cross polar behaviour is not. **Dual polarisation is therefore a gated option, not current architecture.** It would need its own feasibility evidence and a decision that supersedes parts of decision 0003.

52. Demonstration sequence

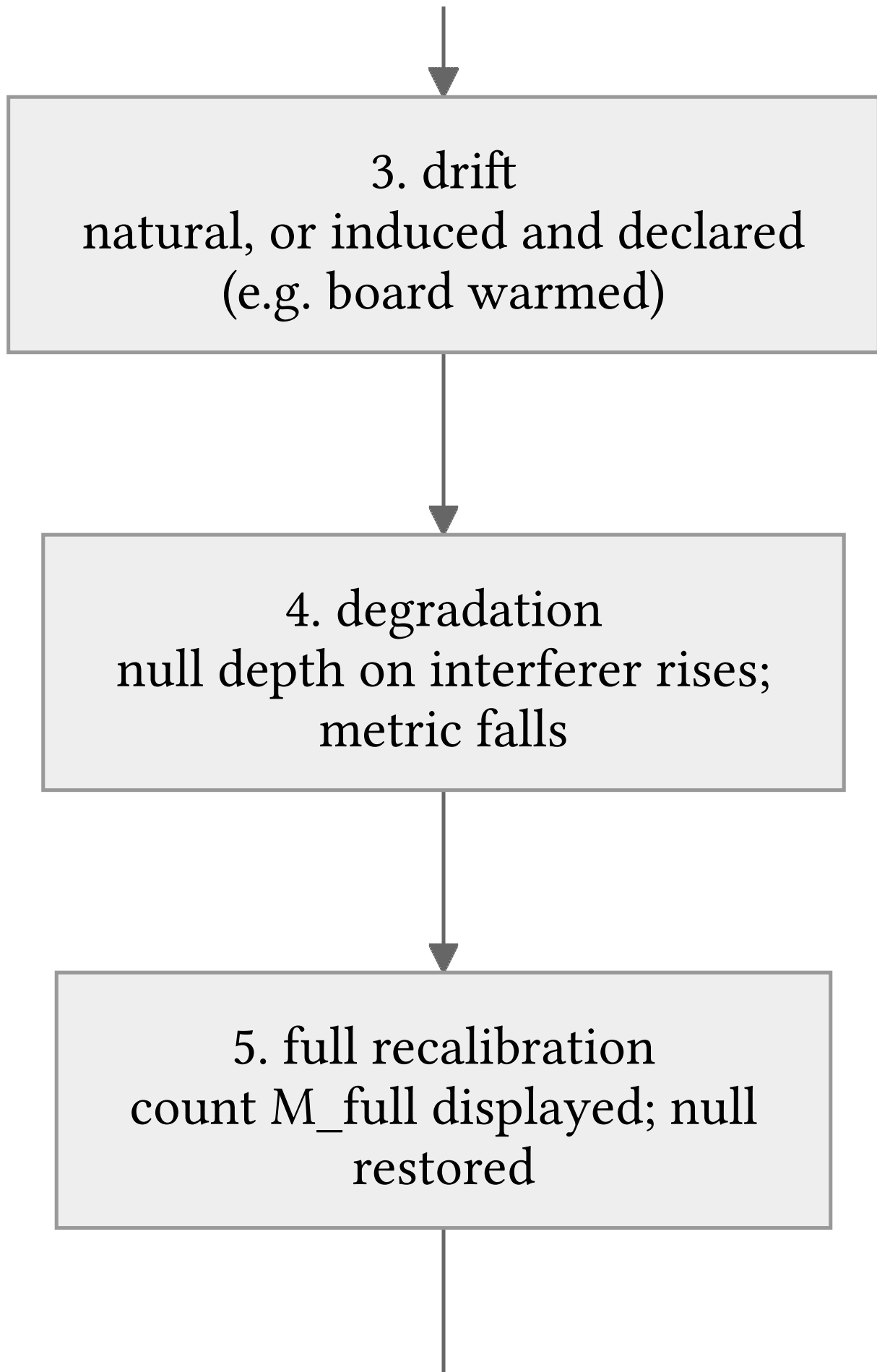


Figure 17: drift, degrade, recalibrate. A narrative, not a measured sequence.

Step	What the audience sees	Honesty condition
1	the calibrated array: estimated pattern, per state powers	the calibration's own measurement count is shown
2	the wanted link held while the interferer is suppressed	the receiver used to separate the sources is named
3	drift	an induced drift is announced as induced; a handling event is not called drift
4	the null filling in, the metric falling	levels are whatever is measured on the day
5	a full recalibration and its count	the full method at its own minimum, not padded
6	a sparse recalibration with the learned prior and its smaller count, if the research supports it	if the prior does not help, the demonstration says so; the comparison is the result
7	sensing on the calibrated array	the task's error rate is stated

The demonstration must stand without the learning extension. Steps 1 to 5 and 7 use only the classical calibration and the exact enumeration; step 6 is the research overlay. If the learned prior is not ready, or does not help, the demonstration still shows an ISAC receiver, its dependence on calibration, and a full recalibration restoring it. The calibration overlay is

```
drift --> COMM and SENSE degrade --> full recalibration baseline (count M_full)
      \--> learned prior sparse recalibration (count M_sparse), research
```

and each recalibration shown must say whether it estimated the hardware state or a room dependent channel (section 10bis.5).

The exact implementation will evolve. No value in this sequence is known today.

Part X. Current project status

Generated entirely from the repository at the baseline. The status words below map onto the repository's own: DONE for a recorded, accepted outcome; READY for a task whose prerequisites are met; IN PROGRESS for partially executed work; BLOCKED for a ready task that failed or waits on an external prerequisite; NOT STARTED; UNRESOLVED for an open question with no route chosen.

Area	Status	Evidence	Blocker	Next action
project architecture	DONE	decisions 0003, 0005, 0009	none	none at architecture level
decision records	DONE, nine accepted	decisions/0001 to 0009	none	a decision on the AP-S direction, not yet written
control architecture	specified, NOT STARTED in hardware	docs/architecture/control-architecture.md	H1 to H5	EXP-005 Phase A, part selection
KiCad schematic	captured, ERC 0 violations, control section superseded	hardware/rev-a/README.md, erc/erc-report.txt	EXP-005 Phase A (R9, H3), H1	re-capture for decision 0005
stack-up	DONE	decision 0009, reva-stackup-r1:6363d8ab0f2b	none	coupons at layout
HFSS automation	implemented, dry run tested in CI, never executed successfully	tools/sim/sim001_hfss.py, results/SIM-001/notes.md	AEDT Student opens no scripting session	open AEDT Student once by hand, rerun step 2
SIM-001	READY, execution BLOCKED; no solver data	experiments/SIM-001-microstrip-50-ohm.md	as above	as above
SIM-002 and later	NOT STARTED; not registered	only SIM-001 exists	SIM-001	register with criteria before data
array simulator, EXP-001	NOT STARTED	experiments/plan.md, "to do"	none	implement; it underpins EXP-002, EXP-012, EXP-013
RF acceptance budget	DONE, provisional-theory-derived	decision 0007, rfsim.thresholds	analyser uncertainty for the simulation against analyser class	EXP-004 O1, O7, then a repeatability measurement
G4 coupling gate	criterion DONE; no data	decision 0008, rfsim.coupling	antenna geometry not designed	SIM-006, then EXP-011 Stage 1
rfsim	implemented; 160 tests pass on synthetic data	tools/rfsim/, pytest at the baseline	no real data yet	first real file: SIM-001 output
VNA audit, EXP-004	IN PROGRESS: 4 complete, 4 partial, 1 not taken	results/EXP-004/README.md	a bench visit	run SCH-001 (READY)
EXP-005 Phase A	READY, NOT STARTED; C1 not run	results/EXP-005/README.md	B1 to B5 confirmations; harness	close B1, build harness, run C1
EXP-005 Phase B	NOT READY	decision 0006; register SCH-003	purchases; decision rules not written	write the rules
purchases	NOT STARTED; nothing bought	decision 0006	O1, O7 for class 2; C1 for the detector	SCH-001
board fabrication	NOT STARTED	decision 0006 gate F1 to F5; F5 half met	F1 to F5	clear the gate

Area	Status	Evidence	Blocker	Next action
antenna design	NOT STARTED; sanity estimates only	decision 0009 patch table	none	SIM-006
FPGA gateware	NOT STARTED	results/EXP-005/README.md P2: Quartus 17.1 installed, no programmer ever attached	none	EXP-005 harness first
detector	part selected (AD8318); not bought, not characterised	decision 0003, V6	C1 with V1 to V4	EXP-005 Phase A
ADC	local converter part not selected; DE1-SoC LTC2308 documented	decision 0005, H3, T6	EXP-005 Phase A	decide H3
machine learning	formalised; NOT STARTED; no data	docs/mathematics/inverse-calibration.md	gate G2, which needs the built array	baselines and simulator first
AP-S proposal	NOT STARTED; not in the repository	none	team, mentor, decision, receiver choice	read the call in full; decide
scaling study, N up to 128	NOT STARTED; not in the repository	none	EXP-001; measured distributions	register it if wanted
national frequency allocation check	NOT STARTED	decision 0004, conditions for reopening	none	a LOCAL lookup
PE4259 datasheet reading	UNRESOLVED	I18, H1, ERC notes	the file is a scanned image	a person reads isolation at 2.44 GHz, thresholds, truth table
licence	UNRESOLVED	LICENSE-NOTES.md	none	a decision before reuse is invited

53. Decision history

Decision, date	Question	Decision	Reason	Consequence
0001, 2026-08-21	hardware first or simulator first?	simulator and hardware in parallel, simulator first	a method can only be validated where the truth is known	the first result is a comparison in simulation; EXP-001 is still to do
0002, 2026-09-17	where can learning reduce measurements at this scale?	a learned drift prior for recalibration , not a first calibration shortcut	at $N = 4$ the classical first calibration has little or no count headroom; recalibration has prior information	the drift experiment becomes central; per element access and temperature telemetry become non retrofittable requirements; gate G2 decides the track
0003, 2026-09-18	which Rev A RF architecture?	two boards, four elements, three switched line bits, phase only, per channel enable, detector and analyser paths	per element access by construction; three bits protect the REV baseline; varactors rejected as a confound	schematic captured; 29 PE4259-63; amplitude control absent
0004, 2026-09-23	which working frequency, on what evidence?	2.44 GHz, band 57a, fixed on direct bench observation	the only licence exempt band below 3 GHz with no duty cycle limit; an observation is the strongest evidence of coverage	gate G1 passes; free space quantities fixed; lengths now wait on the stack-up
0005, 2026-09-23	what drives the array and records the data?	an external DE1-SoC: FPGA for timing, processor for everything else	the measurement needs determinism a microcontroller lacks; state correlated error imitates calibration	registered buffer, local converter as precaution, quiet window; schematic re-capture; Bayesian inverse problem formalised; surrogate pattern synthesis withdrawn
0006, 2026-09-25	when may hardware be bought?	staged in four classes, each gated only by what it depends on	the old rule was circular	finite pre-fabrication gate F1 to F5; drift half of G2 after fabrication
0007, 2026-09-25	what acceptance	derived from the 3-bit quantisation floor with a declared fraction $\eta = 0.10$	no pointing target exists; limits must precede data	2.29 degrees, 0.40 dB, 0.82 dB; simulation against analyser unresolved until U is known

Decision, date	Question	Decision	Reason	Consequence
	limits, before any data?			
0008, 2026-09-26	how is gate G4 decided?	a model adequacy test by full propagation on steered beams	a raw coupling number cannot decide; a calibration residual cannot detect coupling	executable criterion; staged promotion if it fails
0009, 2026-10-03	which stack-ups?	four layer JLC04161H-7628 for the beamformer, two layer 1.6 mm FR-4 for the antennas, calibrated by coupons	physics of the two boards differs by an order of magnitude; RF laminates cost one to two budgets and still need measuring	SIM-001 ready; state dependent loss presses on decision 0007's imbalance allowance

Supersessions recorded in the documents: decision 0003's microcontroller by decision 0005; decision 0003's ordering rule by decision 0006; the "no FPGA" answer to uncertainty I10 by decision 0005; the scheduled surrogate pattern synthesis track of decision 0002's update by decision 0005; the wording of one sentence of decision 0007 by an erratum in decision 0009.

54. What exists physically today

Class	Items	Evidence
owned	three Terasic DE1-SoC boards; one Digilent Zybo; two STM32G0 Nucleo boards; three ESP32 boards; a PC	docs/hardware/inventory-and-needs.md; the DE1-SoC count confirmed
available in the school laboratory, observed	a Rohde and Schwarz ZVL vector network analyser, 9 kHz to 3 GHz, two N female ports, complex S_{21} , source to 0 dBm, USB	[observed] 2026-09-20, results/EXP-004/README.md
historically documented, presence unconfirmed	Agilent N9923A FieldFox, HP 8714C, Agilent N9000A CXA, HP 8562A; 3.5 mm calibration kits belonging to the handheld analyser	[inventory] , docs/hardware/measurement-bench.md section 2.2
unconfirmed	an oscilloscope, a function generator, any software defined radio, any calibration kit or adapter for the ZVL, a ribbon cable of the intended length, a direct voltage source for EXP-005, a room thermometer, any antenna usable as a probe	docs/hardware/inventory-and-needs.md; EXP-005 section 3
installed software	HFSS Student 2025 R2, Quartus 17.1, KiCad (the generator expects version 10) on the project computer; full HFSS and ADS at school	SIM-001 notes; EXP-005 P2; hardware/rev-a/README.md
planned, not bought	40 PE4259-63, an AD8318, two MCP9808, passives, SMA connectors, jumpers, the two Rev A boards, interconnect, antennas	decisions 0003 and 0006
does not exist	any Rev A board, fabricated or assembled; any coupon; any antenna board	this document, Part X

55. What exists in software today

Item	State
rfkit	14 modules, eight command line entry points; 160 tests passing at the baseline on synthetic and analytically constructed data
canonical stack-up	rev-a-stackup-r1, validated by rfkit.stackup, with generated tables in four documents, including this one
SIM-001 builder	tools/sim/sim001_hfss.py, PyAEDT, dry run tested in CI; no successful AEDT execution
SIM-001 analysis	tools/sim/sim001_analyse.py and rfkit.lineparams, tested on an emulated run
error budget	rfkit.budget, the derivation of decision 0007, Monte Carlo checked
coupling study	rfkit.coupling, the executable criterion of decision 0008, and a synthetic chart
schematic generator	hardware/rev-a/tools/generate-schematic.py, producing a schematic with four identical channels by construction
runbook system	tools/runbooks/build.py; one runbook, SCH-001, READY, with its PDF
documentation checks	tools/check-docs.sh
figure script for this document	tools/docs/master_reference_figures.py, three analytical figures
not present	an array simulator (EXP-001), any calibration method implementation (B2 to B6), any gateway, any learning code, any dashboard

56. What has not happened yet

This list exists so that planning maturity is never mistaken for experimental maturity. As of 2026-10-04:

- **No full wave solve has produced a result.** SIM-001's first execution failed before creating geometry.
- **No ADS model exists**, and no circuit model of the switched line channel exists in any tool.
- **No board has been fabricated**, assembled or ordered; no part has been bought.
- **No antenna has been designed.**
- **No Rev A characterisation** on the analyser has taken place; no Touchstone file from any simulator or instrument has been processed by `rfkit`.
- **No calibrated measurement** has been made with the analyser: no calibration kit is confirmed.
- **EXP-005 has not been run**, not even condition C1.
- **No gateway exists**; no programmer has been attached to the project computer.
- **No drift has been measured**; gate G2 is open and cannot be answered before fabrication.
- **No calibration method has been implemented**, classical or learned; the array simulator of EXP-001 does not exist.
- **No learning model exists**, trained or untrained; no dataset exists.
- **No AP-S work exists** in the repository: no decision, no team, no mentor recorded, no receiver chosen, no proposal drafted, no sensing or communication experiment designed.
- **No pointing target, null depth target or sensing accuracy target** has been recorded, so M_{required} cannot yet be evaluated for any method.

Part XI. Validation philosophy

theory	what must be true in principle	sections 1 to 12
analytical estimates	what to expect, and how sensitive it is	seeds, budgets, tables
independent simulation	what the exact geometry gives, two ways	HFSS and a circuit model
fabrication	the real object, with its coupons	Rev A
calibrated measurement	what the object does, at known planes	analyser, after O7
model correction	update material values from coupons	new stack-up revision
repeated experiment	is it reproducible, across sessions	at least 3 sessions

Each level exists because the one above cannot catch a class of error. Theory cannot see a fabrication tolerance; an analytical model cannot see a discontinuity; a simulation cannot see a material value it was given wrongly; a single measurement cannot see its own drift; a single session cannot see what changes between days. The repository's discipline is that a claim may rest on a level only once the levels below it are satisfied, and that the rule for judging each level is written before the data reach it.

57. Acceptance budgets

Decision 0007 derived every acceptance threshold from one anchor and one declared policy number, before any HFSS, ADS or analyser data existed.

The anchor is the quantisation floor of section 8: $\sigma_q = 12.99$ degrees, costing 1.85 degrees of pointing spread at broadside, 0.167 dB of coherent gain and an error sidelobe floor at -18.9 dB.

The policy is $\eta = 0.10$: a validation discrepancy, in its most damaging arrangement, may add at most a tenth of the error variance the floor already imposes:

$$\delta\theta_{\text{worst}}^2 \leq \eta \sigma_{\theta,q}^2, \quad s_\phi^2 + s_a^2 \leq \eta \sigma_q^2 \left(1 - \frac{1}{N}\right)$$

Symbol	Meaning	Unit
$\delta\theta_{\text{worst}}$	beam shift caused by the discrepancy in its worst pattern	rad
$\sigma_{\theta,q}$	pointing standard deviation caused by quantisation	rad
s_ϕ^2, s_a^2	variance across channels of the phase discrepancy and of the relative amplitude discrepancy	rad ² , dimensionless

$\eta = 0.10$ is declared as a policy, not measured: pointing spread may grow by at most 4.9 per cent, gain loss and the error sidelobe floor by at most 10 per cent. Every value scales with $\sqrt{\eta}$, and `python -m rfskit.cli budget` prints them at 0.05, 0.10 and 0.20.

Metric	Comparison class	Value	Derivation	Status
S_{21} phase difference, state dependent part	HFSS against ADS	2.29 deg	$T = \sqrt{\eta} \sigma_q \sqrt{5}/4 = 2.296$	provisional-theory-derived
S_{21} magnitude difference, state dependent part	HFSS against ADS	0.40 dB	$m^2 = \eta \sigma_q^2 (1 - 1/N) - T^2$; $20 \log_{10}(1 + m) = 0.403$	provisional-theory-derived
amplitude imbalance across channels of one state	design	0.82 dB peak to peak	$20 \log_{10} \frac{1+m}{1-m} = 0.826$	provisional-theory-derived
S_{21} phase and magnitude	simulation against analyser	none	needs the analyser's expanded uncertainty U , after O1 and O7	unresolved

Metric	Comparison class	Value	Derivation	Status
S_{11} magnitude difference	both	none	no array level consequence in these units	unresolved
channel phase spread	design	not a limit	it is what calibration removes	not-a-limit
hardware state dependent phase error against nominal	design requirement	2.29 deg at f_0	same derivation; recorded, not wired into a metric	derived requirement

Three features of the derivation are worth understanding.

- **Only the state dependent part is judged.** A discrepancy common to every state of a channel is absorbed by the diagonal array state and costs nothing downstream, so the S_{21} limits apply to state differences, through `compare_states`, not to plain trace differences.
- **The worst pattern, not the typical one.** A comparison of two traces cannot know which steering pattern a discrepancy will meet, so the limit bounds the worst arrangement, 1.79 times stricter than for an independent error of the same rms. Decision 0007 acknowledges this is conservative for proportional discrepancies and names the remedy if it bites: propagate the discrepancy through the array model and judge pointing and gain directly, without moving the budget.
- **Guarded acceptance against the analyser.** For a simulation against a measurement, the observed difference plus the analyser's expanded uncertainty must lie within the limit, so that the measurement's own error is never credited to the model. Until U is known at 2.44 GHz, those limits stay unresolved.

Why the thresholds were set before any discrepancy was seen. A threshold chosen after looking at a discrepancy can always be chosen to pass it. Decision 0007 therefore also fixed how the values may change: only by a new decision record, in the same commit as `rfkit.thresholds`, triggered by a change of a derivation input (bit count, element count, spacing, steering set, η for a stated downstream reason, or a newly measured uncertainty term), and never from the distribution of the discrepancies being judged. The tests fail if a recorded value is not its derivation rounded down.

58. Coupling gate G4

Section 10 explained the physics; this section records the rules of decision 0008 as a validation procedure.

Outcome	Condition, fixed before any data
PASS	the broadside calibration keeps every judged beam inside the budget, the outcome is the same across the last two mesh passes or across every guard matrix, and the far field route has been checked
FAIL	even the best diagonal leaves a judged beam outside the budget, stably, with the route checked
INTERMEDIATE	anything else: the best diagonal holds but the broadside calibration does not find it; the outcome changes across passes or within the uncertainty; the route is unchecked; the two far field routes disagree
UNRESOLVED	the data do not cover band 57a, the matrix is not passive within tolerance, or a required input is missing

The judged set is every frequency point in band 57a, steering from -45 to $+45$ degrees in one degree steps, all eight command origins, with a calibration over the 29 configurations of the rotation family at broadside. For measured data, the guard set is the matrix with every coupling term inflated by U plus 32 reciprocal perturbations of magnitude U from a fixed seed, and the observed reciprocity defect sets a floor under U . Each INTERMEDIATE reason has a named remedy; a FAIL is promoted in stages, a frozen measured coupling matrix first. A further rule waits on EXP-005 Phase B: if the calibration residual exceeds the repeatability floor, a pass becomes intermediate, because the diagonal likelihood is then misspecified at the noise level.

The test is executable, `python -m rfkit.cli g4`, and EXP-011 specifies exactly what each stage must deliver. No coupling data exist.

59. Stack-up sensitivity

59.1 The derivation

A printed line of length L has phase $\varphi = \beta L$ with $\beta = 2\pi f \sqrt{\epsilon_{\text{eff}}}/c$. The length is fixed at layout, so an error in ϵ_{eff} scales every electrical length by the same fraction, exactly:

$$\frac{\delta\varphi}{\varphi} = \sqrt{1 + \frac{\delta\epsilon_{\text{eff}}}{\epsilon_{\text{eff}}}} - 1 \approx \frac{1}{2} \frac{\delta\epsilon_{\text{eff}}}{\epsilon_{\text{eff}}}$$

With the quasi-static model of section 5, $\partial\epsilon_{\text{eff}}/\partial\epsilon_r = (1 + F)/2$, so to first order

$$\delta\varphi \approx \frac{\theta}{2} \cdot \frac{1 + F}{2} \cdot \frac{\delta\epsilon_r}{\epsilon_{\text{eff}}}$$

Symbol	Meaning	Unit
θ	designed electrical length: 45, 90, 180, or 315 for state 7	deg
F	filling factor, about 0.36 on the beamformer construction	dimensionless
$\delta\epsilon_r$	error in the substrate permittivity used at layout	dimensionless

The chain is: small permittivity error, then a proportional error in β , then a phase error proportional to line length. **The longest state is the most exposed:** the 180 degree bit carries four times the error of the 45 degree bit, and state 7, all three bits, seven times. Example from decision 0009: on the beamformer, an error of 0.2 in ϵ_r moves the 315 degree state by about 6.5 degrees in this closed form, against the 2.29 degree requirement; `rfkit.stackup` evaluates the exact line model as well, and the tests hold the two within 10 per cent.

59.2 The tables

Each bound applied alone at the seed width, then the worst of every corner. Phase is the error of a line laid out for the nominal permittivity.

Construction	Tolerance	Status	Z_0 (ohm)	180 degree error (deg)	315 degree error (deg)
beamformer	rf dielectric thickness	assumed	46.9 to 52.8	+0.76 to -0.67	+1.34 to -1.16
beamformer	rf dielectric permittivity	assumed	51.0 to 49.0	-3.62 to +3.55	-6.34 to +6.21
beamformer	rf copper thickness	assumed	50.0 to 49.7	+0.00 to -0.37	+0.00 to -0.65
beamformer	etched width	guaranteed	56.3 to 45.0	-2.10 to +1.78	-3.68 to +3.12
beamformer	worst corner of the bounds above	combined	41.1 to 60.5	6.60	11.56
antenna	rf dielectric thickness	assumed	46.9 to 52.9	+0.70 to -0.60	+1.22 to -1.05
antenna	rf dielectric permittivity	assumed	51.6 to 49.5	-5.57 to +1.82	-9.75 to +3.19
antenna	etched width	guaranteed	56.7 to 44.8	-1.90 to +1.63	-3.33 to +2.86
antenna	rf copper thickness	unbounded	not computed	not computed	not computed
antenna	worst corner of the bounds above	combined	41.4 to 61.6	7.88	13.80

What the beamformer's permittivity bound does to the beam, against decision 0007's pointing budget:

Keeping the 315 degree state within 2.29 degrees needs the beamformer substrate permittivity known to plus or minus 0.073, about 1.7 per cent.

Steering angle (deg)	Pointing budget (deg)	Shift, permittivity bound alone, phase scaled by 2.01 per cent (deg)	Shift, worst corner, phase scaled by 3.67 per cent (deg)
0	0.585	0.000	0.000
15	0.605	0.654	1.191
30	0.675	0.668	1.221
45	0.827	0.194	0.354

Shift is the worst over the eight command origins, from `rfkit.budget.worst_proportional_pointing_deg`; the budget is decision 0007's pointing bound at that angle.

59.3 The 315 degree concern

The error is the same on every channel, so it looks harmless, but it depends on the commanded state, and decision 0007 showed that such a common proportional error is not absorbed by the diagonal array state and steers the beam. At 15 degrees of steering the permittivity bound alone already exceeds the pointing budget, and the worst corner of all bounds roughly doubles it. A beamformer laid out from a datasheet permittivity would therefore fail decision 0007's derived requirement before any modelling error is counted. That is why the effective permittivity must be measured on coupons and the model calibrated to it (section 25), and why the result is an **analytical concern about a design input, not a measured failure**.

59.4 The loss concern

Construction	Conductor loss (dB/m)	Dielectric loss (dB/m)	Extra loss of the 315 degree state, smooth copper (dB)	Same, conductor loss doubled (dB)
beamformer	4.49	5.29	0.59	0.86

Construction	Conductor loss (dB/m)	Dielectric loss (dB/m)	Extra loss of the 315 degree state, smooth copper (dB)	Same, conductor loss doubled (dB)
antenna	0.59	5.60	0.36	0.39

The 315 degree state's extra loss, 0.59 dB with smooth copper and 0.86 dB with conductor loss doubled, approaches or exceeds decision 0007's amplitude imbalance allowance of 0.82 dB **for everything together**, before switch to switch variation, Wilkinson imbalance or connector differences are counted. Dielectric loss alone is about 0.32 dB of it and does not depend on line width. Decision 0009 records this as a known limitation with a reopening trigger: if the coupon attenuation or a simulation of the switched line channel shows the imbalance of one array state exceeding 0.82 dB, a lower loss laminate is reconsidered. It is deterministic and calculable, so it does not corrupt the comparison of models with measurements, but it can fail the design check. **It is an unresolved analytical concern, not a measured failure.**

A further remark, [**proposed here**]: because this loss depends on the state, not on the channel, it is the same for every channel commanded to the same state, and it partly cancels in beams where the channels occupy similar states. Whether the 0.82 dB allowance, which bounds the spread across channels of one array state, is actually exceeded depends on the steering table; a propagation of the state dependent loss through the 512 states would settle it, in the spirit of decision 0007's rule 4.

Part XII. Software and tool table

Only tools the project uses, has decided to use, or has explicitly not chosen are listed.

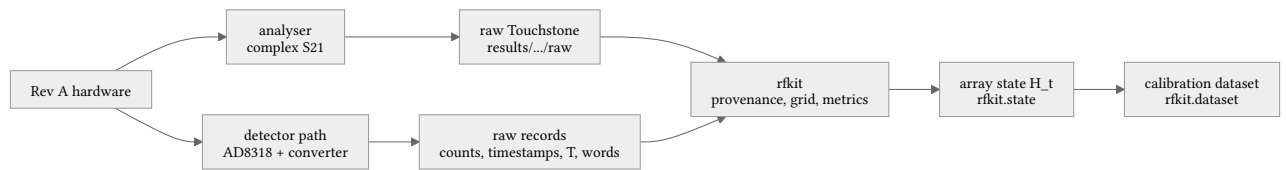
Tool	Role	Why this tool	Input	Output	Alternative	Status, 2026-10-04
Python 3.12	working language of analysis, scripting, tests	one language from solver scripting to inference; free	data, configs	results, figures	MATLAB, listed in the inventory as a cross check	in use; pinned in CI
NumPy, SciPy	arrays, root finding, statistics	the base of scikit-rf; standard	numbers	numbers	none needed	pinned
scikit-rf	common RF data layer: Touchstone, networks, line models	open, tested, reads every source's files	.snp files, line parameters	Network objects, seeds	vendor tools' own analysis	pinned 1.12.0
rfkit	project specific metrics, comparisons, array state, budget, G4, stack-up	encodes the comparison rules once, with tests	traces with provenance, canonical stack-up	verdicts, array states, generated tables	ad hoc notebooks	implemented; 160 tests; synthetic data only
HFSS Student	geometry aware full wave validation, locally	default HFSS path within documented limits	geometry, materials, ports	.s2p, .s4p, fields, patterns	full HFSS; a free full wave solver as a cross check (state of the art O4)	installed 2025 R2; no successful solve
full HFSS	the same, beyond the Student limits	only on escalation	as above	as above	none	at school; not needed so far
PyAEDT	scripted model building from the canonical stack-up	reproducibility, no retyping	canonical stack-up	HFSS projects, exports, sidecars	building by hand	builder written; dry run in CI; not executed successfully
ADS	independent circuit and distributed model	different model form from HFSS, for decision 0007's comparison	line models, component S-parameters	.s2p	scikit-rf media locally	at school, optional; no model exists
KiCad	schematic capture	free; scriptable symbol and sheet files	the generator script	schematic, BOM, ERC report	none considered	schematic captured; ERC clean; version 10 expected
Git and GitHub	version control, timestamps for pre-registration	the commit is the evidence of when a rule was written	everything	history	none	in use
GitHub Actions	continuous checks	runs on every push	repository	pass or fail	none	three workflows: docs, rf, runbooks
Quartus	DE1-SoC FPGA toolchain	the vendor tool for the Cyclone V	gateway sources	bitstream	none for this device	17.1 installed; no gateway; no programmer attached
DE1-SoC FPGA and HPS	deterministic timing and control, not learning; storage and inference on the HPS	decision 0005	sequences	applied states, records	a microcontroller, option A of decision 0005	owned; nothing implemented
R&S ZVL analyser	every real RF measurement: complex S-parameters	the observed instrument; reaches 2.44 GHz with phase	the device, at calibrated planes	Touchstone files	the documented FieldFox, if present	[observed] ; model, kit, options unconfirmed
Touchstone	the bridge between every tool	universal text format	any S-parameter source	.snp	vendor binary formats	the chosen exchange format
matplotlib, reportlab, mistune, pillow, pymupdf	figures; runbook PDFs	deterministic, scriptable	Markdown, data	PDFs, figures	none	pinned; runbook SCH-001 built
scikit-learn, PyTorch	possible learning libraries	not chosen				not dependencies; the model class is chosen on data
Gaussian process or Kalman filter tooling	the drift prior	candidates in section 40				not chosen

Part XIII. Complete data flow

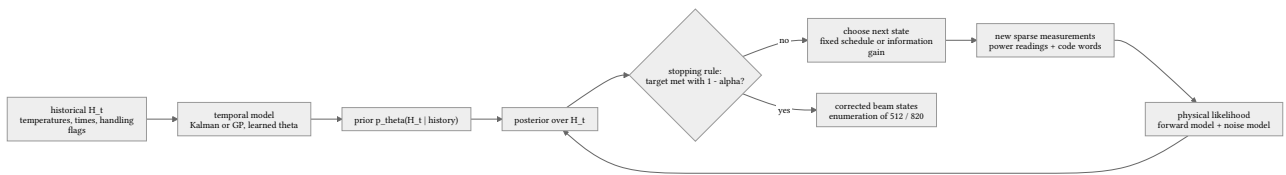
60. Design data flow



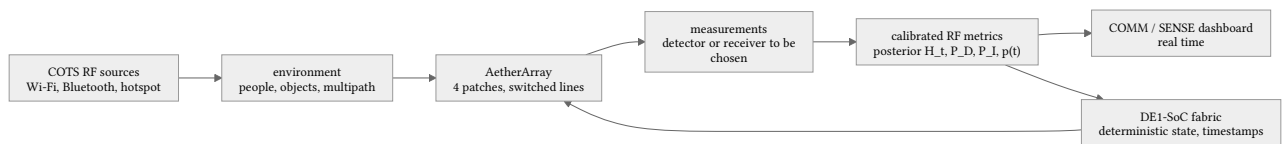
61. Measurement data flow



62. Learning data flow



63. AP-S demonstrator data flow



Part XIV. Why this project is useful

64. Educational usefulness

AetherArray puts, in one object small enough to understand completely, most of what an RF engineer has to learn: electromagnetic waves and transmission lines, microwave circuits and S-parameters, antenna design, phased array theory, PCB stack-up and fabrication tolerance, digital control and FPGA timing, metrology and calibration, signal processing, estimation and Bayesian inference, and the software discipline of reproducible work. Each topic appears because the project needs it, not because a syllabus lists it, and each has a concrete number attached: 8.6 mm for 45 degrees, 2.29 degrees for an acceptance limit, 0.5 mV for a state correlated error. Every decision is written down with its alternatives, so a student can see not only what was chosen but why the obvious alternative was not.

65. Research usefulness

The useful scientific object is not a four element antenna; it is a **controlled physical platform** on which these questions can be studied with ground truth:

- how calibration accuracy trades against the number of physical measurements;
- how a real, cheap RF array drifts with time and temperature, outside a climate chamber;
- how an inverse problem with an explicit forward model behaves on real hardware, power only and complex;
- how measurement design, choosing what to measure next, changes the count;
- whether a learned prior, kept inside a physical inverse problem, earns its place.

66. Industrial relevance, without overclaiming

The same problems appear, at much larger scale and with different hardware, in wireless devices, cellular base stations, satellite terminals, radar, RF production test and adaptive antennas: channels drift, calibration costs time and equipment, and arrays are recalibrated in the field. AetherArray is **not** equivalent to any production system. Industrial arrays use integrated beamformers, active modules with amplifiers, thermal management, factory calibration in chambers and proprietary built in calibration paths, at hundreds or thousands of channels. What carries over is method: the separation of state estimation from beam synthesis, the counting of measurements, pre-registered acceptance rules, and a candid account of which errors calibration can and cannot absorb.

67. Why not simulate everything?

Because a simulator reproduces what it was told (decision 0001, known limitations). Real hardware contains manufacturing tolerances, a laminate from an unspecified brand, glass weave under a 0.37 mm line, connector repeatability, thermal drift, switch nonidealities, measurement noise and coupling paths nobody modelled. The questions this project asks, how a real array drifts and whether its history helps recalibrate it, have no answer inside a simulator, because the drift statistics are exactly what a simulator would have to be given. The physical Rev A exists to supply them.

68. Why not measure everything every time?

Because the cost of measurement is the problem being studied. On four channels a full calibration is a dozen readings; on hundreds of channels, repeated in the field, measurement count and calibration time become expensive in downtime, equipment and energy. If a learned prior can restore the same accuracy with fewer new measurements, that saving recurs at every recalibration for the life of the array. Whether it can, on real hardware, is the experiment.

Part XV. Literature and evidence

XV.1 How the review was done, and its limits

On 2026-10-04 the references already in docs/references/bibliography.md were rechecked and a wider search was made for work on phased array calibration, power only calibration, online and over the air calibration, drift and temperature, Gaussian process and Bayesian calibration, machine learning phase retrieval, experimental design, self calibration and ISAC receiving arrays. **Every direct fetch from a publisher, IEEE Xplore, arXiv, DOI resolvers and bibliographic databases was refused by the network policy of the environment.** Verification is therefore **index level**: authors, titles, venues and abstract wording as they appear in search engine results that index the publisher or preprint pages. No full text was read for this document. A reference marked “index level” in the References may be cited for its existence and its stated claim, and should be read in full before any number from it is used. Where a field could not be confirmed, the References say so.

XV.2 What is already established

Topic	Established result	Sources
power only calibration	rotating element field vector: phase stepping of one element while total power is measured	Mano and Katagi 1982 [A6]; time modulated variant [A20]
self calibration through coupling	adjacent element pairs transmit and receive; no external probe	Aumann, Fenn and Willwerth 1989 [A7]; in situ mutual coupling calibration tracking changes from an initial state on a 64 element panel [L18]
coded calibration	all elements measured at once under orthogonal codes, with variance bounds	Silverstein 1997 [A18]
fast multi element power only methods	about $2N$ power readings with three phase states	Long and co-authors 2017 [L7, L8]; Takahashi and co-authors 2008 [L9]
over the air calibration with complex signals	calibration of a millimetre wave array in beam steering mode from measured complex signals	Wang and co-authors 2021 [L10]
experimental comparison of methods	REV, fast amplitude only and complex methods on one 4 by 8 array; REV and complex comparable, fast amplitude only less accurate	Pan and co-authors 2023 [A22]
learned power only calibration	networks trained on simulated data reduce power readings on a 64 element Ka band array; transfer learning reduces training data	Xie and co-authors 2024 [A21]; Sarayloo and co-authors 2020 [A16]; Zhou and co-authors 2023 [L11]
amplitude only self calibration with learned phase retrieval	on board peak detectors and a network for phase retrieval on a 2 by 4 array at 2.5 GHz, 5.3 degrees average rms phase error	Wu, Syed, Ayling and Hajimiri, IMS 2025 [L13]
phase retrieval theory	$4N - 4$ generic measurements suffice; necessity only in some dimensions; 11 suffice at $N = 4$	[A12], [L1], [L3], [L4], [L5]; stable algorithms [L2], [L39]
online reciprocity calibration in massive MIMO	over the air and coupling based calibration at scale; calibration training spread over time	[L14], [L15], [L16], [L17]
drift and calibration stability	multi day stability of a phased array feed with a calibration update scheme	[L19], authors not confirmed
Gaussian process calibration from sparse data	calibration posed as Bayesian function approximation from near field data	Tambovskiy, Fodor and Tullberg 2023 [A19]
temporal priors with a physical forward model	Kalman filtering of calibration terms; Gaussian process priors on antenna gain time series with learned temporal correlation; both in radio interferometry	Tasse 2014 [L21]; Arras and co-authors 2019 [L22]; Kim and co-authors 2024 [L23]; for antenna arrays, a 2025 preprint infers element errors jointly with channel parameters [L24]
Bayesian experimental design	expected information gain as a design criterion; near optimal greedy selection for Gaussian processes	[L28], [L26], [L27], [L29]
ISAC and Wi-Fi sensing	surveys of integrated sensing and communication; Wi-Fi passive radar; channel state information sensing; phased array processing on commodity Wi-Fi access points	[L32], [L33], [L34], [L35]

Topic	Established result	Sources
array calibration for ISAC	hardware impairments learned end to end in ISAC; calibration of beam patterns for ISAC	[L36]; preprints [L37], [L38]

XV.3 Where AetherArray is similar

- It uses REV, coupling based calibration and orthogonal coding as baselines, exactly as the literature defines them.
- Its sparse recalibration is a Bayesian estimation with an explicit forward model, as in the radio interferometry work [L21, L22, L23].
- Its preference for Gaussian processes or linear Gaussian models on small data follows the precedent of [A19].
- Its use of on board power detection resembles the amplitude only self calibration of [L13], which is at almost the same frequency.

XV.4 Where it differs

- **The prior is temporal and learned from the array's own history**, not a population prior learned from simulated arrays [A21, L11] and not a spatial or weight space model [A19].
- **The forward model is a phased array's**, with a commanded discrete state per measurement, not a radio interferometer's sky model.
- **The figure of merit is the number of new physical measurements at equal accuracy**, against persistence, a non learned temporal baseline and a from scratch baseline at its own minimum.
- **Labels come from hardware**, by an expensive classical calibration run after the cheap one, rather than from a simulator.
- **The scale is deliberately small**, with exact enumeration of beam states, so that beam synthesis error is removed from the comparison.

XV.5 Claims that are not novel

Phased array calibration; power only calibration; REV; coupling based self calibration; orthogonal coding; learned power only calibration; Gaussian process calibration; Kalman filtering of calibration parameters; Bayesian experimental design; Wi-Fi sensing; ISAC as a concept; a reconfigurable receiving array. Each has clear precedent above. The patent US 10,211,527 is close prior art for recalibrating with a subset of excitation patterns on the grounds that drift since the last calibration is limited, though deterministically and without a learned prior [L20], index level.

XV.6 What gap may remain

No paper found in this search combines a learned temporal prior over phased array calibration drift with an explicit physical likelihood to reduce the number of recalibration measurements, measured against non learned temporal baselines on real hardware. Nor was active, information based probe selection for phased array recalibration found. The candidate contribution is therefore the combination, and its experimental test on real hardware.

Potential research differentiation; novelty not yet established.

The search was index level, did not include a full IEEE Xplore keyword search, under-covered paywalled conference proceedings (AP-S/URSI, EuCAP, IMS, radar conferences) and non English journals, and did no citation chasing. Before any novelty is claimed, a systematic review is needed: Xplore and Scholar searches on calibration drift with Kalman filtering, recalibration with a prior, and Bayesian phased array calibration, and forward citation searches from [A19], [L7] and [L13].

Part XVI. IEEE strategy

The IEEE societies relate to the project as sources of mentorship, scrutiny and external validation. Membership in itself is affiliation, not validation.

69. MTT-S

The IEEE Microwave Theory and Technology Society covers microwave and RF hardware, measurement and calibration, the core of layers 1 and 2. Four different things should not be confused:

Thing	What it means
membership	affiliation and access to publications and events
mentoring	an experienced engineer reviewing the work
scholarship selection	a competitive external judgement of a proposal
a conference paper	peer review of a result

The MTT-S undergraduate and pre-graduate scholarships are described, **[snippet only]**, as up to twenty awards a year of US\$2,500, for students with an MTT-S member faculty mentor in microwave or RF, on a two page proposal, with deadlines on 15 April and 15 October [P4]. If accurate, the next deadline falls eleven days after this document's date.

70. AP-S

The IEEE Antennas and Propagation Society covers antennas, arrays and propagation, the subject of layers 1 and 3, and runs the Student Design Contest and, with URSI, the annual symposium. Selection as one of the contest's semi-finalists would be external validation that a jury found the design credible and achievable; a working demonstration at the symposium would be validation of the hardware. AP-S also lists graduate fellowships, doctoral research grants and undergraduate summer research scholarships, **[snippet only]** [P5, P6, P7].

71. AP-S 2027 ambition

Level	Ambition	Condition
primary	submit a technically credible four page preliminary design by 31 December 2026	a team of 2 to 5 students with at least half undergraduates, an AP-S member mentor, a decision recorded, the call read in full
next	be selected among the six semi-finalist teams, if the proposal is competitive	the jury's judgement of achievability, creativity and writing
ultimate	demonstrate the physical system at the symposium in Kyoto, June 2027	Rev A built, calibrated and working, with a receiver and a dashboard, by the final deadline of 24 May 2027

None of these is assured, and this document does not say the project will be selected.

72. What would be meaningful on a CV

Kept short, because this document is technical. In increasing strength of evidence: IEEE student membership; a technical presentation at a student branch or chapter; an MTT-S scholarship; an AP-S Student Design Contest semi-finalist place; a peer reviewed IEEE paper or a published, measured project result. What a hardware or RF reader will look for, beyond any of these, is the work itself: the stack-up decision, the acceptance budget, the pre-registered experiments, and above all measurements that agree, or candidly disagree, with the models.

Part XVII. Risk register

Likelihoods are qualitative, from the evidence named; no numerical probability is supported by data, and none is given.

RF design

Risk	Likelihood	Consequence	Evidence	Mitigation	Reopening gate
state dependent loss exceeds the 0.82 dB imbalance allowance	moderate to high	the design check of decision 0007 fails	0.59 to 0.86 dB for state 7 alone [analytical] , decision 0009	coupon attenuation; channel simulation; propagate through the 512 states (section 59.4)	decision 0009: consider FR408HR (B3) or the hybrid (B4)
phase error from permittivity uncertainty	high before coupons	pointing budget exceeded at some angles	± 0.2 bound gives about 6.3 degrees on state 7 [analytical]	measure ε_{eff} on coupons; calibrate the model	coupons outside the bound reopen decision 0009
channel to channel permittivity from the 7628 glass weave	unknown	a per channel error coupons cannot remove	no source quantifies it (I25)	identical layout; measure per channel state phases (SCH-006)	channels differing by more than the coupon uncertainty reopen decision 0009
switch isolation at 2.44 GHz low	moderate	per channel measurements at the common port corrupted; B2 baseline degraded	20 dB typical at 2 GHz, nothing at 2.44 GHz (I18)	read the datasheet curve; prefer conducted labels at element ports	none; affects one baseline
resonance of the de-selected switched line arm	unknown	a sharp state dependent loss and phase excursion in band	standard switched line hazard [to verify]	choose reference arm length; simulate (SIM-003)	redesign of the 180 degree bit
patch detuning	moderate to high on the first board	the antenna misses band 57a	resonance shift over the permittivity bound exceeds the estimated bandwidth [analytical]	coupons on the antenna board; expect a second order	two misses: RO4350B antenna board (B2)
coupling makes the diagonal model inadequate	unknown	G4 fails; counts reopen	synthetic chart only	staged promotion: frozen coupling matrix first	decision 0008 outcomes
switch state repeatability above the floor	unknown	the drift experiment measures the switches	no measurement possible before fabrication (E6)	measure on the built board before any drift claim	replace the phase control approach (decision 0003)

Digital and interface

Risk	Likelihood	Consequence	Evidence	Mitigation	Reopening gate
logic level incompatibility	unknown	switches misread commands	thresholds unread (H1)	read datasheet; fallback level translator plus bench threshold measurement	blocks board release
state correlated noise from the control path	unknown	a fake calibration coefficient	mechanism plausible, magnitude unmeasured	quiet window, local converter, EXP-005 Phase A	R9 escalation reopens decision 0005
header cannot supply the board	unknown	separate supply, larger ground question	current limit unread (H2)	read the limit	H5 grows
ground loop and EMI over the ribbon	unknown	slow offsets, state correlated pickup	none measured (H5)	interleaved grounds; C5 strap test	H5 becomes a measured requirement
gateway effort dominates the schedule	moderate	delays every experiment	no gateway experience recorded in the repository	phased harness; fallback to a simpler controller for the first campaign	decision 0005 conditions

Measurement

Risk	Likelihood	Consequence	Evidence	Mitigation	Reopening gate
no calibration kit or adapters for the analyser	moderate	no calibrated measurement at the SMA plane	nothing confirmed (O7)	SCH-001; class 2 interconnect; TRL coupon fallback	conducted only fallback (decision 0004)
analyser model specified below 2.44 GHz	low	the frequency reopens	the panel read 3 GHz; model unread (O1)	read the label	decision 0004 reopens immediately
automation unproven	moderate	no unattended analyser runs	nothing ever enumerated (O9)	detector path for unattended work	a dedicated source, class 4

Risk	Likelihood	Consequence	Evidence	Mitigation	Reopening gate
reference plane errors	moderate	wrong labels, wrong G4 inputs	the reference plane is a recurring concern in EXP-011	coupons C3; explicit planes in every file	none
room reflections dominate radiated readings	unknown	patterns and sensing confounded	EXP-005 Phase B not run (I3)	conducted measurement backbone; time gating if K3 installed; absorbers	acoustic route or conducted only (docs/uncertainties.md)
shared analyser access	moderate	scheduling delays	shared with another project (docs/shared-resources.md)	alternate by phase	none

Machine learning

Risk	Likelihood	Consequence	Evidence	Mitigation	Reopening gate
drift too small to learn	unknown	the learning track does not exist	G2 unanswered	answer G2 honestly before modelling	decision 0002 superseded
drift unpredictable, dominated by handling	unknown	no prior helps	I14 open	log handling; model jumps separately	decision 0002
dataset too small	moderate	weak or seed dependent results	about 100 sessions a week, unverified	simple models; long unattended runs	none
no count advantage over the non learned temporal baseline	moderate	the learned contribution is null	baseline C may capture most of the saving	report it as the result	none; a negative result is reported
temporal leakage in evaluation	low if the protocol is followed	optimistic results	standard pitfall	chronological and rolling origin splits	none
overconfident prior stops early on a wrong answer	moderate	silent calibration error	section 39.2	coverage tests; residual check with fallback	none

AP-S demonstrator

Risk	Likelihood	Consequence	Evidence	Mitigation	Reopening gate
four elements cannot meet a suppression requirement robustly	unknown	communication mode weak	feasibility not studied	N = 4 feasibility gate with criteria first	element count reopened by decision
sensing not reproducible at the venue	moderate to high	sensing mode fails on site	venue multipath differs; visitors near the booth	simplest task; ratio features; recalibration on site	choose a simpler task
receiver architecture versus the rules	unknown	a disallowed or missing source separation	SDR and receiver rules not found in excerpts	read the call in full; choose a receiver early	design change before the proposal
schedule	high	no working system by 24 May 2027	nothing fabricated on 2026-10-04; gates F1 to F5 open	propose only what can be built; parallelise	descope to a credible subset
team and mentor not in place	unknown	no eligible submission	not recorded in the repository	recruit; record	none
contest facts wrong	low to moderate	a non compliant proposal	snippet level only	read the official call in full	correct Part IX

Project

Risk	Likelihood	Consequence	Evidence	Mitigation	Reopening gate
HFSS Student does not run on the project computer	moderate	the local simulation path is blocked	SIM-001 failed to open a session twice; low free memory recorded	interactive first launch; free memory; school full HFSS if needed	EITHER escalation
fabrication delays, second antenna board	moderate	schedule slip	decision 0009 expects a possible second antenna order	order early once the gate clears	none
school laboratory availability	unknown	bench work delayed	access terms not recorded (SCH-008)	runbooks make visits short	dedicated source purchase
budget, 50 to 70 EUR rule	moderate	descopeing pressure	BOM about 62 EUR before connectors	neutral levers only; never shrink bits or elements silently	decision 0003

Part XVIII. Roadmap

From 2026-10-04, ordered by the repository’s dependencies; dates are given only where an external deadline exists. Where a task runs follows the repository’s classes.

Phase	Objective	Prerequisites	Output	Gate	Where
0a	finish the analyser audit	none	O1, O6 to O9 recorded	unblocks class 2 purchases; F1	SCHOOL-BENCH, SCH-001 READY
0b	unlock HFSS Student	AEDT Student opened once by hand; memory freed	a working scripted session	none	LOCAL
0c	EXP-005 Phase A, condition C1 first	B1 to B5 confirmed; harness	V1 to V4, the floor; then C2 to C5	R9, H3; detector purchase; F2	LOCAL
0d	read the PE4259 datasheet by hand	the file	isolation at 2.44 GHz, input thresholds, truth table	H1 inputs; I18	LOCAL
0e	decide the AP-S direction	the official call read in full; a team and a mentor	a decision record; a proposal plan	the 31 December 2026 deadline	LOCAL
0f	build the array simulator, EXP-001, then EXP-002 and EXP-012	none	the virtual bench; measured counts for the baselines, including the fast amplitude only method	needed before any learning claim	LOCAL
1	SIM-001, a genuine HFSS run	0b	W_{50} , ε_{eff} , attenuation, with provenance	the SIM-001 criterion	LOCAL
2	phase line and Wilkinson simulations, SIM-002	1	arm geometries	criteria registered first	LOCAL
3	switched line bit and eight states, SIM-003 and SIM-004	2; a switch model	state dependent phase and loss	decision 0007 comparison	EITHER
4	divider, SIM-005	1	balance, isolation, match	criterion registered first	LOCAL
5	antenna, SIM-006	antenna construction	patch geometry	criterion registered first	LOCAL
6	coupling and array, EXP-011 Stage 1 and SIM-008	5	S_A , patterns, G4 verdict; system model	decision 0008	EITHER
7	schematic re-capture with H1 to H5 closed or decided	0c, 0d; part choices	a schematic matching decision 0005	F4	LOCAL
8	EXP-005 Phase B with class 2 purchases	0a, 0c; Phase B rules written	the RF repeatability floor on both routes	F3	SCHOOL-BENCH, SCH-003
9	layout and fabrication	1 to 7 for lengths and widths; F1 to F5	Rev A boards with coupons	F1 to F5	LOCAL, then the fabricator
10	coupon and VNA characterisation	9; the O7 calibration chain	measured ε_{eff} , attenuation, launches; stack-up revision	SCH-012	SCHOOL-BENCH
11	full Rev A characterisation	10	per state phases and losses, divider, reflections, labels, detector; E6	SCH-006; EXP-011 Stage 2	SCHOOL-BENCH
12	radiated experiments	11; Phase B floor	EXP-006 to EXP-009, including the known cable error EXP-007	lifts decision 0007’s provisional status if EXP-007 agrees	SCHOOL-BENCH, SCH-007
13	drift dataset	11; gateway; unattended rig	EXP-010 and EXP-014 data	G2	EITHER
14	simple temporal baselines	13	baselines A to D	none	LOCAL
15	learned prior, EXP-015	13, 14	M_{required} curves against baselines	failure declared by the pre-registered rule	LOCAL
16	active measurement selection	15	information gain schedule against fixed schedules	none	LOCAL
17	AP-S COMM and SENSE demonstrator	11 at least; a receiver; gateway; a dashboard	the contest system	the contest’s final materials, 24 May 2027	EITHER

A plain reading of the calendar. The contest’s preliminary design is due on 31 December 2026, about three months from now. A credible proposal can be written from the design, the decisions, the simulations done by then and a clear plan; it cannot honestly report a learned recalibration result. The final materials are due on 24 May 2027 and must include measurement results. Reaching a working demonstrator by then requires fabrication early in 2027, which in turn requires phases 0 to 9 to clear in the coming months. If that is not achievable, a demonstrator whose calibration is classical, with the learned prior presented as ongoing research, would still be truthful.

Part XIX. Glossary

Term	Definition
ADC	analogue to digital converter: turns a voltage into a number
ADS	Keysight Advanced Design System, a circuit and system RF simulator
AEDT	Ansys Electronics Desktop, the environment that hosts HFSS
AP-S	IEEE Antennas and Propagation Society
array factor	the coherent sum of element contributions for given weights, as a function of direction
array state $\mathbf{H}_t$	the complex errors that turn commanded weights into realised weights at time t
attenuation constant α	rate of amplitude decay along a line, Np/m or dB/m
beamforming	combining element signals with weights to form a directional pattern
beamwidth, half power	the angle between the two directions where the beam is 3 dB below its peak
Bayesian experimental design	choosing measurements to maximise expected information about unknowns
Bayesian optimisation	searching for the input that maximises an expensive function, with a probabilistic surrogate
broadside	the direction normal to the array
calibration	estimating the array state so that commands can be corrected
common mode error	an error identical on all channels; unobservable in beam shape
coupling, mutual	the influence of one element or channel on another
coupon	a test structure fabricated beside a circuit to characterise the board
dB	ten times the base 10 logarithm of a power ratio, or twenty times that of an amplitude ratio
dBi	antenna gain relative to an isotropic radiator
dBm	power relative to 1 mW in decibels
DE1-SoC	Terasic board with an Intel Cyclone V system on chip: FPGA fabric plus an ARM processor
de-embedding	mathematically removing a characterised fixture from a measurement
directivity	radiation intensity in a direction relative to its average over all directions
drift	slow change of the array state over time
effective permittivity ϵ_{eff}	the permittivity of a uniform medium that would give a microstrip's propagation constant
embedded element pattern	the pattern of one element excited inside the array, the others terminated
enable switch	the per channel switch that passes the signal or terminates the channel in 50 ohm
ENIG	electroless nickel immersion gold, a surface finish
ERC	electrical rule check of a schematic
far field	the region where the pattern no longer depends on distance; beyond about $2D^2/\lambda$
FPGA	field programmable gate array: configurable digital logic with deterministic timing
FR-4	glass reinforced epoxy laminate, the standard PCB material
G1 to G5	the project's gates: phase measurability, drift above floor, element and bit count, coupling adequacy, budget
Gaussian process (GP)	a probability distribution over functions, defined by a mean and a covariance kernel
grating lobe	an unwanted full strength lobe caused by spacing that is too large
guided wavelength λ_g	the wavelength inside a transmission line, $\lambda_0/\sqrt{\epsilon_{\text{eff}}}$
HFSS	Ansys High Frequency Structure Simulator, a finite element full wave solver
HPS	hard processor system: the ARM processor inside the Cyclone V SoC
identifiability	whether parameters can in principle be determined from the available measurements
illuminator of opportunity	a transmitter operating for its own purpose whose field a separate receiver uses for sensing
insertion loss	the loss a component adds in a matched system, $-20 \log_{10} S_{21} $
ISAC	integrated sensing and communication: systems sharing spectrum, waveform, hardware, aperture or field between communication and sensing; three families in section 10bis.2
isolation	attenuation of a switch between its common port and its off throw
Kalman filter	the recursive Bayesian estimator for linear Gaussian state space models
likelihood	probability of the observed data given the parameters; here, the physical forward model
microstrip	a strip conductor over a grounded dielectric
multipath	the arrival of one transmitter's signal at the receiver by several propagation paths
passive bistatic radar	sensing with a receiver separated from a transmitter it does not control

Term	Definition
MTT-S	IEEE Microwave Theory and Technology Society
null	a direction of near total cancellation in a pattern
PE4259-63	the pSemi SPDT RF switch used throughout Rev A
phase constant β	phase accumulated per unit length, rad/m
phase retrieval	recovering a complex vector from magnitude measurements
phase shifter	a device that changes the phase of a signal; here, three switched line bits per channel
phasor	a complex number representing the amplitude and phase of a sinusoid
posterior	the probability distribution of the parameters after the data
prior	the probability distribution of the parameters before the new data
quantisation	rounding a continuous command to the nearest reachable state
quiet window	the interval from strobe to end of conversion in which no control line or I2C line changes
reciprocity	the property $S_{ij} = S_{ji}$ of passive networks without nonreciprocal materials
reference plane	the plane at which a calibrated measurement refers
return loss	$-20 \log_{10} \Gamma $, how far below the incident wave a reflection is
REV	rotating element electric field vector, the classical power only calibration method
RF	radio frequency
S-parameters	scattering parameters: ratios of outgoing to incoming waves at a network's ports
S11	reflection at port 1 with the other ports matched
S21	transmission from port 1 to port 2 with the other ports matched
SIM-NNN	a simulation task that fixes a design input
SINR, SIR	signal to interference plus noise ratio; signal to interference ratio
SMA	a threaded coaxial connector family used on the boards
spatial covariance $\mathbf{R}$	the 4×4 matrix of correlations between element signals; all power measurements through any pattern are linear in it
SPDT	single pole double throw switch
stack-up	the layer construction of a printed circuit board
state correlated error	a measurement error that depends on the commanded beam state, and so imitates a calibration coefficient
steering vector	the vector of element phases a plane wave from a given direction produces
switched line phase shifter	a phase shifter selecting between printed lines of different length
Touchstone	the text file format for S-parameters, .sNp
TRL	thru reflect line calibration
VNA	vector network analyser: measures complex S-parameters
VSWR	voltage standing wave ratio, $(1 + \Gamma)/(1 - \Gamma)$
Wilkinson divider	a matched, isolated power divider with quarter wave arms and an isolation resistor

Part XX. What the project is and is not

AetherArray is	AetherArray is not
a real RF hardware platform, designed and partly captured	a 128 element physical array
a calibration testbed with ground truth and logged drift	an AI beamformer: beam synthesis is exact enumeration
a four element phased array with three bit phase control	a neural network antenna
a hybrid problem: explicit physical likelihood, learned temporal prior	a replacement for HFSS, ADS or the analyser
a potential AP-S ISAC demonstrator, proposed and not yet decided: shared aperture plus opportunistic sensing	a production antenna for any company's product
a narrow, receive only member of ISAC families B and C	a joint waveform ISAC system, a monostatic or range and Doppler radar, or a localisation system
a public lab notebook with pre-registered rules	a claim that four elements prove large array behaviour
a platform that will report a negative result if the prior does not help	a completed experimental result: nothing is built or measured

Part XXI. One page synthesis

- 1. What are we building?** A four element, 2.44 GHz phased array with three bit switched line phase control: a passive antenna board, a beamformer board with 29 RF switches, a Wilkinson combiner, an analyser port and an on board power detector, driven by an external DE1-SoC whose FPGA applies beam states deterministically. It is designed, its schematic is captured, and nothing is built.
- 2. What scientific question are we asking?** Whether a prior learned from the array's own drift history lets it be recalibrated with fewer new physical measurements than calibrating from scratch, at the same final accuracy. The figure of merit is M_{required} , a count of measurements.
- 3. Why four elements?** Because four is the smallest array where the learning question has room (two elements leave two parameters), because every state can be enumerated, and because it fits the budget. It is a research instrument, not a demonstration of scale; scaling to larger arrays would be studied in simulation, parameterised by what four channels measure.
- 4. Why an FPGA?** Not to form beams, which an analogue array does without one, but to make the measurement deterministic: all sixteen lines on one clock edge, a fixed settling delay, control lines silent while sampling, one clock for every timestamp. A state correlated error would otherwise imitate a calibration coefficient.
- 5. Why HFSS, ADS and a VNA?** Because each catches errors the others cannot: HFSS solves the real geometry, a circuit model provides an independent model form, and the analyser measures reality at calibrated planes. Their agreement, judged by rules fixed before the data, is the evidence.
- 6. Why machine learning?** Only for what physics cannot supply: the temporal structure of this array's drift. The likelihood stays physical; beam selection stays exact. At $N = 4$, learning cannot honestly reduce first calibration counts, so the claim is restricted to recalibration.
- 7. What does the AP-S demonstrator add?** A narrow but genuine form of ISAC, one receive aperture shared between interference rejection and opportunistic sensing (section 10bis), in which calibration's value is visible to anyone: a null on an interferer that fills when the hardware drifts and returns when the array is recalibrated, with the measurement count shown, plus sensing that is only trustworthy on calibrated hardware. It tests the idea in public; it does not change the research question.
- 8. What has actually been completed?** Nine decisions; the stack-up; the schematic of the RF design; the acceptance budget and the coupling test, both written before any data; the RF data layer and its 160 tests on synthetic data; the SIM-001 model builder and analysis; one school runbook; the frequency fixed from a bench observation. No simulation result, no hardware, no measurement, no model.
- 9. What is the immediate next task?** In parallel: open AEDT Student once by hand and run SIM-001; make the SCH-001 bench visit; run condition C1 of EXP-005; read the PE4259 datasheet by hand; build the array simulator EXP-001; and, if the contest is pursued, read the official call in full and record a decision.
- 10. What result would make AetherArray scientifically successful?** A measured curve, over held out sessions on real hardware, showing that the learned prior reaches the same final accuracy as a full recalibration with measurably fewer new measurements than both the from scratch baseline and the non learned temporal baseline; or, equally valuable, a clean demonstration that it does not, with the reason. Along the way: a measured drift and calibration lifetime for a low cost array outside a climate chamber, and simulated against measured coupling, both of which the literature rarely reports.

Appendices

Appendix A. Rev A parameters

Decided parameters, each from its record:

Parameter	Value	Source
element count	4	decision 0003
working frequency	2.44 GHz, band 57a, 2400 to 2483.5 MHz	decision 0004
free space wavelength	122.9 mm	from f_0
element spacing	$\lambda_0/2$, 61.4 mm	decisions 0003, 0004
aperture across element centres	184 mm	EXP-004
far field distance	0.55 m	EXP-004
phase bits	45, 90, 180 degrees, eight states	decision 0003
relative beam states	512; 820 with enable bits	section 8.3
amplitude control	none; enable or terminate per channel	decision 0003
RF switches	29 PE4259-63; 28 in the channels, 1 path selector	hardware/rev-a/README.md
divider	four way Wilkinson, three stages, 70.7 ohm arms, 100 ohm resistors	hardware/rev-a/layout-constraints.md
detector	AD8318, -25 mV/dB nominal	decision 0003, V6
temperature sensors	two MCP9808 at 0x18 and 0x19, plus the AD8318 die output	decision 0003
controller	external DE1-SoC	decision 0005
beam state word	16 bits, $b = 4c + f$	docs/architecture/control-architecture.md
stack-up	rev-a-stackup-r1	decision 0009
acceptance policy	$\eta = 0.10$; $\eta_c = 0.10$ for coupling	decisions 0007, 0008

Derived dimensions, generated, **not layout values**:

INITIALISATION ONLY. $W_{\text{seed}} \neq W_{50}$. SIM-001 decides the width; every length below is a feasibility estimate, not a layout value.

Construction	Quantity	Value
beamformer	W_{seed} for 50 ohm	0.372 mm
beamformer	closed form check, zero thickness	0.402 mm
beamformer	ϵ_{eff} at f_0	3.191
beamformer	λ_0	122.9 mm
beamformer	λ_g	68.8 mm
beamformer	line for 45 degrees	8.6 mm
beamformer	line for 90 degrees	17.2 mm
beamformer	line for 180 degrees	34.4 mm
beamformer	line for 315 degrees	60.2 mm
beamformer	W_{seed} for 70.7 ohm, Wilkinson arms	0.182 mm
beamformer	width flags	none
antenna	W_{seed} for 50 ohm	2.836 mm
antenna	closed form check, zero thickness	2.876 mm
antenna	ϵ_{eff} at f_0	3.415
antenna	λ_0	122.9 mm

Construction	Quantity	Value
antenna	λ_g	66.5 mm
antenna	line for 45 degrees	8.3 mm
antenna	line for 90 degrees	16.6 mm
antenna	line for 180 degrees	33.2 mm
antenna	line for 315 degrees	58.2 mm
antenna	width flags	none

SIM-001 inputs, generated:

Generated from reva-stackup-r1:6363d8ab0f2b, construction beamformer. W_{seed} is not the final width.

Variable	Value	Origin
sub_h	0.2104 mm	PP1 thickness
cu_t	0.035 mm	L1 finished copper
w_seed	0.372 mm	50 ohm seed, scikit-rf model
l_short	10 mm	fixed choice
l_long	27.2 mm	short line plus a quarter guided wavelength at f_0
port_w	3.72 mm	the larger of ten widths and the width plus ten heights
port_h	2.104 mm	ten substrate heights
sub_er	4.4	PP1 permittivity, nominal
sub_tand	0.015	PP1 loss tangent
cu_sigma	5.8e+07 S/m	copper, assumed
width sweep	0.3348, 0.372, 0.4092 mm	the seed scaled by 0.9, 1.0 and 1.1

Solution frequency 2.44 GHz; sweep 1 to 3 GHz in 5 MHz steps; maximum change in S 0.02 over 2 consecutive passes, at most 20 passes; ports renormalised to 50 ohm.

Appendix B. Stack-up

The canonical definition is hardware/rev-a/stackup/rev-a-stackup.json, identifier reva-stackup-r1, fingerprint reva-stackup-r1:6363d8ab0f2b at the baseline. Its generated tables appear in this document in section 21 (construction and tolerances), section 22 (patch sanity check), section 23 (comparison with the candidates), section 59 (sensitivity, pointing and loss) and Appendix A (seeds and SIM-001 inputs). The alternatives compared are in hardware/rev-a/stackup/candidates.json. How each tool reads the file is in hardware/rev-a/stackup/README.md section 6. To regenerate every table: `cd tools && python -m rfkit.cli stackup --write-docs.`

Appendix C. RF acceptance budget

Item	Value	Record
quantisation floor	12.99 degrees rms; 1.85 degrees pointing at broadside; 0.167 dB gain; -18.9 dB error sidelobe floor	decision 0007
policy fraction	$\eta = 0.10$, declared	decision 0007
pointing budget at broadside	0.585 degrees	python -m rfkit.cli budget
HFSS against ADS, S_{21} phase, state dependent	2.29 degrees	decision 0007
HFSS against ADS, S_{21} magnitude, state dependent	0.40 dB	decision 0007
amplitude imbalance, one array state	0.82 dB peak to peak	decision 0007
hardware state dependent phase against nominal	2.29 degrees at f_0 , a derived requirement, not wired	decision 0007
simulation against analyser	unresolved until the analyser's U is known	decision 0007
solver convergence for a comparison	$\Delta S \leq S_{21} T$, about $0.04 S_{21} $	decision 0007
coupling allocation	$\eta_c = 0.10$, separate; together at most 0.20 of the floor variance	decision 0008
coupling screen, descriptive only	$\rho \leq 0.040$: about -28 dB aggregate, -34 dB per neighbour	decision 0008
how values may change	only by a new decision with a derivation input change; never from the judged discrepancies	decision 0007

Appendix D. Open hardware items, gates and requirements

Open items, docs/architecture/control-architecture.md section 8. H1 logic level compatibility, blocks board release; H2 expansion header supply capability; H3 converter part, decided by EXP-005; H4 buffer part; H5 ground strategy between the boards. All open.

Pre-fabrication gate, decision 0006. F1 EXP-004 closed, O1 checked against the model's datasheet, O7 calibration chain settled: open. F2 EXP-005 Phase A complete with R9 and H3 applied: open. F3 EXP-005 Phase B complete on both routes, by rules written before data: open, rules not written. F4 schematic re-captured, H1 to H4 closed, H5 decided: open. F5 stack-up chosen and line lengths derived: half met, the stack-up is chosen.

Research gates. G1 phase measurable: closed, passed (decision 0004). G2 drift above the repeatability floor: open; floor before fabrication (F3), drift after it (EXP-010). G3 element and bit count: closed, four and three (decision 0003). G4 coupling model adequacy: criterion fixed (decision 0008), no data. G5 budget: at order time.

Requirements, docs/hardware/rev-a-requirements.md. R1 per element connectors; R2 drive one element, receive on another; R3 scalar power sense at the sum port, digitised on the board; R4 temperature sensors near the phase network and the detector; R5 commanded word recorded; R6 at least four elements; R7 a documented reference channel; R8 timestamped logging with the fixed schema; R9 deterministic application and quiet window, adopted as a precaution and tested by EXP-005. R1, R2, R4, R6, R7 and R9 are not retrofittable.

Appendix E. Simulation list

ID	Title	Status	Where
SIM-001	50 ohm microstrip on the beamformer stack-up	registered; READY; execution blocked; no data	LOCAL
SIM-002	Wilkinson line, bends, meandered arms	proposed here	LOCAL
SIM-003	one switched line bit with a switch model	proposed here	EITHER
SIM-004	enable plus three bits, eight states	proposed here	EITHER
SIM-005	four way Wilkinson and path selector	proposed here	LOCAL
SIM-006	single patch	proposed here	LOCAL
SIM-007	four element coupling and embedded patterns	registered as EXP-011 Stage 1; criterion fixed	EITHER
SIM-008	system model by cascade	proposed here	LOCAL

Appendix F. Experiment list

ID	Title	Status	Where
EXP-001	array simulator with injected defects	to do	LOCAL
EXP-002	calibration methods compared in simulation	to do	LOCAL
EXP-003	sensitivity to noise and measurement count	to do	LOCAL
EXP-004	instrument audit and working frequency	running; frequency closed; 4 complete, 4 partial, 1 not taken	SCH00L-BENCH, SCH-001
EXP-005	repeatability floor and control path	planned; Phase A ready, not run; Phase B gated	LOCAL, SCH00L-BENCH
EXP-006	two element array	to do	SCH00L-BENCH
EXP-007	known cable error	to do	SCH00L-BENCH
EXP-008	first real calibration	to do	SCH00L-BENCH
EXP-009	four elements	to do	SCH00L-BENCH
EXP-010	calibration validity over time	to do; gates the learning track	EITHER
EXP-011	coupling and gate G4	planned; criterion fixed; no data	EITHER, SCH00L-BENCH
EXP-012	measurement count against classical baselines	to do	LOCAL
EXP-013	learned first calibration estimator, a control	to do	LOCAL
EXP-014	unattended recalibration rig and schema	to do	EITHER
EXP-015	learned drift prior against from scratch, the central claim	to do; needs G2	LOCAL

Appendix G. Decision index

ID	Title	Status
0001	build the simulator before the array	accepted
0002	learning enters as a drift prior, not as a first calibration shortcut	accepted, four gates named
0003	Rev A is a two board, phase only, switched line array at three bits	accepted; control path superseded by 0005
0004	the working frequency is 2.44 GHz	accepted
0005	the Rev A controller is an external DE1-SoC	accepted; re-capture required
0006	purchases are staged by what they unblock	accepted
0007	a provisional RF acceptance budget from the quantisation floor	accepted; values provisional-theory-derived
0008	gate G4: the diagonal state stays only if neglecting coupling costs less than the budget	accepted; no data
0009	the Rev A stack-up	accepted

Appendix H. Software and tool index

Item	Path
RF data layer	tools/rfkit/, with tools/rfkit/README.md
tests	tools/rfkit/tests/
SIM-001 builder and analysis	tools/sim/sim001_hfss.py, tools/sim/sim001_analyse.py
runbook builder	tools/runbooks/build.py
documentation checks	tools/check-docs.sh
schematic generator	hardware/rev-a/tools/generate-schematic.py
figures of this document	tools/docs/master_reference_figures.py, output in docs/figures/
CI workflows	.github/workflows/
package list	requirements.txt

Part XII gives each tool's role and status.

Appendix I. AP-S requirement checklist

Every item is unchecked on 2026-10-04; each depends on reading the official call in full first.

- ☐ official call read in full and Part IX corrected against it
- ☐ decision recorded to pursue the contest, with scope
- ☐ team of 2 to 5 students, at least half undergraduates by end of May 2027
- ☐ AP-S member mentor, letter agreeing to supervise and advance costs
- ☐ receive only compliance confirmed for the chosen receiver
- ☐ COTS transmitters chosen, certified for use in Japan
- ☐ communication mode: $N = 4$ feasibility gate written and run
- ☐ receiver able to separate wanted and interfering sources chosen
- ☐ sensing task chosen, simplest reproducible first
- ☐ real time dashboard designed
- ☐ whole system bill of materials under US\$1,500
- ☐ four page proposal, 12 point Times New Roman, by 31 December 2026 (GMT-10)
- ☐ if selected: video, replication instructions and five page report by 24 May 2027

Appendix J. Unresolved questions, contradictions and stale documentation

Found while writing this document. None is resolved here; each needs its owner's decision or edit.

Technical points where repository wording should be corrected.

1. *The $4N - 4$ bound is stated as necessary.* docs/architecture/ml-calibration.md section 3 and decision 0002 say power only recovery “generically requires at least $4N - 4$ ”. The literature proves $4N - 4$ sufficient for generic vectors; necessity is proved only for some dimensions; eleven suffice at $N = 4$ (section 11.5). The conclusion survives in substance; the argument should be restated (section 11.6), and uncertainty I16 extended.

2. *A faster classical power only baseline is missing.* The fast amplitude only method of Long and co-authors, about $2N$ readings, is not in the baseline list B1 to B7 of benchmarks/specification.md; EXP-012 should include it, after verification.
3. *Decision 0002's second reopening condition was triggered and not written up.* G1 passed (decision 0004), and the condition asks for the first calibration comparison to be redone; the conclusion is unchanged and arguably strengthened, but the redone comparison is not recorded.
4. *The uniform array's first sidelobe at $N = 4$ is -11.3 dB, not about -13 dB.* The README table, docs/mathematics/formulation.md section 3 ("independent of N ") and the EXP-001 acceptance checklist ("first side lobe near -13 dB") use the large N value; a correct EXP-001 simulator would fail that check at $N = 4$.
5. *Common gain unobservability.* The repository says a common gain scaling leaves the sum port reading unchanged; strictly it changes absolute power, and is unobservable because it is confounded with the unknown probe path gain (section 11.4).
6. *"Degrades rather than failing silently"* in docs/mathematics/inverse-calibration.md section 3 holds only for a prior whose uncertainty is calibrated (section 39.2).
7. *The label reference plane is not fixed in one place:* conducted labels at element ports avoid enable switch leakage but exclude the antenna board; radiated per channel labels include everything but suffer leakage (section 16).
8. *No target for M_{required} :* no pointing, null depth or sensing target and no risk level α are recorded (section 0.2).
9. *Patch bandwidth against band 57a:* the analytical patch bandwidth, about 26 MHz, is about a third of the 83.5 MHz band over which decisions 0007 and 0008 judge (section 6.4).
10. *The AP-S direction is not recorded,* and its receiver requirement, separating wanted from interfering sources, is not met by the AD8318 alone (section 47). Demonstration time calibration would have to be power only and over the air against a commercial source, since no analyser travels to the venue.
11. *ansys-aedt-core is not pinned* in requirements.txt, so the SIM-001 environment is recorded only in session notes.
12. *Over the air calibration in a room is not a hardware calibration.* Found in the ISAC pass, version 0.2. The power forward model of docs/mathematics/inverse-calibration.md section 2 and the G4 test of decision 0008 assume a single far field probe at broadside. A demonstration calibration against an ambient transmitter at another angle, with multipath, estimates the product $h_n s_n$ of hardware and incident field (section 10bis.5). It would also trigger decision 0008's reopening condition "the probe position of the bench changes from broadside". How the demonstrator calibrates, and what its calibration means, needs a decision before it is designed.
13. *One experiment's noise is the other's signal.* EXP-005 Phase B treats the effect of a person moving as part of the repeatability floor; in the sensing mode that effect is the signal. The floor used to judge drift must be measured in a controlled, unoccupied configuration, and the two uses of the same measurement kept apart.

Stale or superseded text outside decision records. Decision records are kept as written by convention; these other documents could carry a supersession note.

Document	Stale statement	Superseded by
README.md	"104 tests"; "PyAEDT, planned, not used yet"; "Line lengths and antenna sizes wait for the board stack up"; the "Where it stands" table, last reviewed 2026-09-28	160 tests at the baseline; the SIM-001 builder; decision 0009
README.md, docs/mathematics/formulation.md section 3	side lobe level about -13 dB for four elements	item 4 above
docs/scope.md, last reviewed 2026-08-21	"Smallest credible prototype: two elements ... going to four elements adds no new question"; Bayesian optimisation described as choosing maximally informative measurements	decision 0002; docs/architecture/ml-calibration.md sections 4 and 8
docs/mathematics/formulation.md section 7	Bayesian optimisation as "the most defensible way to bring learning into this project"	decision 0002
docs/architecture/ml-calibration.md section 4 update; docs/architecture/rev-a-rf-architecture.md section 8 and 8.1; docs/hardware/rev-a-requirements.md section 6	the pattern synthesis track scheduled or "worth running" because $Q^{N-1} = 512$	decision 0005; ml-calibration.md section 6, ML-D; inverse-calibration.md section 5
experiments/EXP-004-instrument-audit.md status line	"five of nine observations recorded"	results/EXP-004/README.md: four complete, four partial, one not taken
ROADMAP.md	"Two element array built" as a phase 3 milestone	decision 0002 makes four elements the smallest prototype for the learning track;

Document	Stale statement	Superseded by
		two remain valid for the physics demonstration
docs/architecture/rev-a-rf-architecture.md section 5.5	converter resolution computed for 12 bits over 3.3 V	decision 0005 and control-architecture.md section 4.1
experiments/EXP-005-repeatability-floor.md section 7.3	0.5 mV called “about one fortieth” of the detector’s temperature figure	± 0.5 dB is ± 12.5 mV, so the ratio is nearer one twenty fifth of the half span; the rule itself is unaffected

Open questions carried by the repository, in docs/uncertainties.md: I1 to I27, with I12 (drift above the floor, gate G2) the most consequential, and I16, I18, I19, I24 to I27 directly relevant to this document.

Appendix K. Notation

Symbol	Meaning	Unit
f, f_0	frequency; working frequency 2.44 GHz	Hz
λ_0, λ_g	free space and guided wavelength	m
$k = 2\pi/\lambda_0, \beta, \alpha, \gamma$	free space wavenumber; phase, attenuation and propagation constants	rad/m, Np/m, 1/m
$\epsilon_r, \epsilon_{\text{eff}}, \tan \delta$	relative and effective permittivity; loss tangent	dimensionless
h, W, t	substrate height, trace width, copper thickness	m
Z_0, Z_L, Γ	characteristic and load impedance; reflection coefficient	ohm, ohm, dimensionless
a_i, b_i, S_{ij}	incident and outgoing waves; scattering parameters	$\sqrt{W}$, dimensionless
N, d, θ, θ_0	element count, spacing, observation and steering angles	count, m, rad
$AF(\theta)$	array factor	dimensionless
$\mathbf{a}(\theta)$	steering vector	dimensionless
$\mathbf{x}, \mathbf{w}$	commanded state; weights	16 bit word; complex
$\mathbf{H}_t, h_n = g_n e^{j\psi_n}$	array state; per channel gain and phase	complex
$\mathbf{z}_t$	identifiable relative parameters, six at $N = 4$	mixed: log ratio, rad
$\mathbf{S}_A, \mathbf{S}_{Bo}, \mathbf{C}$	antenna board scattering matrix; beamformer reverse path; coupling operator	dimensionless
σ_q	quantisation residual, 12.99 degrees	deg
η, η_c	budget fractions	dimensionless
T	phase agreement threshold, 2.29 degrees	deg
U	analyser expanded uncertainty	per S term
y_k, ϵ_k	measurement and noise	V or complex
$\mathcal{D}$	history of sessions	data
p_θ	learned prior	probability density
M_{required}	new measurements needed to reach the target	count
$\mathbb{I}, \mathbb{H}$	mutual information; entropy	nats

Appendix L. Where to find canonical information

README.md	short overview (partly stale, Appendix J)
CONVENTIONS.md	how documents, numbers and evidence are written
ROADMAP.md, BACKLOG.md	early plans (partly stale)
decisions/0001 ... 0009	every decision, with reopening conditions
docs/	
aetherarray-master-reference.md	this document
scope.md, uncertainties.md	scope (stale in places); I1 to I27
architecture/	ml-calibration, control-architecture, rev-a-rf-architecture, rf-data-layer, options
mathematics/	formulation, inverse-calibration
hardware/	requirements R1 to R9, measurement bench, inventory, BOM proposal
runbooks/	school task classes, register, SCH-001 and its PDF
references/	bibliography (verified entries), glossary
verification/strategy.md	checks before a result is trusted
figures/	generated figures of this document
benchmarks/	metrics, methodology, the benchmark contract

experiments/	plan, EXP-004, EXP-005, EXP-011, SIM-001
results/	what was actually measured: EXP-004 readings, empty tables otherwise
hardware/rev-a/	schematic, generator, BOM, ERC, layout constraints
hardware/rev-a/stackup/	reva-stackup.json (canonical), candidates.json, README
research/	state of the art, research plan
tools/	rfkit, sim, runbooks, check-docs, docs figures

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Verification levels: **full** means the source was read for the repository's bibliography as recorded there; **abstract** means abstract only, as recorded there; **index level** means bibliographic details and abstract wording seen in search engine results on 2026-10-04, with no full text read; **snippet only** means the official page itself could not be opened. Identifiers beginning A, V, T and R are those of docs/references/bibliography.md, whose entries give full details and consultation dates.

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Review from five reader perspectives

A self review, written so that the weak sections are named rather than hidden.

Reader	Can they get what they need?	Where the document is weak
first year RF student	Part I builds from fields to S-parameters, microstrip, antennas and arrays, with a worked number in every section	Part I is dense; some sections, especially 10 and 11, assume comfort with complex matrices; there are no exercises
RF professor	equations are standard and sourced; conventions stated; every project number traced to a decision or a generated table; switched line hazards named	no simulated or measured RF result exists to examine; switch behaviour at 2.44 GHz rests on typical spot values; the off arm resonance and leakage analyses are qualitative; the textbook citations are index level
ML professor	the learned object, likelihood and prior are separated; baselines include a non learned temporal model; temporal splits, leakage and prior calibration are addressed	no data exist, so the model class, the data rate and the drift magnitude are all unknown; the target δ and risk α are undefined; the power only likelihood needs approximations not yet specified
AP-S jury	Part IX states the official requirements separately from the response, shows why nulls make calibration visible, and names the receiver gap and the schedule risk	contest facts are snippet level; no feasibility numbers, sensing data or receiver choice exist; the team and mentor are not recorded; schedule is the dominant risk
hardware recruiter	real skills are visible and attributable: stack-up selection from fabricator data, error budgeting, pre-registered experiments, a scripted schematic, PyAEDT tooling, a tested RF library, FPGA timing design	nothing has been fabricated, assembled, brought up or measured; gateway and layout skills are specified, not yet demonstrated

The ISAC pass, read from five angles (version 0.2).

Reader	Question	Answer, and remaining weakness
RF professor	does the ISAC explanation have electromagnetic meaning?	yes: the field is a sum of paths, the array samples it, every power is a quadratic form of the spatial covariance (10bis.3, 10bis.4); weakness: narrowband model, no delay spread, no measured multipath
signal processing professor	is the sensing formulation defensible?	the observability statement and the confound $\tilde{\mathbf{R}} = \mathbf{H}\mathbf{R}\mathbf{H}^H$ are exact for the diagonal model; weakness: no detector, no false alarm model, no data, bursty sources not modelled
ML professor	is learning solving a genuine temporal inference problem?	yes: estimating a slowly drifting hidden state from sparse measurements, with a physical likelihood; the ISAC setting adds the attribution problem of 10bis.5 but no new reason for a learned beamformer
AP-S reviewer	is the contest implementation linked to the official problem?	section 46 traces every official requirement to a physical function, an implementation and a metric; weakness: snippet level rules, no receiver choice, no feasibility numbers
skeptical engineer	does every claim map to evidence or a labelled future experiment?	section 10bis.8 lists each ISAC claim with its status and the experiment that would establish it; none is established today

Sections where current evidence is insufficient, by design of this document: 6.4 and 22 (antenna performance), 15.3 (switch behaviour), 17 (detector chain performance), 37 (drift protocol), 40 to 43 (learning), 10bis.4 to 10bis.6 and 45 to 52 (ISAC and the demonstrator), and every row of Part X that reads NOT STARTED.